

SHARING COORDINATES: IDENTIFIABILITY AND DECISIONS FROM OPTIMIZER RESPONSE

Author information pending

ABSTRACT

When layers share learned bases B through a simplex router A , the effective weights $\Theta = AB$ need not determine the factors or their training response. We ask whether optimizer response identifies these factors and how much of that information is needed to choose a hard-sharing partition. For positive full-rank softmax factors on the same product fiber, complete response under fixed known positive Euclidean block rates and temperature is identical if and only if the factors differ by a common permutation. On uniformly nondegenerate domains we give an explicit inverse bound and, when $D - K \geq L$, a whole-feasible-set diameter bound from a noisy product and K designed matrix-valued response queries, under a stated product-noise threshold. Choosing a partition requires less information than recovering the factors: balanced router-vector clustering minimizes full Hilbert–Schmidt response error, and off-diagonal router Gram entries suffice to select a minimizer. It can conflict with weight clustering, even across factorizations with the same effective product. We construct a fixed squared loss whose gradient flow separates coefficient assignment from weight clustering, and give conditional agreement and trajectory-error bounds. In controlled experiments on nine pretrained language models, factor recovery error was below 0.034 in all 1,080 cases meeting the specified rank conditions. A separate three-model study found decision crossings during training. Local SGD from a selected real-model state and a controlled regression experiment reproduced crossings across numerical precisions and step sizes. The results establish identification under the specified optimizer and show that training can separate sharing decisions. They leave open how often crossings occur and whether full factor recovery improves downstream performance; they do not identify historical sharing graphs or extend the identification theorem to AdamW.

1 INTRODUCTION

When several layers share a learned bank of parameters, their effective weights can be written as $\Theta = AB$: each row of a router A selects a convex combination of the rows of a jointly learned basis matrix B . The router is useful for optimization and compression, but its entries are not uniquely determined by the effective weights. To identify the *current* factors, we therefore need an additional observation. Even then, a decision based on those factors may differ from one obtained by clustering the effective weights.

This ambiguity is known in simplex-structured matrix factorization (SSMF) theory (Vu Thanh et al., 2023). For example, the invertible transform $M_s = (1 - s)I + s\mathbf{1}\mathbf{1}^\top/K$, $0 < s < 1$, preserves the product under $(A, B) \mapsto (AM_s, M_s^{-1}B)$ while moving each router row toward uniformity. This increases the entropy of nonuniform rows. It preserves coordinate ordering and argmax assignments; other feasible, nonuniform changes of basis can change the argmax partition (Appendix C). An effective model alone therefore need not determine coordinate-based summaries or decisions.

We study an additional observation supplied by a specified optimizer: the instantaneous response of the effective parameters to an arbitrary effective gradient. With $A = \text{softmax}_{\text{row}}(Z/\tau)$, fixed temperature, and known positive Euclidean learning rates for Z and B , this is a linear map $\mathcal{R}_{A,B} : G \mapsto -\dot{\Theta}$. We observe this map through controlled interventions at a fixed checkpoint, using the

same optimizer metric throughout. It contains more information than the response to a single task gradient; observing a passive training trajectory does not supply the same map.

Under positivity and rank assumptions, the response resolves the static ambiguity. For strictly positive, full-column-rank routers and full-row-rank bases with the same product, equal response operators imply a common basis permutation, and conversely. The proof recovers a Gram matrix and a family of router covariances from the response, then reduces the remaining ambiguity to simultaneous diagonalization. On uniformly nondegenerate domains, the same chain gives an explicit inverse-stability bound.

The complete-operator premise can also be replaced by finite, imperfect observations. If $D - K \geq L$, we design K unit-norm queries from a noisy observed product. Every pair of factors consistent with the product and response error budgets is close modulo permutation, with an explicit bound on the diameter of the *entire* feasible set. The true basis need not lie in the estimated product subspace, and candidates need not belong to one local solution branch. The bound describes the information available on the specified domain. Finding a feasible solution efficiently and verifying the domain’s rank and interior margins from data remain separate problems.

A structural decision may need less information than the full factors. For equal-capacity hard tying, minimizing full Euclidean-response error reduces exactly to clustering the router rows; off-diagonal Gram information suffices. This differs both from clustering the effective weights and from assigning each row to a fixed router coordinate. We construct strictly interior, full-rank examples in which these decisions disagree, including equal products with different response-optimal partitions. A fixed squared-loss gradient flow can separate coefficient assignment from weight clustering; finite margins can also protect agreement for a specified interval.

We establish the following results:

1. A complete Euclidean-response stabilizer characterization, an explicit inverse on nondegenerate domains, and a noisy-product, finite-query bound for the entire feasible factor set (Section 4).
2. An exact balanced response-decision objective and its sufficient Gram information, explicit conflicts with weight-based decisions, and a constructive gradient-flow crossing with conditional stability bounds (Section 5).
3. Recovery and training experiments on pretrained models: a fixed recovery protocol on nine pretrained models (160M–8B); a separate 18-run, three-model language-training study; a selected real-model local SGD reproduction; and controlled regression/precision checks (Section 6).

The analysis builds on existing results for response operators, diagonalization, robust inverses, Gram-based clustering, and trajectory perturbations. We use these tools to derive the identification guarantees and decision consequences of the present observation model. The original nine-checkpoint structural test shows no incremental benefit from recovery over effective-weight clustering. The subsequent decision study asks whether training can reach a disagreement. No crossing was detected in its ordinary SGD runs under the specified schedule, although local SGD from a selected state did cross. This establishes possibility while leaving its frequency open. Throughout, identification concerns the current optimizer-dependent factorization; the historical process that produced it is outside the observation model.

2 RELATED WORK

Static factorization and coordinate dependence. Under $X = \Theta^\top$, $W = B^\top$, and $H = A^\top$, our static model is unconstrained SSMF. The uniformizing path of Vu Thanh et al. (2023, Sec. IV-A) is exactly the one above after $s = \delta/(1 + \delta)$; positive static recovery requires additional geometric conditions (Fu et al., 2018; Abdolali & Gillis, 2021). Low-rank factor gauges, ordinary-update dependence on coordinates, and gauge-sensitive merging are also established (Yen et al., 2025; Zheng & Wu, 2026; Gong & Xu, 2026). Our auxiliary ASLoRA experiment applies this perspective to its documented first-merge rule (Hu et al., 2024).

Optimizer response and identifiability. Jacobian–Gram operators and induced factor flows are standard (Jacot et al., 2018; Marcotte et al., 2026). Varre et al. (2026, Lem. D.2) already derive the single-router squared-covariance softmax flow; Euclidean factor isometries and compatible softmax permutation/shift updates are studied by Singh (2026); Lau & Su (2026). Function-level and conditional-density softmax-MoE identifiability and inverse bounds use different observations (Tran et al., 2025; Nguyen et al., 2023). Our question is a converse at two fixed factorizations of the same finite product: does equal complete response exclude every non-permutation gauge? It is metric-specific; natural-gradient or quotient geometries can instead remove chart dependence (van Oostrum et al., 2023; Mishra et al., 2014).

The tools behind the inverse. Orthogonal-to-permutation mechanisms, joint-diagonalizer uniqueness and sensitivity, and robust tensor uniqueness precede our final algebraic steps (Dorrell et al., 2026; Afsari, 2008; Bhaskara et al., 2014). In particular, Bhaskara et al. (2014, Thm. 5) already give a linear robust inverse when their conditioning parameters are fixed. That work establishes linear dependence and global robust uniqueness. Our quantitative analysis tracks the stable reduction from response observations to diagonal data, eliminates scaling through the simplex constraints, and controls both original factors with explicit constants. Matrix–vector and structured operator probing are likewise established (Novak et al., 2022; Chiu & Demanet, 2012; Halikias & Townsend, 2024; Amsel et al., 2026). The noisy-product result combines a uniform bound outside the observed subspace with this response inverse. Whether K queries are optimal remains open. Appendix H maps the closest results, the wider sharing literature, and unresolved full-text access limitations.

Clustering and preserving training dynamics. Kernel k -means already expresses within-cluster distortion as a Gram-matrix trace objective (Dhillon et al., 2004). Neural Tangent Transfer matches both initial outputs and empirical training kernels to construct trainable sparse networks (Liu & Zenke, 2020). These works establish Gram-based clustering, kernel preservation, and the tradeoff between matching initial states and training dynamics. Here we reduce the *complete softmax-factor response* exactly to balanced router clustering: its local covariance blocks contribute a partition-independent term. The product-preserving decision counterexample and the explicit transverse crossing delimit the information and dynamical consequences for this shared-basis model. A standard Gronwall bound relates the two distortion objectives along training trajectories.

3 MODEL AND OBSERVATION

Let L be the number of layers, K the number of shared bases, and D the number of mixed parameters per layer. Write

$$\Theta = AB, \quad A \in \Delta_K^L, \quad B \in \mathbb{R}^{K \times D}, \quad (1)$$

where Δ_K^L denotes nonnegative, row-stochastic $L \times K$ matrices. The architecture mixes parameters: layer i uses the product row $a_i B$. Mixtures of nonlinear expert *outputs* require a different analysis. Two factors are product-equivalent when

$$(A, B) \sim_{\Theta} (A', B') \iff AB = A'B'. \quad (2)$$

We measure factor discrepancy modulo the unavoidable basis labels:

$$d_{\text{perm}}((A, B), (A', B')) = \min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2}. \quad (3)$$

Here $\mathfrak{S}_K$ is the set of permutation matrices and $\|S\|_{F \rightarrow F} = \sup_{\|G\|_F=1} \|S(G)\|_F$.

Controlled Euclidean observation. Set $A = \text{softmax}_{\text{row}}(Z/\tau)$ with fixed $\tau > 0$. For a loss $\mathcal{L} = \ell(\Theta)$ and known positive block rates, let

$$\dot{Z} = -\eta_Z \nabla_Z \mathcal{L}, \quad \dot{B} = -\eta_B \nabla_B \mathcal{L}, \quad (4)$$

$$\mathcal{R}_{A,B}(G) = -\dot{\Theta}. \quad (5)$$

where $G = \nabla_{\Theta} \ell$. Every $G \in \mathbb{R}^{L \times D}$ is admissible, using the test loss $\ell_G(\Theta) = \langle G, \Theta \rangle$. The comparison fixes τ, η_Z, η_B across charts. Direct factor penalties, adaptive preconditioning, and unknown optimizer state define different observation laws.

Target and scope. We identify the *current factors* under this law. An effective-parameter partition instead groups equal rows of Θ ; a functional partition groups equal transition functions on a stated input distribution; a historical graph describes the process that generated the model. Each definition refers to a different object. A coordinate-derived action r is product invariant only if

$$r(A, B) = r(A', B') \quad \text{whenever } (A, B) \sim_{\Theta} (A', B'). \quad (6)$$

Identification relative to an optimizer does not itself imply this property, small action regret, or stable argmax decisions without a coordinate gap. The auxiliary decision experiments evaluate each selected action in the same canonical model, without assuming a historical ground-truth partition.

4 IDENTIFICATION AND STABILITY FROM OPTIMIZER RESPONSE

4.1 EXACT RESPONSE IDENTIFIES THE CURRENT FACTORS

Writing router and gradient rows as columns a_i, g_i , and setting $C_i = \text{Diag}(a_i) - a_i a_i^\top$, the chain rule gives

$$[\mathcal{R}_{A,B}(G)]_i = \eta_B \sum_{j=1}^L \langle a_i, a_j \rangle g_j + \frac{\eta_Z}{\tau^2} B^\top C_i^2 B g_i. \quad (7)$$

The first term couples layers through the router Gram matrix; the second records how the softmax coordinates move. Together, these terms provide the information used in the following theorem.

Theorem 1 (Exact Euclidean-response stabilizer). *Let $A, A' \in \Delta_K^L$ be strictly positive with column rank K , and let $B, B' \in \mathbb{R}^{K \times D}$ have row rank K . Suppose $AB = A'B'$, and use the same $\tau, \eta_Z, \eta_B > 0$ in both parameterizations. Then*

$$\mathcal{R}_{A,B}(G) = \mathcal{R}_{A',B'}(G) \quad \text{for every } G \in \mathbb{R}^{L \times D} \quad (8)$$

if and only if there is a permutation matrix P such that

$$A' = AP, \quad B' = P^\top B. \quad (9)$$

The corresponding logits remain unidentified under the independent shifts $z_i \mapsto z_i + c_i \mathbf{1}$.

Proof idea. Full rank and equal products imply $A' = AM$, $B' = M^{-1}B$, with $M\mathbf{1} = \mathbf{1}$. Off-diagonal response blocks determine the off-diagonal router Gram entries. A diagonal block is a scalar identity plus $\lambda B^\top C_i^2 B$, where $\lambda = \eta_Z/\tau^2$ and the latter matrix is positive semidefinite and singular, including when $D = K$. Its smallest eigenvalue therefore separates the scalar part. Equal responses give $AA^\top = A'A'^\top$ and hence $MM^\top = I$. Cancelling the full-row-rank basis from the remaining blocks and taking the unique positive semidefinite square root yields $C_i = MC_i' M^\top$. The rank-one parts cancel, leaving $\text{Diag}(a_i) = M \text{Diag}(a_i') M^\top$ for every i . Because the router rows span $\mathbb{R}^K$, this forces M to be a signed permutation; $M\mathbf{1} = \mathbf{1}$ makes every sign positive. The converse follows by substitution. Appendix D.2 gives the complete proof and the boundary examples.

For $K \geq 3$, Gram information alone is insufficient: non-permutation orthogonal transformations fixing $\mathbf{1}$ can preserve positivity. The covariance blocks resolve the remaining ambiguity. Dropping either rank assumption admits counterexamples, as does replacing the full response by one realized gradient. The conclusion depends on the specified Euclidean observation; different optimizers need not select the same chart.

4.2 QUANTITATIVE INVERSE AND IMPERFECT PRODUCTS

Corollary 1 (Stable response identification). *Fix $K \geq 2$, $L \geq K$, and $D \geq K$. Restrict both factorizations to a uniformly nondegenerate set: every router entry is at least $\alpha > 0$, the smallest singular values of both routers and both basis matrices are at least $s_A, s_B > 0$, respectively, and their Frobenius norms are at most L_A, L_B . If $AB = A'B'$ and both response operators use the same $\tau, \eta_Z, \eta_B > 0$, then a finite explicit constant C , depending only on $K, L, \alpha, s_A, s_B, L_A, L_B, \tau, \eta_Z, \eta_B$, satisfies*

$$\min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2} \leq C \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}. \quad (10)$$

Appendix D.3 gives an explicit conservative choice of C , the proof, and families showing why the interior and two rank margins enter; the norm bounds control the nonlocal diameter. The constant establishes inverse stability but can be loose in practical models.

Why a linear inverse is possible. Near equality of the response first controls the Gram and squared-covariance data. A general positive semidefinite square-root estimate would lose a square root in the error. Here the compared router covariances share the kernel $\text{span}(\mathbf{1})$, and the interior/rank margins give a uniform spectral gap on its orthogonal complement. A Sylvester equation therefore controls their difference linearly. The resulting approximate diagonal relations round to a permutation; a bounded-domain diameter handles large discrepancies. These steps bound the discrepancy across the stated domain, including pairs far from a preselected solution.

The assumptions have distinct roles. Appendix D.3 gives families with a vanishing router rank margin, basis rank margin, or interior margin for which a uniform linear inverse fails. In the last family both factors remain well-conditioned while the factor discrepancy is of order t and the response discrepancy is of order t^2 . The constants are explicit but deliberately conservative.

Allowing a small product discrepancy. Equal products can be relaxed without assuming the two factors are already close. Write $e_\Theta = \|AB - A'B'\|_F$ and $e_R = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}$. For two factors in the same nondegenerate domain, if $e_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$, then Corollary 6 gives

$$d_{\text{perm}}((A, B), (A', B')) \leq C_* e_R + \kappa(1 + C_* L_R) e_\Theta. \quad (11)$$

Here $\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}$; C_* is the explicit inverse constant on the enlarged domain $(\alpha/2, s_A/2, s_B/2, 2L_A, 2L_B)$, and L_R is its explicit forward Lipschitz bound. All constants are specified in Appendix E. The proof projects A' onto $\text{col}(A)$ and constructs a nearby intermediate basis with product exactly AB . It applies the same-product inverse there and then returns to the original pair. The smallness condition concerns the product discrepancy alone; the factors may lie in different neighborhoods or local solution branches.

4.3 FINITE OBSERVATIONS, INCLUDING A NOISY PRODUCT

The response is structured: its shared part is a layer Gram matrix, and each local block is supported in the product row space. This permits finite reconstruction when that space is known exactly.

Corollary 2 (Finite structured response tomography). *Under the hypotheses of Theorem 1, suppose $D - K \geq L$. Let $S = \text{row}(\Theta)$, choose an orthonormal basis $u_1, \dots, u_K$ of S , and choose orthonormal $v_1, \dots, v_L \in S^\perp$, all from Θ . Define K effective-gradient probes by*

$$g_j^{(1)} = v_j + u_1, \quad g_j^{(r)} = u_r \quad (2 \leq r \leq K, 1 \leq j \leq L).$$

Their K response matrices determine the complete operator $\mathcal{R}_{A,B}$. Hence two exact full-rank charts of the same Θ agree on these probes if and only if they differ by a simultaneous basis permutation (with the same residual logit row shifts as above).

Appendix D.4 gives the decoder and its component noise bounds with an exact product. Natural task gradients can replace these designed probes when the complementary shared system and every layer's local system have full rank (Proposition 15). The number of gradients alone does not ensure these ranks; they must be checked at the checkpoint.

With a noisy product, its estimated row space generally excludes part of the true basis. The following result controls that missing part uniformly over all feasible factors. Its queries repeat the complement code in every probe, unlike Corollary 2.

Corollary 3 (Global diameter under finite noisy observations). *Let $L \geq K \geq 2$ and $D - K \geq L$. Fix the known positive rates and temperature, and restrict factors to row-stochastic routers with entries at least $\alpha > 0$, singular-value margins $s_A, s_B > 0$, and $\|B\|_F \leq M$ for $M > 0$. From an arbitrary observed product $\hat{\Theta}$, choose its top- K right-singular frame U and an orthonormal L -row complement frame W . There are exactly K unit-Frobenius probes $G_p = (W + \mathbf{1}_L u_p^\top)/\sqrt{2L}$, where $u_p^\top$ is row p of U . For nonnegative error budgets, let $\mathcal{F}$ consist of all domain members whose product residual is at most ε_Θ and whose stacked response residual on these probes is at most ε_R . If $\varepsilon_\Theta \leq \min\{\alpha s_B/4, s_A s_B/8\}$, then*

$$\text{diam}_{d_{\text{perm}}}(\mathcal{F}) \leq C_\Theta \varepsilon_\Theta + C_R \varepsilon_R.$$

Appendix F constructs the probes, expands every constant, and proves the bound for the entire feasible set in the original basis space. No true-subspace or local-branch premise is imposed.

Here the residuals are $\|AB - \hat{\Theta}\|_F$ and $(\sum_{p=1}^K \|\mathcal{R}_{A,B}(G_p) - \hat{R}_p\|_F^2)^{1/2}$. Each query returns a full $L \times D$ response, not a scalar. The constants depend only on the dimensions, declared domain margins, and known optimizer parameters. Noise can be deterministic, adversarial, or dependent on the chosen probes, provided these budgets hold.

Why the bound covers the whole feasible set. Let $\hat{P} = U^\top U$. Every feasible rank- K product competes with the best rank- K approximation of $\hat{\Theta}$, so

$$\|B(I - \hat{P})\|_F \leq 2\varepsilon_\Theta/s_A \quad \text{for every } (A, B) \in \mathcal{F}. \quad (12)$$

Replacing B temporarily by $B\hat{P}$ changes the complete response by at most $4\lambda M\varepsilon_\Theta/s_A$. For two such projected operators, multiplication of each response by $W^\top$ recovers their shared Gram difference. Subtracting the recovered shared term and multiplying by $U^\top$ then recovers their local blocks. This is a stable decoder on a common linear operator family. Adding back both projection errors gives, for any two feasible pairs,

$$\|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F} \leq 2h\varepsilon_R + \beta\varepsilon_\Theta, \quad (13)$$

where $h = \sqrt{2L}(K^{-1/2} + 1 + \sqrt{L})$ and $\beta = 8\lambda M(h\sqrt{K} + 1)/s_A$. Their original products differ by at most $2\varepsilon_\Theta$; Equation 11 then proves the diameter bound for the original, unprojected factors. Appendix F expands all constants and proves each step.

Interpreting the bound. If the true factors lie in the specified domain and satisfy the noise budgets, every feasible estimate lies within the displayed bound of them, up to a permutation. The theorem does not ensure that a solver finds a feasible estimate, infer domain membership from data, or provide numerically tight constants. It requires no smallness assumption on ε_R . For $K = 1$, the router is fixed and the product alone gives diameter at most $2\varepsilon_\Theta/\sqrt{L}$ without response queries. By comparison, the numerical check in Proposition 16 gives a weaker, local guarantee within one feasible component.

5 FROM IDENTIFIED COORDINATES TO SHARING DECISIONS

Once the factors are known, choosing which layers to tie still requires an objective. We compare three choices and ask whether their preferred partitions can diverge during training. Let $L = Km$ and let $\mathcal{H}_m$ contain binary $L \times K$ assignment matrices with one entry per row and m per column. For $H \in \mathcal{H}_m$, $Q_H = HH^\top/m$ projects onto group-constant rows. Partitions are compared without group labels. Define

$$S_A(H) = \langle A, H \rangle, \quad J_\Theta(H) = \|(I - Q_H)AB\|_F^2, \quad J_A(H) = \|(I - Q_H)A\|_F^2. \quad (14)$$

Coefficient assignment maximizes S_A : it assigns to fixed simplex vertices, since $\|A - H\|_F^2 = \|A\|_F^2 + L - 2S_A(H)$. *Weight clustering* minimizes J_Θ , refitting each group center; it is router clustering in the metric $BB^\top$. *Router clustering* minimizes J_A with free centers. These are different objectives, even when the original factors are known exactly.

5.1 WHAT RESPONSE INFORMATION IS SUFFICIENT FOR A DECISION?

After fixing H , optimize only the hard-shared bases with rate η_B . Their effective response is $\mathcal{R}_H = \eta_B(HH^\top) \otimes I_D$; it does not depend on their initial fitted values. Here $\|\cdot\|_{\text{HS}}$ denotes the Frobenius norm of the full $LD \times LD$ operator, distinct from $\|\cdot\|_{F \rightarrow F}$.

Theorem 2 (Response-optimal balanced tying). *For the soft response in Equation equation 7 and hard responses with the same $\eta_B > 0$, there is a scalar $c(A, B)$ independent of $H \in \mathcal{H}_m$ such that*

$$\|\mathcal{R}_{A,B} - \mathcal{R}_H\|_{\text{HS}}^2 = c(A, B) + 2\eta_B^2 Dm J_A(H). \quad (15)$$

Thus the global minimizers are exactly the balanced router-clustering partitions. Off-diagonal entries of $AA^\top$ suffice to select them.

Proof. Set $\gamma = \eta_Z/\tau^2$ and $T_i = B^\top C_i^2 B$. Blockwise expansion gives

$$\begin{aligned} \|\mathcal{R}_{A,B} - \mathcal{R}_H\|_{\text{HS}}^2 &= \eta_B^2 D \|AA^\top - HH^\top\|_F^2 \\ &\quad + 2\eta_B \gamma \sum_i (\|a_i\|^2 - 1) \text{tr } T_i + \gamma^2 \sum_i \|T_i\|_F^2. \end{aligned}$$

The last two terms are partition independent. Use $\|HH^\top\|_F^2 = Lm$ and $\langle AA^\top, HH^\top \rangle = \|H^\top A\|_F^2 = m(\|A\|_F^2 - J_A(H))$. Diagonal Gram entries contribute the same sum for every partition, proving the final assertion. $\square$

The reduction uses standard Gram-based clustering algebra (Dhillon et al., 2004). Under the equal-capacity and learning-rate assumptions, the full softmax covariance contribution is independent of the partition. The metric averages squared response error over isotropic effective gradients. It need not select the best partition for actual task gradients, operator norm, AdamW, or final loss. Unlike full factor identification, this decision does not require recovering A : the off-diagonal response blocks already provide the sufficient Gram entries. Response-Gram clustering is therefore a simpler sufficient alternative for this objective.

Equivalence and strict conflicts. For $K = 2$ and $b_1 \neq b_2$, coefficient assignment and global balanced weight clustering have exactly the same optimal unlabeled partitions. Writing $a_i = (t_i, 1 - t_i)$, both select the top $L/2$ values of t_i and their complement. This includes ties and is a global-objective statement. For arbitrary K , if $P_0 B B^\top P_0 = \lambda P_0$, where $P_0 = I - \mathbf{1}\mathbf{1}^\top/K$ and $\lambda > 0$, then $J_\Theta = \lambda J_A$. This equates the two free-center clustering objectives, not coefficient assignment.

For $K = 3$, strictly interior full-rank examples separate all these claims (Appendix A). One example with $L = 6, D = 9$ and basis condition number four has a unique weight-optimal partition with $(J_\Theta, J_A) = (65, 50)/529$, whereas the unique response-optimal partition has $(119, 14)/529$. A separate, exact product-preserving change of factors gives two identical effective models with *different unique* response-optimal partitions. Consequently no decision using only Θ can always be response optimal. Response information can therefore matter even when full coordinate reconstruction is unnecessary.

5.2 TRAINING REACHABILITY AND FINITE-TIME LIMITS

The static examples leave open whether training can move from agreement to disagreement. The following existence result shows that it can.

Theorem 3 (A realizable transverse decision crossing). *There is a fixed squared loss $\ell(\Theta) = \frac{1}{2}\|\Theta - T\|_F^2$ and a nonempty open set of initial softmax/shared-basis states with $K = 3, L = 6, D = 9$ such that, under Euclidean factor gradient flow with $\eta_B = \eta_Z = \tau = 1$, coefficient assignment and global balanced weight clustering start with the same unique partition and subsequently disagree. The weight partition stays fixed through the crossing; both factors remain full rank and the router stays strictly positive on the considered interval.*

Mechanism. Appendix A.3 gives rational factors at a tie between exactly two coefficient assignments, H_p, H_q , with a strict weight-clustering gap. Put $E = H_p - H_q$ and set the fixed loss gradient at that point to $g_i = B^\top C_i^2 e_i$ by choosing $T = AB - G$. Then

$$\left. \frac{d}{dt} \langle A, H_p - H_q \rangle \right|_{t=0} = - \sum_i \|B^\top C_i^2 e_i\|^2 < 0. \quad (16)$$

Local ODE existence and the strict remaining gaps give a crossing while the weight optimum stays fixed. Taking a nearby earlier point as the initial state produces ordinary forward training under the same loss. Continuous dependence yields an open neighborhood of such initial states. The constructed target proves that the crossing is reachable. Its frequency and attraction properties under natural data remain open.

Positive margins can also protect agreement for a finite time. Appendix A.4 gives explicit bounds using initial assignment/clustering gaps and bounds on $\|B\|_2$ and $\|\nabla \ell\|_F$. Under these conditions, agreement persists for the bounded interval; it need neither hold indefinitely nor inevitably break down.

Why static and dynamic objectives can both matter. Consider soft and hard trajectories of the same loss. Assume its gradient is L_ℓ -Lipschitz between the trajectories, $\|\nabla \ell(\Theta_s(t))\|_F \leq M$, and

$\|\mathcal{R}_s(t) - \mathcal{R}_H\|_{F \rightarrow F} \leq \delta$ throughout the interval. With $c = \eta_B m L_\ell$ and group-mean initialization,

$$\|\Theta_s(t) - \Theta_h(t)\|_F \leq e^{ct} \sqrt{J_{\Theta}(H)} + M\delta \frac{e^{ct} - 1}{c}. \quad (17)$$

The last quotient is t at $c = 0$. This standard trajectory perturbation bound, proved in Appendix A.4, separates initial weight distortion from continuing response distortion. Matching training kernels is already a compression objective (Liu & Zenke, 2020). Our instantaneous Hilbert–Schmidt optimum only bounds operator mismatch at that time; it does not supply a uniform-in-time δ or guarantee lower downstream loss.

6 EMPIRICAL EVIDENCE: IDENTIFICATION AND DECISION DYNAMICS

We evaluate factor recovery in trained modules attached to pretrained language models, varying the observation noise and query design. We then test whether the recovered coordinates improve a structural decision. These experiments measure numerical behavior on selected factorizations; the global guarantees and their constants come from the theory. Earlier synthetic and byte-LM studies, including large errors and failed rejection rules, remain in Appendix J.

6.1 FACTOR RECOVERY ACROSS PRETRAINED MODELS

Nine models, a fixed module, and held-out gradient domains. We use three Pythia sizes (160M, 1B, 2.8B), four dense Qwen3 sizes (0.6B, 1.7B, 4B, 8B), and two SmolLM2 sizes (360M, 1.7B) (Biderman et al., 2023; Yang et al., 2025; Allal et al., 2025). Each frozen BF16 backbone receives a $K = 4$ softmax-shared additive module on 64 fixed output channels of every attention output projection. Its bases are direct trainable matrices, not LoRA products. A fixed isometric embedding preserves the factor-response model (Proposition 17). We train only these modules for 256 AdamW updates on WikiText-2, using three seeds per model and the final checkpoint. The resulting 27 checkpoints share nine fixed pretrained backbones, with three adapter-training replicates per backbone.

Natural queries are measured next-token-loss gradients from disjoint general-text, MBPP code, and GSM8K math documents (Merity et al., 2016; Austin et al., 2021; Cobbe et al., 2021). The code and math domains are not used for adaptation or calibration. Responses are controlled reverse-gradient/forward-JVP observations of the factor map in FP64, treating the measured BF16-backbone gradients as fixed inputs. These measurements require internal access to the product and matrix-valued responses. They measure the controlled Euclidean response, rather than the AdamW updates used to train the adapters. The full protocol, model revisions, rejected access requests, development corrections, and data/source hashes are in Appendix I.

Finite recovery across model sizes and query types. We compare the repeated-complement design of Corollary 3, Gaussian queries, and three natural gradient families, with $q \in \{1, 4, 8\}$, four independent held-out directions, and relative noise $0, 10^{-6}, 10^{-4}, 10^{-2}$ added to both product and responses. The noisy product subspace is re-estimated and orthogonal residuals are retained. Three estimators use observations only; permutation-aligned ground truth is supplied afterward for scoring.

All 1,080 cases with $q \geq 4$ met the observed rank conditions. Joint-fit factor error was below 0.034 in every case, with median 1.86×10^{-5} . At $q = 4$ without noise, the worst error across all models and query families is 8.31×10^{-13} . Table 1 reports maxima rather than selected seeds. All 540 cases with $q = 1$ failed the sufficient local-rank check and were rejected before joint fitting. This failure concerns the specified reconstruction condition; it leaves open what other one-query estimators could recover.

Acceptance rules after calibration. Four separate Pythia adapter checkpoints, with general-text validation queries only, fix acceptance thresholds before held-out recovery scoring. Defining an error above 0.10 as bad, the combined joint-fit rule accepted all 1,080 sufficient-rank cases without accepting a bad output. The residual-only rule accepted 868, also without a bad output. The absence of bad outputs in these cases prevents a comparison of the rules’ ability to detect them. For projected spectral recovery, the combined rule accepts 43 bad outputs among 1,056, versus 20 among 1,037 for residual-only; its largest accepted error is 6.19. Raw spectral recovery has an all-reject calibrated

Table 1: Joint-fit factor error with $q = 4$: maxima over three adaptation seeds, and (for natural queries) three text domains. Noise affects both product and response. D is the adapter basis dimension, not the backbone parameter count. Each reported maximum includes all declared cases.

Model	L	D	Natural, 0	Natural, 10^{-4}	Designed, 10^{-4}
Pythia 160M	12	49152	6.5×10^{-14}	1.6×10^{-4}	1.4×10^{-4}
Pythia 1B	16	131072	2.8×10^{-13}	5.7×10^{-5}	5.7×10^{-5}
Pythia 2.8B	32	163840	8.3×10^{-13}	3.9×10^{-5}	3.8×10^{-5}
Qwen3 0.6B	28	131072	6.6×10^{-14}	4.9×10^{-5}	4.9×10^{-5}
Qwen3 1.7B	28	131072	9.2×10^{-14}	4.2×10^{-5}	4.2×10^{-5}
Qwen3 4B	36	262144	1.0×10^{-13}	3.7×10^{-5}	3.7×10^{-5}
Qwen3 8B	36	262144	2.6×10^{-13}	3.7×10^{-5}	3.6×10^{-5}
SmolLM2 360M	32	61440	7.8×10^{-14}	3.8×10^{-5}	3.8×10^{-5}
SmolLM2 1.7B	24	131072	6.4×10^{-14}	4.5×10^{-5}	4.4×10^{-5}

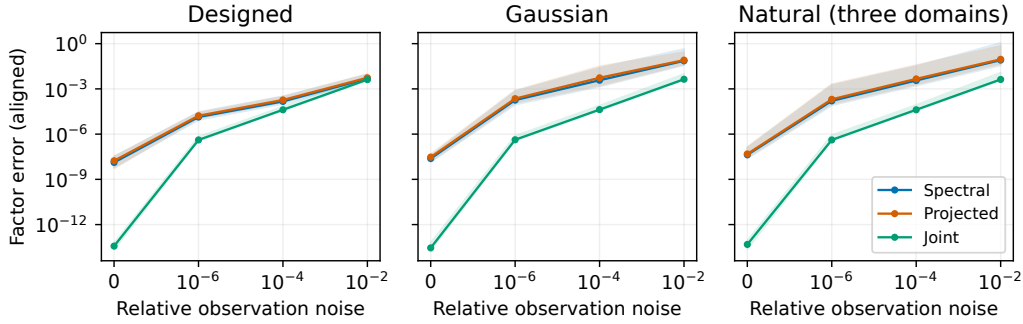


Figure 1: Recovery with four queries. Lines show medians and shaded bands the 10th–90th percentiles over all 27 checkpoints; the natural panel also includes all three domains. These are descriptive spreads across correlated cases, not confidence intervals. Noiseless results use controlled FP64 factor responses, not full-LLM FP64 inference.

policy under the strict simplex gate, even though many of its factor errors are small. Table 7 retains all denominators. The earlier synthetic study also retains a combined-rule accepted-bad fraction of 5.16%, above its 5% target. Neither calibration nor candidate-local diagnostics replace the theorem’s externally specified domain assumptions.

Finite precision and decision quality. The controlled response formula agrees with autodiff within 1.62×10^{-16} relative error. Finite-step subtraction is less reliable: at step 10^{-5} , FP32 errors range from 0.040 to 0.102, and BF16 errors from 1.00 to 1.95. Separately, shrinking the router-rank margin in learned factors amplifies noisy recovery error by roughly three orders of magnitude in the registered secondary stress test (Figure 4).

At the nine prespecified checkpoints, balanced coefficient assignment using recovered routers at noise 10^{-4} and 10^{-2} reproduced every native router partition. The fixed balanced Lloyd heuristic applied to effective parameters produced the same nine partitions. With equal-budget refitting and 64 recovery updates, coordinate recovery therefore showed no advantage over this simpler control. Random balanced groups worsen general-text NLL after recovery at all nine checkpoints, while code/math differences have mixed signs. These are token-loss results, not generation accuracy or realized inference speedups. Detailed action results, repeated-action numerical controls, costs, and independent CPU replays are in Appendix I.

6.2 CAN REAL TRAINING ENTER A DECISION DISAGREEMENT?

The agreement at the original nine checkpoints did not extend to the full saved set. A post hoc analysis of all 27 checkpoints found seven coefficient/weight-Lloyd disagreements. Exhaustive enumeration

Model	Optimizer	Coefficient	Router only	Response
Pythia-160M	AdamW	1/3	0/3	3/3
Qwen3-0.6B	AdamW	1/3	0/3	3/3
SmolLM2-360M	AdamW	2/3	1/3	3/3
All three	SGD	0/9	0/9	0/9

Table 2: Runs with at least one strict agreement-to-disagreement transition against global weight clustering, over the entire 256-step trajectory. Coefficient means S_A assignment; router only is its subset with unchanged weight partition; response means J_A clustering under the Euclidean-response objective, not an AdamW-optimal decision. All runs use 12 selected projections, $K = 4$, and 64 channels. Counts are descriptive, not prevalence estimates.

on Pythia-160M also separates global weight and response optima in two of its three checkpoints (Appendix B). These observations motivated a separate trajectory experiment. The original decision test and its fixed evaluation set remain as reported.

Ordinary initialization and natural-language losses. We run Pythia-160M, Qwen3-0.6B and SmolLM2-360M with three adapter seeds and two optimizers, AdamW and momentum-free SGD, for 18 complete 256-step runs. The loss is ordinary WikiText next-token cross entropy, with no grouping term or constructed target. Each run trains a $K = 4$, 64-channel shared module on 12 uniformly selected attention output projections; the backbone is frozen. Restricting the selected depth makes all 15,400 balanced partitions enumerable at *every* update. This differs from the all-eligible-layer recovery study. Models, seeds, steps and decision criteria were fixed before this grid; the grid itself is exploratory, motivated by previous checkpoint observations.

Table 2 reports all runs. Four of nine AdamW runs entered coefficient/weight disagreement; one changed only the coefficient partition. All nine entered router-clustering/weight disagreement, and eight remained in disagreement for at least five recorded states. None of the nine SGD runs crossed under the same, unretuned learning-rate schedule. These are dependent finite-run counts, not estimates of a population frequency or an optimized AdamW–SGD comparison.

In SmolLM2 seed 1, coefficient and weight partitions agree at update 70. At update 71, the old-minus-new coefficient score changes from 2.61×10^{-5} to -1.66×10^{-3} while the weight partition stays fixed, with global runner-up gaps 1.84 and 1.86. A fresh pretrained-backbone forward/backward pass and restored AdamW update reproduce the stored loss, gradients and update exactly in the same environment. This reproduced a crossing under the natural task loss and AdamW. The Euclidean-flow theorem concerns a different optimizer.

A controlled local reproduction on the real model. We then select that real update-70 state and its next recorded text batch, fix the batch, and optimize the unchanged next-token loss with equal-rate, no-momentum, unclipped SGD. No factors or targets are constructed. The state was selected *post hoc*, so this local experiment is separate from the ordinary-initialization grid. All four step sizes 0.02, 0.01, 0.005, 0.0025 cross while preserving the global weight optimum. A zero-rate FP32 control is unchanged. FP32 initially obscures refinement through accumulated rounding; an explicit precision follow-up retains all runs and evaluates the shared injection and RMS normalization without downcasting. In FP64, adjacent-resolution endpoint distances are 2.06×10^{-6} , 1.03×10^{-6} and 5.14×10^{-7} , showing empirical first-order refinement (Figure 2). The refinement supports the local crossing mechanism in this real model, although the trajectories remain numerical approximations. The absence of crossings in the ordinary-SGD grid is unchanged.

Discrete optimization of the constructed loss. For Theorem 3, automatic-differentiation SGD on the fixed squared loss reproduces crossing in all eight step-size/precision settings, with a zero-gradient control and 60 retained initialization perturbations. The FP64 endpoint errors against a forward ODE reference halve with step size. Appendix B records the noneligible perturbations, numerical precision corrections, all natural-run outcomes, and exact reproduction settings. Together, the studies show that training can reach disagreement in the specified shared module. Whether crossings are common, or full factor recovery improves task performance, remains open.

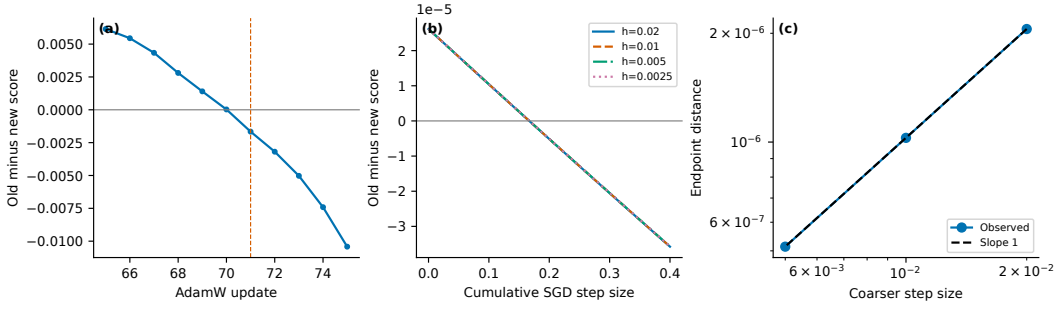


Figure 2: Complementary real-model evidence. (a) Natural-language AdamW training of SmolLM2-360M crosses a coefficient boundary at update 71 while the global weight partition stays fixed. (b) Starting from the selected real update-70 state, fixed-real-batch FP64 SGD crosses at all four step sizes, preserving the weight partition. Training time is the sum of step sizes, not seconds. (c) Adjacent-resolution final parameter distances approximately halve with step size. These are deterministic trajectory checks, not confidence intervals or evidence of prevalence.

7 DISCUSSION AND LIMITATIONS

Optimizer response can distinguish factorizations that have the same effective weights. For positive full-rank softmax/shared-basis factors, complete Euclidean response identifies the current coordinates up to permutation on a common product fiber. Explicit inverse bounds extend this observation to noisy products and finitely many matrix-valued queries on declared nondegenerate domains. A decision can require less information: balanced Hilbert–Schmidt-response-optimal tying only needs off-diagonal router Gram entries. Thus response-optimal tying can differ from weight clustering while requiring less than full factor recovery.

The identification guarantees require known positive Euclidean rates and temperature, internal access to the effective product and responses, and rank/interior margins; the specified finite-query construction additionally requires $D - K \geq L$ and a domain-dependent product-noise threshold. Constants are conservative, query optimality is unproved, and no efficient globally feasible recovery algorithm is supplied. Empirical rejection rules cannot certify these assumptions, and the experiments include false acceptances. The results do not identify an AdamW trajectory, historical sharing graph, or nonlinear output-mixture model.

Decision equivalence has narrower conditions than identification. The response objective uses equal group capacities, matching hard/soft basis rates, fixed hard routes and an isotropic full-operator error. It need not minimize error for observed task gradients or an adaptive optimizer. The trajectory bound requires response control throughout an interval, which a single-checkpoint optimum does not provide. The proof and experiments establish that crossings are possible. Their frequency and attraction properties under natural training remain open: the evidence comes from a constructed loss, a selected real-model state, and a finite exploratory grid. In particular, ordinary SGD produced no crossings in the grid with its unretuned learning-rate schedule.

The recovery experiments cover a fixed-rank 64-channel additive module in nine dense pretrained models up to 8B. The trajectory grid covers three models up to 0.6B with 12 selected projections; local refinement uses one selected SmolLM2 state and one fixed real batch. These are frozen-backbone adapter studies, not full-model training or architectural universality. Model-family comparisons confound architecture, pretraining and, for the selected Qwen models, post-training. In the original matched decision test, factor recovery provided no incremental benefit over effective-weight clustering. The later crossing experiments address how decisions change during training; downstream performance and compression gains remain unestablished. The results separate the information needed to identify current factors from that needed for a particular decision, and establish conditions under which training can move the decisions apart.

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AI ASSISTANCE DISCLOSURE

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A BALANCED DECISIONS: EQUIVALENCE, CONFLICT, AND DYNAMICS

This appendix supplies proofs and exact certificates for Section 5. Every partition has K groups of size $m = L/K$. All displayed layer indices are one-based. For row vectors x_i and a group U of size m ,

$$\sum_{i \in U} \|x_i - \bar{x}_U\|^2 = \frac{1}{m} \sum_{i < j, i, j \in U} \|x_i - x_j\|^2. \quad (18)$$

This identity follows by expanding both sides; it also provides finite integer certificates when all rows have a common rational denominator.

A.1 EQUIVALENCE CASES

Proposition 1 (Two-basis equivalence). *Let $K = 2$, $L = 2m$, and $b_1 \neq b_2$. The globally optimal unlabeled partitions for maximizing S_A and minimizing J_Θ coincide, including ties.*

Proof. Write $a_i = (t_i, 1 - t_i)$ and $\theta_i = b_2 + t_i(b_1 - b_2)$. Assign a size- m subset U to the first basis, put $s = \sum_{i \in U} t_i$ and $T = \sum_i t_i$. Its coefficient score is $m - T + 2s$, maximized by the top- m sum. Removing the common translation and positive scale from the weight SSE, minimizing it is equivalent to maximizing $s^2 + (T - s)^2$, hence $|s - T/2|$. The smallest and largest feasible sums are the bottom- m and top- m sums; they are complementary and thus correspond to the same unlabeled partitions. A convex quadratic attains its maximum over these sums only at those endpoints, unless all feasible sums coincide, in which case all partitions are tied. All endpoint realizations are precisely the tied sorting optima, proving the claim. $\square$

The condition on b_1, b_2 matters: identical bases make all weight partitions optimal. Nor does the proposition certify a heuristic clustering algorithm's output. For $m = 1$ every unlabeled partition is the singleton partition, independently of K .

Proposition 2 (Isotropy on the simplex tangent space). *If $P_0 B B^\top P_0 = \lambda P_0$, $\lambda > 0$, then $J_\Theta(H) = \lambda J_A(H)$ for every balanced H .*

Proof. For $X = (I - Q_H)A$, stochasticity gives $X\mathbf{1} = 0$, so $XP_0 = X$. Thus $\text{tr}(X B B^\top X^\top) = \lambda \|X\|_F^2$. $\square$

A.2 EXACT INTERIOR EXAMPLES AND SUFFICIENT INFORMATION

Set $A = N/23$, $B = [I_3 \ 0_{3 \times 6}]$, where

$$N = \begin{pmatrix} 8 & 11 & 4 \\ 8 & 5 & 10 \\ 10 & 9 & 4 \\ 5 & 11 & 7 \\ 6 & 10 & 7 \\ 11 & 5 & 7 \end{pmatrix}. \quad (19)$$

Every router entry is at least $4/23$, and the determinant of the first three rows of N is $276 \neq 0$; B has three unit singular values. The unique coefficient optimum has labels $(2, 3, 1, 2, 3, 1)$, partition $\mathcal{C} = \{\{1, 4\}, \{2, 5\}, \{3, 6\}\}$, and score $60/23$. For an explicit optimality certificate on the integer scale, subtract column prices $(1, 7/2, 0)$ from each row of N . Each row now has a *strict* unique maximum at the assigned column, and those maxima use each column twice. Adding the fixed price contribution to their sum gives 60. Swapping the assignments of rows 4 and 5 gives score 59; all scores are integers, so the runner-up gap is exactly $1/23$.

The unique weight and router-clustering optimum is instead $\mathcal{D} = \{\{1, 3\}, \{2, 6\}, \{4, 5\}\}$, cost $14/529$; its runner-up costs $31/529$, while $\mathcal{C}$ costs $41/529$. Table 3 exhausts the 15 unlabeled partitions. Thus isotropic bases do not equate coefficient assignment with free-center clustering. Strict gaps imply an open neighborhood of disagreement. Replacing A by $(1 - r)\mathbf{1}\mathbf{1}^\top/3 + rA$, $0 < r \leq 1$, multiplies coefficient-score differences by r and clustering-cost differences by r^2 . The same example can therefore approach the uniform router; its rank margin then shrinks. Proximity to uniformity alone is not an agreement condition.

For $K = 4$, append a fourth column and two rows: let $\tilde{N}$ have first six rows $(16N_{i1} + 23, 16N_{i2} + 23, 16N_{i3} + 23, 23)$ and last two rows $(23, 23, 23, 391)$. Then $\tilde{A} = \tilde{N}/460$ and $\tilde{B} = [I_4 \ 0_{4 \times 8}]$ are positive and full rank. Enumerating its 105 pair partitions gives $\mathcal{C}$ and $\mathcal{D}$ respectively, with the pair $\{7, 8\}$ added. The $K = 3$ example already establishes non-equivalence; this extension matches the experimental number of bases.

Weight–response tradeoff. In Equation equation 19, replace B by $[\text{Diag}(4, 1, 1) \ 0]$. Router clustering still uniquely selects $\mathcal{D}$. Weight clustering uniquely selects $\mathcal{E} = \{\{1, 2\}, \{3, 6\}, \{4, 5\}\}$, with

$$(J_\Theta(\mathcal{D}), J_A(\mathcal{D})) = (119, 14)/529, \quad (J_\Theta(\mathcal{E}), J_A(\mathcal{E})) = (65, 50)/529.$$

For $\eta_B = 1$, Theorem 2 gives response-error increase $2Dm(50 - 14)/529 = 1296/529$ when choosing $\mathcal{E}$. Neither objective uniformly controls the other.

Corollary 4 (Effective weights do not determine the response optimum). *There exist two strictly positive full-rank routers and full-row-rank bases with the same product but different unique balanced Hilbert–Schmidt–response-optimal partitions.*

Proof. Use the isotropic example above and

$$M = \frac{1}{10} \begin{pmatrix} 2 & 1 & 7 \\ 4 & 4 & 2 \\ 5 & 2 & 3 \end{pmatrix}, \quad A' = AM, \quad B' = M^{-1}B.$$

The matrix is positive and row stochastic, with determinant $-7/100$. Both factors retain their ranks, A' is strictly positive, and $A'B' = AB$. Router-clustering costs now have denominator 230^2 : $\mathcal{E}$

Partition	N	$N \text{ Diag}(4, 1, 1)$	$N M_{\text{int}}$	N_0
12 34 56	80	455	950	1070
12 35 46	85	475	1015	636
12 36 45	50	65	200	1358
13 24 56	56	341	830	758
13 25 46	59	389	917	732
13 26 45	14	119	212	1190
14 23 56	62	347	668	842
14 25 36	41	146	347	842
14 26 35	31	286	457	578
15 23 46	71	401	809	636
15 24 36	47	152	401	662
15 26 34	35	320	533	806
16 23 45	56	161	374	1202
16 24 35	67	322	781	506
16 25 34	65	350	803	914

Table 3: Complete integer SSE certificates. A label such as 12|34|56 denotes three pairs. Divide columns by 23^2 , 23^2 , 230^2 , and 92^2 , respectively, for the stated router/effective-weight costs. Here $M_{\text{int}} = 10M$. Each column is computed by Equation equation 18; no numerical optimizer is used.

uniquely costs 200/52900, and runner-up $\mathcal{D}$ costs 212/52900 (Table 3). Theorem 2 therefore selects $\mathcal{D}$ for the first factorization and $\mathcal{E}$ for the second. Any rule observing only their common product has the same input and cannot always return the unique optimum in both. $\square$

Both worlds use their stated Euclidean factor metric. This is not a passive coordinate substitution accompanied by a transformed optimizer. The result does not make full reconstruction necessary: the partition-dependent term in Theorem 2 is a sum of within-group off-diagonal entries of $AA^\top$. Those entries are already specified by off-diagonal response blocks. This observation is consistent with the insufficiency of Gram information for recovering all factors in Section 4.1: deciding and identifying are distinct targets.

A.3 A FIXED LOSS THAT CROSSES A DECISION BOUNDARY

For Theorem 3, take $B_0 = [I_3 \ 0_{3 \times 6}]$ and $A_0 = N_0/92$, with

$$N_0 = \begin{pmatrix} 40 & 24 & 28 \\ 24 & 44 & 24 \\ 32 & 28 & 32 \\ 43 & 37 & 12 \\ 24 & 20 & 48 \\ 37 & 31 & 24 \end{pmatrix}. \quad (20)$$

Its rows sum to 92 and are positive. The first three rows are linearly independent, so A_0 has column rank three. Define assignments $p = (1, 2, 3, 2, 3, 1)$ and $q = (1, 2, 3, 1, 3, 2)$. They alone maximize the coefficient score, both attaining $238/92$. Indeed, subtract integer column prices $(6, 0, 0)$ from N_0 . Rows 1,2,3,5 have strict maxima at columns 1,2,3,3; only rows 4 and 6 tie between columns 1 and 2. Capacity two forces one to each column, leaving exactly p, q . The next distinct integer assignment score is 228; only positivity of the remaining gap is needed. Table 3 shows that global weight clustering uniquely selects the unlabeled partition of p , costing 506/8464 versus 578/8464 for the runner-up.

Set $E = H_p - H_q$ with rows $e_i^\top$, and define fixed matrices by

$$(G_0)_i = B_0^\top C_i^2 e_i, \quad T = A_0 B_0 - G_0, \quad \ell(\Theta) = \frac{1}{2} \|\Theta - T\|_F^2, \quad (21)$$

where rows of G_0 are written as columns in the first expression and C_i is evaluated at A_0 . Initialize the reference solution at $Z_0 = \log A_0$, B_0 , at time zero. The loss is fixed, smooth and bounded below; its gradient there is G_0 . The exact chain rule is

$$\dot{z}_i = -C_i B_0 g_i, \quad \dot{a}_i = -C_i^2 B_0 g_i, \quad \dot{B} = -A_0^\top G_0.$$

Consequently Equation equation 16 holds. Its strict sign follows because $\ker C_i = \text{span}(\mathbf{1})$, every nonzero e_i is orthogonal to $\mathbf{1}$, and $B_0^\top$ is injective. At least one e_i is nonzero.

The smooth vector field has a solution on a two-sided neighborhood of zero. By continuity, all other assignment gaps, the unique clustering gap, positive entries and the two rank margins persist in a smaller neighborhood. The strict negative derivative gives $S_p > S_q$ just before zero and $S_p < S_q$ just after it. Weight clustering keeps p . Choose a sufficiently small negative-time point as a forward initial condition: the same fixed squared loss now trains from agreement into disagreement. Choose two such endpoints with strict gaps. Continuous dependence on initial conditions preserves their decisions and preserves the weight optimum on the intervening compact interval for a small open neighborhood of starting states. This proves Theorem 3.

The numerical experiment uses an explicit negative tangent displacement from (Z_0, B_0) rather than inverse integration. Its observed crossing checks that particular discrete initialization; the theorem's proof does not rely on the numerical solver.

Local router velocities are not structurally restricted. At any positive router and full-row-rank basis, every row velocity v_i with $\mathbf{1}^\top v_i = 0$ is realized at that point by choosing

$$g_i = -\frac{\tau^2}{\eta_Z} B^\top (BB^\top)^{-1} (C_i^2)^+ v_i.$$

Substitution into $\dot{a}_i = -(\eta_Z/\tau^2) C_i^2 B g_i$ gives v_i , because $C_i^2 (C_i^2)^+$ projects onto $\mathbf{1}^\perp$. A fixed squared loss with target $AB - G$ realizes that gradient at the point. The induced $\dot{B}$ is not independently controlled. Natural tasks do not provide arbitrary gradients, so this lemma supplies no typicality claim.

A.4 SEPARATION, MARGINS, AND TRAJECTORY ERROR

Proposition 3 (A sufficient agreement regime). *Suppose a balanced labeling c satisfies $a_{i,c(i)} \geq 1 - \epsilon_A$, $0 \leq \epsilon_A < 1/2$. Let d be the minimum separation of distinct basis rows and D_B their maximum separation. If $d > 4\epsilon_A D_B$, coefficient assignment and global balanced weight clustering uniquely select c .*

Proof. Each own coefficient strictly exceeds the others and its pointwise best labeling is balanced, so coefficient assignment is unique. Every θ_i is within $\epsilon = \epsilon_A D_B$ of $b_{c(i)}$. Within-group distances are at most 2ϵ ; between-group distances are at least $d - 2\epsilon > 2\epsilon$. All equal-capacity partitions have the same number of within-group pairs. A different partition loses and gains the same positive number of pairs; every new pair is strictly more expensive than every lost pair. Equation equation 18 therefore proves strict SSE optimality. $\square$

More generally, for any given balanced partition, minimum between-group squared distance exceeding maximum within-group squared distance certifies its global SSE optimality by the same argument. Failure of this sufficient condition is not evidence of non-optimality. Under the proposition's separation, farthest-first seeding selects one center from each true group; pointwise nearest-center assignment is then unique and balanced. Recomputed centers stay within their respective ϵ balls, so balanced Lloyd iterations retain that partition. This condition was not shown to hold for most trained checkpoints.

Proposition 4 (Finite-time margin protection). *Initially let the unique labeled coefficient winner have gap $\Gamma_A > 0$ and the unique unlabeled global weight-clustering winner have SSE gap $\Gamma_\Theta > 0$, with identical partitions. Let $X_0 = (I - \mathbf{1}\mathbf{1}^\top/L)\Theta(0)$. Both decisions remain unchanged under perturbations satisfying*

$$\sqrt{2L}\|\Delta A\|_F < \Gamma_A, \quad 2\|X_0\|_F \|\Delta\Theta\|_F + \|\Delta\Theta\|_F^2 < \Gamma_\Theta. \quad (22)$$

Proof. The difference of two assignment matrices has norm at most $\sqrt{2L}$; Cauchy–Schwarz bounds every score-gap perturbation. The difference of two orthogonal projections has operator norm at most one. Expanding each competing quadratic SSE difference bounds its change by $2\|X_0\|_F \varepsilon + \varepsilon^2$, where $\varepsilon = \|\Delta\Theta\|_F$. Centering is valid because every Q_H preserves $\mathbf{1}$ and centering does not increase perturbation norm. $\square$

If $\|B(t)\|_2 \leq \beta$ and $\|\nabla\ell(\Theta(t))\|_F \leq M$ on an interval, then

$$\|\dot{A}\|_F \leq v_A := \frac{\eta_Z \beta M}{4\tau^2}, \quad \|\dot{\Theta}\|_F \leq v_\Theta := \left(\eta_B L + \frac{\eta_Z \beta^2}{4\tau^2} \right) M. \quad (23)$$

Indeed, Gershgorin’s bound gives $\|C_i\|_2 \leq \max_k 2a_{ik}(1 - a_{ik}) \leq 1/2$, and $\|A\|_2^2 \leq \|A\|_F^2 \leq L$. Substituting $v_A t$ and $v_\Theta t$ for the perturbations in Equation equation 22 yields an explicit protected interval. This is conditional stability, not attraction toward agreement.

Proof of Equation equation 17. Write $E(t) = \Theta_s(t) - \Theta_h(t)$ and subtract the two ODEs:

$$\dot{E} = -(\mathcal{R}_s - \mathcal{R}_H)\nabla\ell(\Theta_s) - \mathcal{R}_H(\nabla\ell(\Theta_s) - \nabla\ell(\Theta_h)).$$

The hypotheses give an upper Dini derivative of $\|E\|_F$ bounded by $M\delta + c\|E\|_F$, since $\|HH^\top\|_2 = m$ and $c = \eta_B m L_\ell$. Integrating this scalar inequality yields $\|E(t)\|_F \leq e^{ct}\|E(0)\|_F + M\delta(e^{ct} - 1)/c$; the continuous $c = 0$ limit is $\|E(0)\|_F + M\delta t$. Group-mean initialization gives $\|E(0)\|_F = \sqrt{J_\Theta(H)}$. The bound requires response control over the entire interval, not just at its start, and supplies no lower bound or ranking of final task losses.

B DECISION DYNAMICS: PROTOCOLS, COMPLETE OUTCOMES, AND PRECISION

The examples in Appendix A establish that crossings are possible. Here we test the construction with discrete regression updates, look for crossings during pretrained-model adaptation without a grouping objective, and reproduce a local Euclidean crossing from a selected real-model state with a fixed batch. The selected-state experiment does not estimate crossing frequency. Downstream benefits of full factor recovery are outside these experiments.

B.1 POST HOC GEOMETRY OF THE ORIGINAL CHECKPOINTS

The frozen nine-checkpoint structural test in Appendix I uses coefficient assignment and a deterministic balanced Lloyd control. Both produce the same partitions there; that outcome remains unchanged. A later post hoc analysis of all 27 saved evaluation checkpoints finds seven coefficient/Lloyd differences: Pythia-160M seeds 0 and 1, Pythia-2.8B seed 0, Qwen3-0.6B seeds 0,1,2, and Qwen3-1.7B seed 2. The two analyses use different evaluation sets, which accounts for their different outcomes. Router-vector Lloyd differs from weight Lloyd on four checkpoints. Neither count establishes global clustering optima.

Pythia-160M has 12 layers, so we enumerate all $12!/(3!^4) = 15,400$ unlabeled partitions into four groups of three. The coefficient partition’s excess global weight SSE is respectively 0.365105, 0.364774, 0.198803 for seeds 0,1,2. Global weight and router-clustering optima differ for seeds 0 and 2, with weight runner-up gaps 0.007875, 0.141154 and router runner-up gaps 0.014359, 0.106921. Seed 1 has the same global optimum for the two clustering objectives. Only one of the 27 coefficient partitions passes the sufficient pairwise separation certificate from Appendix A.4. Failure of that sufficient certificate does not imply non-optimality. These FP64 enumerations are not interval-arithmetic proofs for model bytes; the positive margins distinguish the reported winners from numerical ties. The small rational examples have separate integer certificates.

B.2 CONTROLLED REGRESSION REPRODUCES THE CROSSING MECHANISM

Use A_0, B_0, T from Equations equation 20 and equation 21. Regard nine canonical input vectors as a regression dataset, with six output tasks and target matrix $T^\top$. The predictor is $(XB^\top)A^\top$, $X = I_9$, and the loss is half the *sum* of 54 squared errors, not their mean. Compute the analytical flow velocity (V_Z, V_B) at $(\log A_0, B_0)$ and initialize $(Z, B) = (\log A_0 - 0.1V_Z, B_0 - 0.1V_B)$. This is an explicit tangent displacement, not backward integration; the target stays fixed throughout training.

PyTorch automatic differentiation and vanilla SGD update both factors, without momentum or weight decay. Eight primary runs combine FP32/FP64 with $h = 0.02, 0.01, 0.005, 0.0025$ to common time 0.4 (20,40,80,160 updates). Every state enumerates all 90 labeled coefficient assignments and

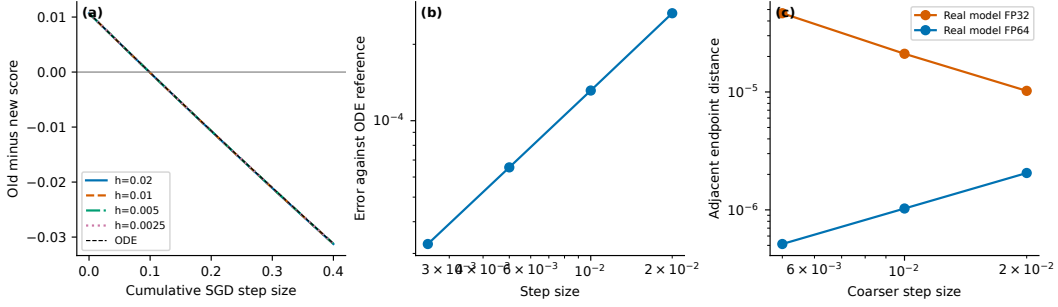


Figure 3: Controls and precision. (a) The fixed quadratic construction crosses under four FP64 SGD step sizes and the forward ODE reference. (b) Its endpoint error decreases approximately linearly with step size. (c) At the selected real-model state, FP32 adjacent-resolution endpoint differences increase as small updates accumulate; the dtype-preserving FP64 follow-up restores first-order refinement. The figure includes the earlier FP32 results as well as the FP64 follow-up.

15 unlabeled pair partitions. Both objective gaps must exceed 10^{-8} . An event starts from strict agreement and reaches strict disagreement with the original weight partition unchanged; sustained means five consecutive recorded post-update states. All eight runs met this event criterion and sustained disagreement, first crossing at updates 5,10,20,40 respectively. Loss is nonincreasing under absolute tolerances 10^{-12} (FP64) and 10^{-7} (FP32).

Analytical and automatic-differentiation parameter gradients agree within 5.56×10^{-17} absolute error at initialization. Forward `solve_ivp` from the same state uses relative/absolute tolerances 10^{-11} , 10^{-13} . FP64 SGD endpoint errors against this numerical reference are 2.6511×10^{-4} , 1.3149×10^{-4} , 6.5485×10^{-5} and 3.2678×10^{-5} . The reference is not an exactly integrated solution.

Perturb every initial logit and basis entry independently with Gaussian standard deviations 10^{-4} , 10^{-3} , 10^{-2} , seeds 0–19 at each scale, fixed target, FP64, $h = 0.005$, time 0.4. At the first two scales all 20 runs start eligible and cross. At 10^{-2} , 16/20 start with strict agreement and all 16 cross; the other four remain in the denominator as initially ineligible. A zero-gradient control uses the initial model’s own outputs as targets and remains stationary. These are perturbations of a deliberate construction, not independent natural-training samples.

B.3 ORDINARY NATURAL-LANGUAGE TRAINING: ALL 18 RUNS

The models use the cached revisions, tokenizers and WikiText training corpus in Appendix I. Freeze every backbone. Select 12 attention output projections uniformly across depth using rounded evenly spaced indices, adapt 64 fixed output rows per projection, and use four shared bases. Pythia-160M uses all layers; Qwen3-0.6B and SmolLM2-360M use subsets. Initialize router logits as standard normal and basis entries with standard deviation $0.01/\sqrt{d_{\text{in}}}$, with seeds 0,1,2. Repeated router initializations across models introduce dependence.

All 18 runs use 256 updates, batch size four and 128 next-token positions per block. AdamW uses zero decay and default moments; SGD uses no momentum or decay. Peak basis/router rates are 10^{-3} , 10^{-2} , with 16-step linear warmup and cosine decay to one tenth of the peaks. Clip global factor-gradient norm at one. Backbone/autocast arithmetic is BF16, factors FP32. SGD is not separately tuned. The batch generator seed is 4900000+adapter seed. No grouping loss, artificial label or constructed boundary initialization is used.

At initialization and after every update record $A, BB^\top$ in FP64. Compute global SSE by Equation equation 18; use exact capacity-constrained assignment for the coefficient objective. A strict winner has runner-up gap greater than $10^{-8} \max(1, |\text{winning objective}|)$. An event requires strictness of both objectives immediately before and after an agreement-to-disagreement transition. Also record persistence for five consecutive post-event states. “Router only” keeps the weight optimum fixed and changes the coefficient optimum. “Response” refers to the Euclidean objective of Theorem 2, including when assessed along AdamW; it does not mean AdamW-optimal tying.

Model	Optimizer	Seed	Coeff.	Router only	Response	Valid NLL before/after
Pythia-160M	adamw	0	—	—	10	4.3082/4.3030
Pythia-160M	adamw	1	—	—	61	4.3090/4.2026
Pythia-160M	adamw	2	6	—	6,52	4.3165/4.2819
Pythia-160M	sgd	0	—	—	—	4.3082/4.3291
Pythia-160M	sgd	1	—	—	—	4.3090/4.2969
Pythia-160M	sgd	2	—	—	—	4.3165/4.2986
Qwen3-0.6B	adamw	0	—	—	8,80	3.5664/2.8878
Qwen3-0.6B	adamw	1	—	—	156	3.5667/2.9090
Qwen3-0.6B	adamw	2	12	—	12	3.5667/2.8973
Qwen3-0.6B	sgd	0	—	—	—	3.5664/3.5517
Qwen3-0.6B	sgd	1	—	—	—	3.5667/3.5510
Qwen3-0.6B	sgd	2	—	—	—	3.5667/3.5474
SmolLM2-360M	adamw	0	—	—	5	3.0372/2.8057
SmolLM2-360M	adamw	1	71	71	97	3.0359/2.8102
SmolLM2-360M	adamw	2	26	—	26	3.0363/2.8040
SmolLM2-360M	sgd	0	—	—	—	3.0372/3.0357
SmolLM2-360M	sgd	1	—	—	—	3.0359/3.0352
SmolLM2-360M	sgd	2	—	—	—	3.0363/3.0369

Table 4: All 18 exploratory runs. Event columns give every transition step, not independent samples. Validation uses 32 fixed WikiText validation blocks per model, not a full-corpus benchmark. Identical seeds reuse the same initial router across models; architecture, pretraining, and selected layers differ. SGD uses the original schedule without retuning.

All runs complete. The Pythia-160M seed-1 AdamW response event does not sustain five states; each other AdamW run has a sustained response event. SGD has no detected event of any declared type. Validation NLL in Table 4 averages 32 fixed validation blocks, not the official full-corpus benchmark. Nine AdamW and seven SGD runs improve this diagnostic; no significance or optimizer-quality claim follows. Training NLL stored at an update is measured on that minibatch *before* the update; the geometric record is the after-update state.

For the first occurrence of each event type, save both factor states, preceding optimizer state, clipped gradients and batch indices. There are 14 records, including one physical update saved under multiple event types. CPU optimizer replay reproduces all within 2×10^{-6} absolute error. Independently recomputing 120 initial/final/event-boundary clustering costs using projector traces rather than pair distances gives the same winners and maximum cost difference 7.11×10^{-14} . These are implementation checks, not additional training replicates.

B.4 SELECTED REAL-MODEL LOCAL EUCLIDEAN REPRODUCTION

In SmolLM2 seed-1 AdamW, coefficient and weight partitions initially disagree, later agree, and diverge again at update 71. The local initial state is its actual update-70 state, not the random initialization. A fresh GPU pass on the recorded update-71 batch reproduces loss, gradients and restored AdamW update exactly in the same environment. The coefficient-gap derivative under its clipped negative logit gradient is -1.57367×10^{-4} . This motivates the explicitly *post hoc* follow-up.

Keep the trained state, pretrained weights and original four text blocks. Fix the batch and optimize the same mean next-token cross entropy, using equal basis/logit rates, no momentum, clipping or decay, and no autocast or TF32. Rates 0.02, 0.01, 0.005, 0.0025 run to time 0.4. All four FP32 runs cross with the weight partition fixed; a 20-step zero-rate FP32 control does not move. Adjacent-resolution endpoint distances 1.0221×10^{-5} , 2.1078×10^{-5} , 4.6522×10^{-5} instead increase with refinement, so FP32 does not establish numerical convergence.

We kept the FP32 results and ran two precision follow-ups. First convert the backbone and factors to FP64 and remove the shared injection’s FP32 cast. Source inspection reveals that RMSNorm still computes in FP32. Its crossings and approximate first-order refinement are retained as outer-FP64 results. Second, evaluate the identical RMS formula in the input dtype at all 65 normalization modules, preserving their weights and epsilon. Fixed rotary position constants may originate from FP32 but do

Step size	Updates	First crossing	Training time	Adjacent endpoint distance
0.0200	20	9	0.180	2.06×10^{-6}
0.0100	40	17	0.170	1.03×10^{-6}
0.0050	80	34	0.170	5.14×10^{-7}
0.0025	160	68	0.170	—

Table 5: Selected real-model local SGD with FP64 factors, backbone arithmetic and dtype-preserving RMSNorm. Training time is update count times step size. Endpoint distances compare each run to the next finer step size at time 0.4, not to an exact flow solution. All four runs keep the weight partition fixed.

not depend on trained factors. Cached BF16 pretrained weight values are merely converted to FP64; no extra pretrained information is introduced.

In dtype-preserving FP64, the initial coefficient-gap velocity is -1.57929×10^{-4} , predicting linearized crossing time 0.16556. Observed first crossing times are 0.18, 0.17, 0.17, 0.17; Table 5 reports empirical refinement. The weight optimum stays fixed at every recorded state. Minimum router entry is 0.006781; minimum singular values of A, B are 0.856727, 2.054078. Fixed-batch NLL decreases from 3.288429 to about 3.286897. This is local training/arithmetic evidence, not a generalization gain attributable to a sharing decision.

The experiment connects a state reached during language training to a local Euclidean crossing under the same language loss. Because the state was selected, it cannot estimate crossing probability or resolve the absence of crossings in the nine ordinary-SGD runs. It also leaves downstream decision quality untested. We use native factors to study the crossing; the effect of recovery error near the boundary remains open.

B.5 ARTIFACT AND COMPUTE BOUNDARIES

The local machine has eight RTX 5090 GPUs with approximately 32 GiB each. Ordinary trajectory runs use three GPUs concurrently; recorded per-job elapsed times sum to 419.12 seconds, including loading/evaluation. This is not wall-clock makespan or total GPU active time. Precision follow-ups use separate single GPUs; constructed regression uses CPU. We do not claim a complete research-compute total.

Protocols, every run, all precision variants and source/input hashes are retained. Experiments-directory entry points include `train_routing_crossing.py`, `train_llm_routing_trajectory.py`, `replay_llm_routing_backward.py`, and `train_llm_local_flow_fp64_norm.py`. `generate_dynamics_paper.py --check` reconstructs tables and vector figures. Frozen recovery sources/raw results are preserved; training refuses to overwrite completed results, so rerun in a fresh output/worktree. The record verifier distinguishes historical manuscript hashes from the current integrated paper. Artifact checks and AI-assisted self-review are not external human peer review.

C SECONDARY THEORETICAL STATEMENTS

This section preserves the auxiliary static, recovery, probe, and optimization statements referenced by the main text. They are separated from the central optimizer-response theorem because the static change of basis is SSMF prior art and the positive results use standard additional assumptions. Complete proofs and failure cases follow in Appendix D.

C.1 STATIC ROUTER-COORDINATE CONSEQUENCES

The rows $b_1, \dots, b_K$ are affinely independent when $[B, \mathbf{1}]$ has row rank K .

Proposition 5 (Local form of the SSMF gauge). *Assume $K \geq 2$, A is strictly positive and has column rank K , and the rows of B are affinely independent. For every nonzero $H \in \mathbb{R}^{K \times K}$ satisfying*

$H\mathbf{1} = 0$, and every sufficiently small $t \neq 0$,

$$M_t = I + tH, \quad A_t = AM_t, \quad B_t = M_t^{-1}B \quad (24)$$

is a valid factorization with $A_t \in \Delta_K^L$, affinely independent bases, and $A_t B_t = AB$. It is not permutation equivalent to (A, B) . The local equivalence class therefore has at least $K(K-1)$ continuous directions.

The condition $H\mathbf{1} = 0$ preserves row sums; positivity and invertibility survive a small perturbation; and $[B_t, \mathbf{1}] = M_t^{-1}[B, \mathbf{1}]$ preserves affine independence. Full column rank of A rules out a hidden permutation.

For $0 < s < 1$, the gauge

$$M_s = (1-s)I + s\mathbf{1}\mathbf{1}^\top / K \quad (25)$$

moves every router row toward uniform and strictly increases its entropy unless it was already uniform, while $M_s^{-1}B$ keeps Θ fixed. This uniformizing path preserves argmax assignments. They are not invariant under other feasible gauges, however. For example,

$$A = \begin{bmatrix} .1 & .9 \\ .3 & .7 \\ .4 & .6 \\ .9 & .1 \end{bmatrix}, \quad M = \begin{bmatrix} .9 & .1 \\ .4 & .6 \end{bmatrix} \quad (26)$$

gives argmax labels $(2, 2, 2, 1)$ before and $(2, 1, 1, 1)$ after multiplication by M . Yet $(AM)(M^{-1}B) = AB$ for every B .

This ambiguity does not erase every partition statistic. Because M is invertible, $a_i = a_j$ iff $a_i M = a_j M$; affine independence of the rows of B already makes this equivalent to $\theta_i = \theta_j$. Indeed, $qB = 0$ and $q\mathbf{1} = 0$ imply $q = 0$ when $[B, \mathbf{1}]$ has row rank K .

Proposition 6 (End-to-end impossibility). *Suppose an architecture and its observation law depend on (A, B) only through $\Theta = AB$. If two gauge-equivalent factorizations have different router summaries, no estimator based on end-to-end input–output observations can recover the correct summary in both worlds. This remains true with oracle access to the predictor on every input.*

This result concerns router coordinates; it neither equates effective-weight equality with functional equivalence nor asserts that either equals a planted generative graph. The latter is independently impossible under composition: a two-layer linear teacher with $W_1 = W_2 = I$ (tied) and one with $W_1 = 2I$, $W_2 = I/2$ (untied) both implement $x \mapsto x$.

C.2 POSITIVE RECOVERY UNDER ADDITIONAL INFORMATION

Call a factorization *separable* if every basis has an anchor layer i_j with $a_{i_j} = e_j$.

Proposition 7 (Standard separable recovery). *If two K -separable factorizations of the same Θ have affinely independent bases, they are permutation equivalent. Consequently, if routing is one-hot with discrete label $c_i \in \{1, \dots, K\}$, every basis is used, and the bases are affinely independent, then*

$$c_i = c_{i'} \iff \theta_i = \theta_{i'}, \quad (27)$$

so the sharing partition is recoverable up to basis relabeling.

This is standard separable-SSMF geometry specialized to layer rows, not a new simplex-identification result (Abdolali & Gillis, 2021). Separability makes $\text{conv}\{\theta_i\}$ equal the simplex whose vertices are the bases; vertices are unique, and affine independence gives unique barycentric coordinates. The restriction to the separable model class is essential: an unrestricted soft factorization may place a larger simplex outside the observed convex hull.

Theorem 4 (Stable noisy one-hot recovery). *Assume $K \geq 2$. Suppose $c_i \in \{1, \dots, K\}$ and $\hat{\theta}_i = b_{c_i} + e_i$, $\|e_i\|_2 \leq \varepsilon$, and $\gamma = \min_{j \neq k} \|b_j - b_k\|_2 > 4\varepsilon$. Connecting two layers when their observed parameters are at distance at most 2ε recovers the exact one-hot partition. Each cluster mean estimates its basis to error at most ε .*

Parameter access is not always available. Shared-input probes instead target the clean functional partition defined in Section 3. Let $Y_i(H)$ be the possibly noisy observation of layer i 's residual update on a common probe. The result below applies only when the distances induced by that observation law retain a positive gap for the clean target partition; layer-dependent noise can destroy this condition. Define

$$d_{ij}^2 = \mathbb{E}\|Y_i(H) - Y_j(H)\|_2^2, \quad \widehat{d}_{ij}^2 = \frac{1}{n} \sum_{s=1}^n \|Y_i(H_s) - Y_j(H_s)\|_2^2. \quad (28)$$

Theorem 5 (Finite-probe partition recovery). *Let $L \geq 2$, let $0 < \eta < 1$, and suppose the target partition has at least two classes. Let it have population within-class radius $w = \max_{i \sim_f j} d_{ij}^2$ and between-class separation $b = \min_{i \not\sim_f j} d_{ij}^2$, with gap $\delta = b - w > 0$. Suppose the centered squared-distance observations satisfy, uniformly over layer pairs, the Bernstein tail*

$$\Pr\left(\left|\widehat{d}_{ij}^2 - d_{ij}^2\right| > u\right) \leq 2 \exp\left[-n \min\left\{\frac{u^2}{2\nu^2}, \frac{u}{2c}\right\}\right]. \quad (29)$$

If, for failure probability η ,

$$n \geq \max\left\{\frac{32\nu^2}{\delta^2}, \frac{8c}{\delta}\right\} \log \frac{2\binom{L}{2}}{\eta}, \quad (30)$$

then thresholding $\widehat{d}_{ij}^2$ at $(w + b)/2$ recovers the functional partition with probability at least $1 - \eta$. The same strict separation guarantees recovery by single, complete, or average agglomerative linkage when the number K_f of functional equivalence classes is known. In general K_f need not equal the number K of router bases.

The result follows by a union bound and certifies equivalence only under the specified probe distribution; behavior off its support remains unidentified.

C.3 OPTIMIZATION AND INITIALIZATION EFFECTS

Let $G = \nabla_{\Theta} \mathcal{L}$.

Proposition 8 (Factorization-induced dynamics). *For Euclidean gradient flow on unconstrained factors, without factor-specific regularizers,*

$$\dot{\Theta} = -G(B^\top B) - (AA^\top)G. \quad (31)$$

Gauge-equivalent initializations therefore generally have different effective parameter trajectories despite identical initial Θ , predictions, and loss.

Proposition 9 (Softmax saturation). *Let $a_i = \text{softmax}(z_i/\tau)$ and suppose $\max_j a_{ij} = 1 - \varepsilon$. With $g_i = \nabla_{\theta_i} \mathcal{L}$,*

$$\|\nabla_{z_i} \mathcal{L}\|_2 \leq \frac{2\varepsilon}{\tau} \|Bg_i\|_2. \quad (32)$$

Thus the instantaneous logit gradient is attenuated as the current router row approaches the simplex boundary. This pointwise bound alone does not imply that an initial argmax graph remains fixed over a finite training trajectory; the loss gradients, basis norm, and probability margin can all change.

These results justify testing gauge-equivalent initializations, final-initial router movement, and recovery under deliberately wrong initial graphs. They do not prove that gradient descent prefers a universal smooth, banded, or low-entropy structure. Nor do reconstruction loss or end-to-end accuracy by themselves identify router coordinates, the effective-weight equality partition, functional equivalence, or a latent generative graph.

D THEORY PROOFS, EXTENSIONS, AND SCOPE

This appendix gives complete proofs for Section 4. It also records edge cases that are easy to lose when a soft router is visualized as a discrete sharing graph.

D.1 STATIC SSMF GAUGE DETAILS

The non-identifiability conclusion and the symmetric uniformizing path are the unconstrained SSMF result of Vu Thanh et al. (2023, Section IV-A) under $X = \Theta^\top$, $W = B^\top$, and $H = A^\top$; see Appendix H. The proof below only records the direct local row-sum-preserving extension used to delimit which router summaries change.

Proof of Proposition 5. The constraint $H\mathbf{1} = 0$ implies

$$M_t\mathbf{1} = (I + tH)\mathbf{1} = \mathbf{1}. \quad (33)$$

The determinant is continuous and $\det(M_0) = 1$, so M_t is invertible for all sufficiently small $|t|$. Strict positivity is also an open condition: because A has finitely many entries and $A_{ij} > 0$, there is a neighborhood of zero in which every entry of AM_t remains positive. Therefore

$$A_t\mathbf{1} = AM_t\mathbf{1} = A\mathbf{1} = \mathbf{1}, \quad A_t > 0, \quad (34)$$

and $A_t \in \Delta_K^L$. The effective parameters are unchanged because

$$A_t B_t = (AM_t)(M_t^{-1}B) = AB. \quad (35)$$

Equation 33 also gives $M_t^{-1}\mathbf{1} = \mathbf{1}$. Hence the augmented basis matrix satisfies

$$[B_t, \mathbf{1}] = [M_t^{-1}B, M_t^{-1}\mathbf{1}] = M_t^{-1}[B, \mathbf{1}]. \quad (36)$$

Left multiplication by an invertible matrix preserves row rank, so affine independence is preserved exactly, not merely infinitesimally.

It remains to rule out a permutation. If $A_t = AP$ for a permutation matrix P , then $A(M_t - P) = 0$. Since A has column rank K , it has a left inverse, and therefore $M_t = P$. The identity is the only permutation matrix in a sufficiently small neighborhood of I . For nonzero t and nonzero H , $M_t \neq I$, yielding a contradiction. Finally, the linear constraint $H\mathbf{1} = 0$ supplies K independent equations on K^2 entries, so its solution space has dimension $K(K - 1)$. Full column rank makes these local directions distinct at A . $\square$

Any strictly positive router can be written as a softmax: for fixed $\tau > 0$, take $z_{ij} = \tau \log A_{ij}$, up to an arbitrary additive constant in each row. Thus softmax normalization does not eliminate the orbit.

Entropy-changing gauges. For M_s in equation 25, the eigenvalue along $\mathbf{1}$ is one and every eigenvalue orthogonal to $\mathbf{1}$ is $1 - s$; hence it is invertible for $0 < s < 1$. Since

$$A \frac{\mathbf{1}\mathbf{1}^\top}{K} = \frac{\mathbf{1}_L \mathbf{1}^\top}{K} = U, \quad (37)$$

we have $AM_s = (1 - s)A + sU$. Strict concavity of Shannon entropy gives

$$H((1 - s)a_i + su) \geq (1 - s)H(a_i) + sH(u) > H(a_i) \quad (38)$$

for every nonuniform a_i , because u is the unique entropy maximizer. The effective parameters remain exactly unchanged after replacing B by $M_s^{-1}B$. As $s \uparrow 1$, the router can approach uniform while the basis norm may grow. A norm penalty can prefer one member of this orbit, but that is a gauge-fixing optimization convention rather than identification by the data.

Verification of the argmax counterexample. The matrix M in equation 26 is positive, row stochastic, and has determinant 0.5. Direct multiplication gives

$$AM = \begin{bmatrix} 0.45 & 0.55 \\ 0.55 & 0.45 \\ 0.60 & 0.40 \\ 0.85 & 0.15 \end{bmatrix}. \quad (39)$$

Thus two layer assignments flip. If B has two distinct rows, those rows are affinely independent; equation 36 shows that $M^{-1}B$ remains so. The example therefore does not rely on collapsed bases.

Proof of Proposition 6. Let (A, B) and (A', B') be the two factorizations. Gauge equivalence gives $AB = A'B' = \Theta$. By assumption, for every input x ,

$$F(x; A, B) = F(x; AB) = F(x; A'B') = F(x; A', B'). \quad (40)$$

The complete input–output process, including its distribution under any design over inputs, is therefore identical in the two worlds. Every possibly randomized estimator has the same output distribution in both worlds. If the two target router summaries differ, the estimator cannot equal both with probability one. The argument already grants the estimator all function values, so increasing the sample size cannot repair the failure. $\square$

The proposition assumes that factors enter the architecture only through the parameter product. If all nonlinear basis modules are executed separately and their outputs are mixed, an arbitrary linear change of basis need not remain in that model class. No claim for such output-space mixtures follows from this proof.

D.2 OPTIMIZER-RESPONSE STABILIZER

We use the convention of Section 4.1. Set

$$\lambda_Z = \eta_Z / \tau^2, \quad C_i = \text{Diag}(a_i) - a_i a_i^\top, \quad H_i = B^\top C_i^2 B. \quad (41)$$

With rows of the input and output represented as column vectors, the (i, j) block of $\mathcal{R}_{A,B}$ is

$$[\mathcal{R}_{A,B}]_{ij} = \eta_B (a_i^\top a_j) I_D + \mathbf{1}\{i = j\} \lambda_Z H_i. \quad (42)$$

Proof of Theorem 1. Because both factorizations have rank K , their column spaces equal the column space of their common product. Standard full-rank factorization algebra therefore gives a unique $M \in \text{GL}(K)$ satisfying

$$A' = AM, \quad B' = M^{-1}B. \quad (43)$$

Both routers have unit row sums, so $A(M\mathbf{1} - \mathbf{1}) = 0$. Full column rank of A gives

$$M\mathbf{1} = \mathbf{1}. \quad (44)$$

Equality of the response on every G is equality of every block in equation 42. For $i \neq j$, it gives

$$a_i^\top a_j = (a'_i)^\top a'_j. \quad (45)$$

For a diagonal block, define $H'_i = (B')^\top (C'_i)^2 B'$. Its equality can be rearranged as

$$H_i - H'_i = c_i I_D, \quad c_i = -\frac{\eta_B}{\lambda_Z} (\|a_i\|_2^2 - \|a'_i\|_2^2) \quad (46)$$

for a scalar c_i . Every simplex covariance satisfies $C_i \succeq 0$ and $C_i \mathbf{1} = 0$; full row rank of B therefore gives $H_i \succeq 0$ and $\text{rank}(H_i) \leq K - 1 < D$. The same holds for H'_i . If $c_i > 0$, then $H_i = H'_i + c_i I_D$ would be positive definite; if $c_i < 0$, then $H'_i = H_i - c_i I_D$ would be positive definite. Both contradict singularity, so $c_i = 0$. The remaining scalar part of the diagonal block then gives $\|a_i\|_2^2 = \|a'_i\|_2^2$. Together with equation 45,

$$AA^\top = A'A'^\top = AMM^\top A^\top. \quad (47)$$

Multiplication by a left inverse of A and its transpose yields

$$MM^\top = I_K. \quad (48)$$

Since $H_i = H'_i$, substituting equation 43 and cancelling B with a right inverse gives

$$C_i^2 = M^{-\top} (C'_i)^2 M^{-1}. \quad (49)$$

By equation 48, this is

$$C_i^2 = M(C'_i)^2 M^\top = (MC'_i M^\top)^2. \quad (50)$$

Both C_i and $MC'_i M^\top$ are positive semidefinite. Uniqueness of the positive-semidefinite square root therefore gives

$$C_i = MC'_i M^\top. \quad (51)$$

In column convention, $a'_i = M^\top a_i$. Expanding equation 51, the two rank-one terms cancel and leave

$$\text{Diag}(a_i) = M \text{Diag}(a'_i) M^\top. \quad (52)$$

For distinct r, s , the (r, s) entry says

$$\sum_{\ell=1}^K M_{r\ell} M_{s\ell} a'_{i\ell} = 0 \quad \text{for every } i. \quad (53)$$

The rows of A' span $\mathbb{R}^K$, hence $M_{r\ell} M_{s\ell} = 0$ for every ℓ . Thus each column of the orthogonal M has exactly one nonzero entry, equal to $+1$ or -1 : M is a signed permutation. Equation 44 removes the negative signs, so $M = P$ is a permutation matrix.

Conversely, if $A' = AP$ and $B' = P^\top B$, then $a'_i = P^\top a_i$, $C'_i = P^\top C_i P$, the router Gram matrix is unchanged, and

$$(B')^\top (C'_i)^2 B' = B^\top C_i^2 B. \quad (54)$$

Every block in equation 42 is therefore unchanged. This proves both directions. $\square$

D.3 A QUANTITATIVE INVERSE FOR THE RESPONSE

The exact stabilizer has a linear inverse on any uniformly interior, well-conditioned, norm-bounded set. The following constants are explicit so that the claim is checkable; they are not intended to be numerically tight. For a linear map between matrix spaces, $\|\cdot\|_{F \rightarrow F}$ denotes the operator norm induced by the Frobenius norm.

Theorem 6 (Quantitative optimizer-response stability). *Let $K \geq 2$, $L \geq K$, and $D \geq K$. Let $A, A' \in \mathbb{R}^{L \times K}$ be row-stochastic and let $B, B' \in \mathbb{R}^{K \times D}$. Suppose*

$$\min_{i,j} \{A_{ij}, A'_{ij}\} \geq \alpha > 0, \quad (55)$$

$$\sigma_{\min}(A), \sigma_{\min}(A') \geq s_A > 0, \quad \sigma_{\min}(B), \sigma_{\min}(B') \geq s_B > 0, \quad (56)$$

$$\|A\|_F, \|A'\|_F \leq L_A, \quad \|B\|_F, \|B'\|_F \leq L_B, \quad (57)$$

$$AB = A'B'. \quad (58)$$

Both response operators are defined with the same $\tau, \eta_Z, \eta_B > 0$. Put $\lambda_Z = \eta_Z/\tau^2$ and

$$\delta = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}. \quad (59)$$

Then

$$\min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2} \leq C\delta, \quad (60)$$

where the following is one finite, fully explicit choice. First define the factorization and Gram constants

$$L_M = \max\{1, L_A/s_A\}, \quad c_G = \frac{L}{\eta_B s_A^2}, \quad c_H = \frac{2}{\lambda_Z s_B^2}. \quad (61)$$

Next define the covariance and diagonal-algebra constants

$$\mu_* = \alpha \left(1 + \frac{1}{4L_M^2}\right), \quad c_J = \frac{L_M^2(c_G + c_H)}{\mu_*}, \quad c_D = \frac{\sqrt{LK}}{s_A} c_J. \quad (62)$$

Finally define the row-rounding and global constants

$$c_{\text{row}} = [Kc_D^2 + (c_G + Kc_D^2)^2]^{1/2}, \quad (63)$$

$$\delta_0 = \min \left\{ 1, \frac{1}{2c_G}, \frac{1}{2\sqrt{K} c_{\text{row}}} \right\}, \quad (64)$$

$$C_{\text{loc}} = \sqrt{L_A^2 + (L_M L_B)^2} \sqrt{K} c_{\text{row}}, \quad C = \max \left\{ C_{\text{loc}}, \frac{2\sqrt{L_A^2 + L_B^2}}{\delta_0} \right\}. \quad (65)$$

Proof. Except where a Frobenius norm is displayed explicitly, all matrix norms in this proof are spectral norms. We divide the argument into four steps.

Step 1: recovering the router Gram matrix and an almost-orthogonal gauge. Injecting a vector into input block j and projecting the output onto block i are contractions for the Frobenius-induced operator norm. Therefore each block of $\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}$ has spectral norm at most δ . For $i \neq j$, equation 42 gives

$$|(AA^\top - A'A'^\top)_{ij}| \leq \frac{\delta}{\eta_B}. \quad (66)$$

For a diagonal block, write

$$E_i = t_i I_D + \lambda_Z (H_i - H'_i), \quad t_i = \eta_B (\|a_i\|_2^2 - \|a'_i\|_2^2), \quad (67)$$

where $H'_i = (B')^\top (C'_i)^2 B'$. Both H_i and H'_i are singular positive-semidefinite matrices: their ranks are at most $K - 1 < D$. If $t_i \geq 0$, evaluating E_i on a unit vector in $\ker(H'_i)$ gives a quadratic form at least t_i . If $t_i < 0$, evaluating it on a unit vector in $\ker(H_i)$ gives a quadratic form at most t_i . Since $\|E_i\|_2 \leq \delta$, in either case

$$|t_i| \leq \delta, \quad \|H_i - H'_i\|_2 \leq \frac{2\delta}{\lambda_Z}. \quad (68)$$

The diagonal entries of the router Gram difference consequently satisfy the same bound as equation 66, and hence

$$\|AA^\top - A'A'^\top\|_F \leq \frac{L\delta}{\eta_B}. \quad (69)$$

The common product has rank K , so full-rank factorization algebra gives a unique $M \in \text{GL}(K)$ with

$$A' = AM, \quad B' = M^{-1}B, \quad M\mathbf{1} = \mathbf{1}. \quad (70)$$

Moreover,

$$M = A^\dagger A', \quad M^{-1} = (A')^\dagger A, \quad \|M\|_2, \|M^{-1}\|_2 \leq L_A/s_A \leq L_M. \quad (71)$$

Multiplying equation 69 on the left by $A^\dagger$ and on the right by $(A')^\top$ yields

$$\|I_K - MM^\top\|_2 \leq c_G \delta. \quad (72)$$

Step 2: recovering the simplex covariance matrices. Let $Q = B^\top (BB^\top)^{-1}$. Then $BQ = I_K$ and $\|Q\|_2 \leq 1/s_B$. Substitute $B' = M^{-1}B$ in H'_i , and multiply equation 68 by $Q^\top$ and Q , to obtain

$$\|C_i^2 - M^{-\top} (C'_i)^2 M^{-1}\|_2 \leq c_H \delta. \quad (73)$$

Set $Y_i = M^\top C_i M$. The identity

$$\begin{aligned} Y_i^2 - (C'_i)^2 &= M^\top C_i (MM^\top - I_K) C_i M \\ &\quad + M^\top \{C_i^2 - M^{-\top} (C'_i)^2 M^{-1}\} M \end{aligned} \quad (74)$$

together with $\|C_i\|_2 \leq 1$, equation 71, equation 72, and equation 73 gives

$$\|Y_i^2 - (C'_i)^2\|_2 \leq L_M^2 (c_G + c_H) \delta. \quad (75)$$

Strict interiority now supplies the spectral gap that is unavailable near the simplex boundary. If a is a probability vector with $\min_j a_j \geq \alpha$, then

$$C(a) = \text{Diag}(a) - aa^\top \succeq \alpha \Pi_{\mathbf{1}^\perp}. \quad (76)$$

Indeed, write $a = \alpha \mathbf{1} + r$, where $r \geq 0$ and $\mathbf{1}^\top r = 1 - K\alpha \leq 1$. For $x \perp \mathbf{1}$, weighted Cauchy–Schwarz gives

$$x^\top C(a)x = \alpha \|x\|_2^2 + \sum_j r_j x_j^2 - \left(\sum_j r_j x_j \right)^2 \geq \alpha \|x\|_2^2.$$

Both Y_i and C'_i have kernel exactly $\text{span}\{\mathbf{1}\}$: for the former, use $M^{-1}\mathbf{1} = \mathbf{1}$. They are symmetric, so they preserve $\mathbf{1}^\perp$. For a unit $x \perp \mathbf{1}$, decompose $Mx = c\mathbf{1} + y$, with $y \perp \mathbf{1}$. Since $x = c\mathbf{1} + M^{-1}y$, orthogonality of x and equation 71 imply

$$|c|\sqrt{K} \leq L_M \|y\|_2, \quad 1 \leq |c|\sqrt{K} + L_M \|y\|_2, \quad (77)$$

and therefore $\|y\|_2 \geq 1/(2L_M)$. It follows from equation 76 that, on $\mathbf{1}^\perp$,

$$Y_i \succeq \frac{\alpha}{4L_M^2} I, \quad C'_i \succeq \alpha I. \quad (78)$$

On this subspace put $D_i = Y_i - C'_i$. The Sylvester equation

$$Y_i D_i + D_i C'_i = Y_i^2 - (C'_i)^2 \quad (79)$$

has the integral solution

$$D_i = \int_0^\infty e^{-tY_i} \{Y_i^2 - (C'_i)^2\} e^{-tC'_i} dt. \quad (80)$$

The two lower bounds in equation 78 show that the norm of this inverse is at most $1/\mu_*$. Both sides vanish on $\text{span}\{\mathbf{1}\}$, so equation 75 proves on the whole space that

$$\|M^\top C_i M - C'_i\|_2 \leq c_J \delta. \quad (81)$$

The shared exact kernel is what permits a linear estimate here; a generic perturbation bound for square roots would only be Hölder of order one half.

Step 3: approximately normalizing the diagonal algebra. Because $a'_i = M^\top a_i$, expanding the two covariance matrices in equation 81 cancels their rank-one terms and gives

$$\|M^\top \text{Diag}(a_i) M - \text{Diag}(a'_i)\|_2 \leq c_J \delta. \quad (82)$$

Let $\text{Off}(X)$ set the diagonal of X to zero, let $E_{jj} = e_j e_j^\top$, and define $X_j = \text{Off}(M^\top E_{jj} M)$. Taking the off-diagonal part is contractive in Frobenius norm, while $\|X\|_F \leq \sqrt{K} \|X\|_2$. Thus equation 82 implies

$$\left\| \sum_{j=1}^K A_{ij} X_j \right\|_F \leq \sqrt{K} c_J \delta \quad (1 \leq i \leq L). \quad (83)$$

Apply row j of $A^\dagger$ to this system in the direct-sum Frobenius space. That row has Euclidean norm at most $1/s_A$; Cauchy–Schwarz over the L rows gives

$$\|\text{Off}(M^\top E_{jj} M)\|_F \leq c_D \delta \quad (1 \leq j \leq K). \quad (84)$$

Step 4: rounding the gauge to a permutation. Write $M^\top E_{jj} M = r_j r_j^\top$, where $r_j^\top$ is row j of M , and choose q maximizing $|r_{jq}|$. Suppose first that $\delta \leq \delta_0$. From equation 72, $\|r_j\|_2^2 \geq 1/2$, so $r_{jq}^2 \geq 1/(2K)$. The squared Frobenius norm in equation 84 contains $2r_{jq}^2 \sum_{\ell \neq q} r_{j\ell}^2$. Consequently

$$\sum_{\ell \neq q} r_{j\ell}^2 \leq K c_D^2 \delta^2, \quad (85)$$

$$|r_{jq}^2 - 1| \leq (c_G + K c_D^2) \delta. \quad (86)$$

The second inequality also uses $\delta \leq 1$. Equations equation 85–equation 86 show that row j is within $c_{\text{row}} \delta$ of a signed coordinate vector.

Collect the selected signed coordinate rows into a matrix S . By equation 64,

$$\|M - S\|_2 \leq \|M - S\|_F \leq \sqrt{K} c_{\text{row}} \delta \leq \frac{1}{2}. \quad (87)$$

If two rows selected the same coordinate, S would be singular and $\sigma_{\min}(M) \leq 1/2$. This contradicts equation 72, which gives $\sigma_{\min}(M) \geq 1/\sqrt{2}$. The selected coordinates are therefore distinct. Each row of M sums to one by equation 70. Its distance from any negative coordinate vector is consequently at least $2/\sqrt{K}$, whereas its selected-row distance in equation 87 is at most $1/(2\sqrt{K})$. Every sign is positive, so $S = P$ is a permutation matrix and

$$\|M - P\|_F \leq \sqrt{K} c_{\text{row}} \delta. \quad (88)$$

Now

$$\|A' - AP\|_F \leq L_A \|M - P\|_F.$$

Using $M^{-1} - P^\top = M^{-1}(P - M)P^\top$ also gives

$$\|B' - P^\top B\|_F \leq L_M L_B \|M - P\|_F.$$

These two bounds and equation 88 prove equation 60 with C_{loc} whenever $\delta \leq \delta_0$. If $\delta \geq \delta_0$, choosing the identity permutation and using the parameter-set diameter gives

$$(\|A' - A\|_F^2 + \|B' - B\|_F^2)^{1/2} \leq 2\sqrt{L_A^2 + L_B^2} \leq \frac{2\sqrt{L_A^2 + L_B^2}}{\delta_0} \delta.$$

Taking the larger of the local and diameter constants completes the proof. $\square$

Why the quantitative assumptions appear. The exact product constraint equation 58 cannot be replaced by response closeness alone: (A, B) and $(A, -B)$ have identical responses but opposite effective matrices. The two singular-value margins keep the factorization gauge and the right inverse of B bounded. The interior margin creates the uniform covariance gap in equation 76; without it, the inverse can become arbitrarily ill-conditioned near a simplex face. Norm bounds make the nonlocal diameter argument finite. Finally, using the same temperature and block rates is part of specifying the optimizer metric rather than a technical normalization. These assumptions are sufficient and transparent, not asserted to be jointly minimal.

Here are three explicit degenerating families showing that a uniform linear constant cannot ignore the interior or either rank margin. They also make clear that these are stability failures, not counterexamples to the exact theorem. Set $K = L = D = 3$, $U = \mathbf{1}\mathbf{1}^\top/3$, and

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & 2 \\ 2 & 2 & -1 \end{bmatrix}. \quad (89)$$

Then Q is a non-permutation orthogonal matrix and $Q\mathbf{1} = \mathbf{1}$.

Router-rank margin. For $0 < \varepsilon \leq 1/4$, let

$$A_\varepsilon = U + \varepsilon(I - U), \quad B_\varepsilon = I, \quad A'_\varepsilon = A_\varepsilon Q, \quad B'_\varepsilon = Q^\top.$$

Both routers are strictly positive, the basis singular values equal one, and all norms and entrywise interior margins remain uniformly bounded. However, $\sigma_{\min}(A_\varepsilon) = \sigma_{\min}(A'_\varepsilon) = \varepsilon$. The products and router Gram matrices agree exactly. At $\varepsilon = 0$, every router row is uniform and $C_i = (I - U)/3$, which commutes with Q ; hence the local response terms also agree. Define the permutation-orbit distance d_ε by

$$d_\varepsilon^2 = \min_P \{ \|A'_\varepsilon - A_\varepsilon P\|_F^2 + \|B'_\varepsilon - P^\top B_\varepsilon\|_F^2 \}.$$

Polynomial dependence on ε gives

$$\|\mathcal{R}_{A_\varepsilon, B_\varepsilon} - \mathcal{R}_{A'_\varepsilon, B'_\varepsilon}\|_{F \rightarrow F} = O(\varepsilon), \quad (90)$$

$$d_\varepsilon \longrightarrow \min_P \|Q - P^\top\|_F > 0. \quad (91)$$

Thus no uniform inverse constant survives as $s_A \downarrow 0$.

Basis-rank margin. Keep Q from equation 89, put

$$A = 0.3\mathbf{1}\mathbf{1}^\top + 0.1I, \quad A' = AQ, \quad B_\varepsilon = \mathbf{1}e_1^\top + \varepsilon I, \quad B'_\varepsilon = Q^\top B_\varepsilon.$$

The routers have fixed positive entries and fixed full rank; all norms remain bounded, while $\sigma_{\min}(B_\varepsilon) \rightarrow 0$. Because $C_i \mathbf{1} = C'_i \mathbf{1} = 0$, all cross terms with $\mathbf{1}e_1^\top$ vanish and

$$B_\varepsilon^\top C_i^2 B_\varepsilon = \varepsilon^2 C_i^2, \quad (B'_\varepsilon)^\top (C'_i)^2 B'_\varepsilon = \varepsilon^2 Q(C'_i)^2 Q^\top.$$

The Gram parts again agree exactly, so the response difference is $O(\varepsilon^2)$. The parameter distance tends to the strictly positive distance between $A' = AQ$ and the finite permutation orbit of the full-rank A . Hence the constant must also deteriorate as $s_B \downarrow 0$.

Interior margin. Let

$$S = \frac{1}{\sqrt{3}} \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}, \quad Q_t = \exp(tS),$$

and, for sufficiently small $t > 0$, set

$$A_t = t\mathbf{1}\mathbf{1}^\top + (1 - 3t)I, \quad B_t = I, \quad A'_t = A_t Q_t, \quad B'_t = Q_t^\top.$$

Here $S^\top = -S$ and $S\mathbf{1} = 0$, so Q_t is orthogonal and fixes $\mathbf{1}$. Moreover $A'_t = I + t(\mathbf{1}\mathbf{1}^\top - 3I + S) + O(t^2)$; its off-diagonal first-order coefficients are $1 \pm 1/\sqrt{3} > 0$. Thus both routers are strictly positive for all sufficiently small t , while their minimum entry is $\Theta(t)$. Their smallest singular values are $1 - 3t$, and both basis singular values equal one, so every rank and norm margin stays uniform. The products and Gram responses agree exactly. Each row tends to a simplex vertex, so $C(a_{t,i}) = O(t)$ and $C(a'_{t,i}) = O(t)$; consequently the remaining squared-covariance response difference is $O(t^2)$. For the first row, direct expansion gives $\text{tr}(C(a_{t,1})^2) = 10t^2 + O(t^3)$ and $\text{tr}(C(a'_{t,1})^2) = 12t^2 + O(t^3)$. Orthogonal conjugation preserves trace, so this block's difference has spectral norm at least $(2/3)t^2 + O(t^3)$ before the fixed positive λ_Z multiplier. The full response gap is therefore $\Theta(t^2)$. On the other hand, the identity is the nearest permutation for small t , and $\|Q_t - I\|_F = \sqrt{2}t + O(t^2)$, so the parameter-orbit distance is $\Theta(t)$. The ratio of parameter distance to response error therefore diverges like $1/t$. This supplies the claimed simplex-face ill-conditioning while s_A, s_B, L_A, L_B remain controlled.

D.4 FINITE STRUCTURED RESPONSE TOMOGRAPHY

We now prove Corollary 2. Write

$$Q = AA^\top, \quad T_i = \lambda_Z B^\top C_i^2 B, \quad S = \text{row}(\Theta).$$

Under the rank hypotheses, $\text{rank}(\Theta) = K$ and

$$S = \text{row}(B), \quad \text{range}(T_i) \subseteq S, \quad T_i x = 0 \text{ for every } x \in S^\perp. \quad (92)$$

Indeed, a left inverse of A gives $B = A^\dagger \Theta$, while $\Theta = AB$ gives the reverse row-space inclusion; the remaining claims follow from the two outer factors $B^\top$ and B in T_i .

Choose $u_1, \dots, u_K$ and $v_1, \dots, v_L$ as in the corollary and define the rows of the K probes by

$$g_j^{(1)} = v_j + u_1, \quad g_j^{(r)} = u_r \quad (2 \leq r \leq K). \quad (93)$$

Let $y_i^{(r)} = [\mathcal{R}_{A,B}(G^{(r)})]_i$. Then

$$Q_{ij} = \eta_B^{-1} \langle y_i^{(1)}, v_j \rangle, \quad (94)$$

$$T_i u_1 = y_i^{(1)} - \eta_B \sum_{j=1}^L Q_{ij} (v_j + u_1), \quad (95)$$

$$T_i u_r = y_i^{(r)} - \eta_B \left(\sum_{j=1}^L Q_{ij} \right) u_r, \quad 2 \leq r \leq K. \quad (96)$$

Proof of Corollary 2. Project the first response onto $S^\perp$. By equation 92,

$$\Pi_{S^\perp} y_i^{(1)} = \eta_B \sum_{j=1}^L Q_{ij} v_j.$$

Taking its inner product with v_j gives equation 94. Subtracting the now-known shared term from the unprojected response leaves $T_i(v_i + u_1) = T_i u_1$, proving equation 95. For probe $r \geq 2$, the same subtraction leaves $T_i u_r$, which proves equation 96.

For every $x \in \mathbb{R}^D$, Equation 92 gives

$$T_i x = \sum_{r=1}^K \langle u_r, x \rangle T_i u_r.$$

Thus Equations 94–96 recover Q and every complete local block T_i , and equation 7 then determines the whole response operator. Two exact charts of the same rank- K product have the same S , so the same probes recover both operators. Equality on those probes is therefore equivalent to equality of the complete responses, and Theorem 1 gives the permutation conclusion and its converse. The Z -level row shifts remain unobserved. $\square$

The complement-code solve in equation 94 and the S -coordinate solves use orthonormal systems and have condition number one. That statement concerns recovery of $Q, T_1, \dots, T_L$ from response observations; it does not remove the rank and simplex-boundary conditioning in the inverse from these components back to the factor chart.

Proposition 10 (Deterministic observation-noise propagation). *Suppose the observed response to probe r is $Y^{(r)} + E^{(r)}$, where $\|E^{(r)}\|_F \leq \epsilon_r$, and use Equations 94–96 for reconstruction. Then*

$$\|\hat{Q} - Q\|_F \leq \epsilon_1 / \eta_B. \quad (97)$$

If $T^{(S)}$ stacks the $K \times K$ matrices representing every T_i in the u basis, then

$$\|\hat{T}^{(S)} - T^{(S)}\|_F \leq \left[(1 + \sqrt{L+1})^2 \epsilon_1^2 + \sum_{r=2}^K (\epsilon_r + \sqrt{L} \epsilon_1)^2 \right]^{1/2}. \quad (98)$$

Proof. Right multiplication by the orthonormal-row complement-code matrix is nonexpansive, which proves equation 97. The first probe has $G^{(1)}(G^{(1)})^\top = I_L + \mathbf{1}\mathbf{1}^\top$, hence $\|G^{(1)}\|_2 = \sqrt{L+1}$. The error in the reconstructed first local column is therefore at most

$$\epsilon_1 + \eta_B \|\hat{Q} - Q\|_F \|G^{(1)}\|_2 \leq (1 + \sqrt{L+1}) \epsilon_1.$$

For every later probe, the erroneous shared subtraction contributes at most $\eta_B \|(\hat{Q} - Q)\mathbf{1}\|_2 \leq \sqrt{L} \epsilon_1$, in addition to ϵ_r . Projection into the orthonormal u basis is nonexpansive; squaring and summing these K column bounds proves equation 98. $\square$

This gives deterministic error radii, not a noisy “if and only if.” Two noisy reconstructions can be separated when their component gap exceeds the sum of the applicable radii. Biased, unknown, or optimizer-state-dependent observation error requires a different model.

Constructive exact factor recovery. The diagonal-algebra reduction also yields an explicit reconstruction once Q and the local blocks are known. Let $U \in \mathbb{R}^{K \times D}$ contain the orthonormal row-space basis, factor $Q = XX^\top$ with X full column rank, and put $C = X^\dagger \Theta U^\top$. There is an orthogonal O with $X = AO$, so $C = O^\top B U^\top$ is invertible. Writing $T_i^{(S)} = U T_i U^\top$ and $x_i = X_{i,:}^\top$, define

$$D_i = \left[C^{-\top} (T_i^{(S)} / \lambda_Z) C^{-1} \right]^{1/2} + x_i x_i^\top = O^\top \text{diag}(a_i) O,$$

where the square root is the PSD one. A weighted sum of these symmetric matrices has distinct eigenvalues whenever the weighted sums of the columns of A are distinct. Full column rank makes those columns distinct, so continuous Gaussian weights satisfy this condition almost surely. If V is the resulting orthogonal eigenvector matrix, the diagonals of $V^\top D_i V$ recover router rows up to one common permutation; eigenvector signs cancel. The basis then follows from $\hat{B} = \hat{A}^\dagger \Theta$. This is a constructive use of established joint-diagonalization algebra (Afsari, 2008; Bhaskara et al., 2014), not an additional uniqueness claim. Its numerical implementation symmetrizes observations and clips negative eigenvalues before taking square roots. Such clipping, and the conditioning of a chosen weighted eigensystem, require separate analysis under noise; the exact argument is not a stability guarantee for that estimator.

Proposition 11 (Generic probes for the structured response). *Let $p = D - K \geq 1$, retain the exact product and rank assumptions above, and draw q effective-gradient matrices independently with iid standard Gaussian entries. If*

$$q \geq \max \left\{ K, \left\lceil \frac{L}{p} \right\rceil \right\}, \quad (99)$$

then their full response vectors determine $Q, T_1, \dots, T_L$, and hence the complete structured response, with probability one.

Proof. Express each probe in fixed orthonormal coordinates for $S \oplus S^\perp$. Concatenate the $S^\perp$ input coordinates over all probes into $X_\perp \in \mathbb{R}^{L \times qp}$. The corresponding projected responses obey $Y_\perp = \eta_B Q X_\perp$. When $qp \geq L$, the Gaussian matrix $X_\perp$ has row rank L almost surely, so it determines Q . After subtracting that shared term, the q input coordinates in S for any fixed layer form a Gaussian $K \times q$ matrix. It has row rank K almost surely when $q \geq K$, and therefore determines that layer’s local block. Taking a finite union over layers preserves probability one. $\square$

At the released dimensions $(L, K, D) = (12, 4, 787, 712)$, four complete Gaussian probe/response vectors would satisfy equation 99 almost surely if the rank solves were actually retained and checked. The older eight-probe artifacts retain norm comparisons rather than a tomographic reconstruction, so they remain a fixed-alternative test. The original four-probe artifact instead executes the explicit design equation 93, evaluates the analytic full-dimensional response, and records the reconstruction residuals. It does not run four full-model autograd/JVP queries; the analytic formula is validated separately on the declared 4×32 extraction. The additive audit in Table 10 measures all four full-dimensional responses by reverse automatic differentiation followed by a forward-mode JVP of the packed factor map. It checks reconstruction on two unseen directions per chart and records deterministic noise-bound diagnostics. These queries use linear objectives on the packed parameters; they do not differentiate the language-model task loss or an AdamW update.

This finite observation result is a structured operator-probing corollary, not an independent central innovation. In particular, Novak et al. (2022, Sec. 3.3, Eqs. (7)–(9)) reconstruct a finite NTK from identity NTK–vector products; generic and structured matrix probing is developed by Chiu & Demanet (2012), Halikias & Townsend (2024), and Amsel et al. (2026). Relative to a generic LD -dimensional identity reconstruction, Equations 93–96 only exploit the present model’s known support and shared-block structure to compress the queries.

Several boundaries are essential. The packing condition $D - K \geq L$ is sufficient for the explicit construction, not a lower bound for all finite designs. When $K = 1$, the single probe recovers the shared response and the softmax local blocks vanish. Exact common product is required because both charts must share S ; a small numerical product residual alone gives no row-space perturbation theorem. The implementation accordingly activates the common-design conclusion only for charts produced by an explicit exact invertible gauge, with the floating-point product residual recorded separately. Known $\eta_B > 0$ is used in equation 94; known $\eta_Z > 0$ and temperature remain part of the chart theorem. Finally, these interventions are executable through linear checkpoint objectives $\mathcal{L}_G(\Theta) = \langle G, \Theta \rangle$, but they need not occur in a natural minibatch trajectory.

Proposition 12 (Finite Gaussian probes for a fixed response alternative). *Let $m = LD$, represent a fixed nonzero response difference by $D \in \mathbb{R}^{m \times m}$, and draw independent $x_1, \dots, x_q \sim \mathcal{N}(0, I_m)$. For every $0 < t < 1$,*

$$\Pr \left[\max_{\ell \leq q} \|Dx_\ell\|_2 < t \|D\|_2 \right] \leq [2\Phi(t) - 1]^q. \quad (100)$$

Thus failure probability at most β is obtained when

$$q \geq \frac{\log(1/\beta)}{-\log(2\Phi(t) - 1)}. \quad (101)$$

At $t = 1/2$, the coefficient is approximately 1.04. This is a fixed-alternative, absolute-threshold statement. It neither calibrates a relative numerical threshold nor uniformly certifies equality of arbitrary response operators when $q < m$.

Proof. Let v be a unit right singular vector for the largest singular value of D . An SVD gives $\|Dx\|_2 \geq \|D\|_2 |v^\top x|$, and $v^\top x \sim \mathcal{N}(0, 1)$. Independence across probes proves equation 100; solving the right-hand side for q gives equation 101.

For the uniform limitation, after any $q < m$ realized probes there exists a nonzero linear map whose kernel contains their span, so all its observed responses vanish. Conversely, m noiseless Gaussian probes form an invertible $m \times m$ probe matrix almost surely and therefore determine an arbitrary response operator. Finally, a reported ratio such as $\|Dx\|_2 / \max\{\|\mathcal{R}x\|_2, \|\mathcal{R}'x\|_2\}$ has a random denominator; equation 100 does not calibrate it without additional bounds on those response norms. $\square$

Corollary 5 (Pointwise probe detection from inverse stability). *Fix two exact product-equivalent charts in one nondegenerate set from Corollary 1, fix them before drawing the probes, and let C be that set’s common inverse constant. Define*

$$d_{\text{orb}} = \min_{P \in \mathfrak{S}_K} (\|A' - AP\|_F^2 + \|B' - P^\top B\|_F^2)^{1/2}$$

and $D = \mathcal{R}_{A,B} - \mathcal{R}_{A',B'}$. If $d_{\text{orb}} \geq \Delta > 0$, then independent standard-Gaussian probes satisfy, for every $0 < t < 1$,

$$\Pr \left[\max_{\ell \leq q} \|Dx_\ell\|_2 \geq t\Delta/C \right] \geq 1 - [2\Phi(t) - 1]^q. \quad (102)$$

Proof. The inverse bound gives $\|D\|_2 \geq d_{\text{orb}}/C \geq \Delta/C$. The complement of the event in equation 102 is therefore contained in the small-ball event of Proposition 12, which proves the claim. $\square$

At $t = 1/2$ and $q = 8$, the upper bound on the miss probability is about 4.62×10^{-4} . This number does not calibrate the released relative response ratios: the threshold $t\Delta/C$ is absolute, C may be loose, the alternative and common nondegenerate set must be fixed before the draws, and the statement is not simultaneous over probe-selected charts. If each observed difference has deterministic error at most ϵ , the corresponding observable lower threshold is only $t\Delta/C - \epsilon$ when positive.

Why the hypotheses and target matter. The following examples separate exact theorem hypotheses from broader interpretations.

- Equality of the effective model is indispensable. With $A' = A$ and $B' = -B$, the router Gram and every quadratic matrix $B^\top C_i^2 B$ are unchanged, but $A'B' = -AB$. The response alone does not identify the represented effective matrix.
- Router rank cannot be dropped. For $K = D = 3$, let every row of A be $\mathbf{1}^\top/3$, let $B = I_3$, and let Q be a non-permutation orthogonal rotation on $\mathbf{1}^\perp$ with $Q\mathbf{1} = \mathbf{1}$. Then $A' = AQ = A$, $B' = Q^\top$, and both the product and response are unchanged. The two factors are not permutation equivalent.
- Basis rank cannot be dropped. Let $K = D = 3$, $A = 0.3\mathbf{1}\mathbf{1}^\top + 0.1I_3$, and $B = \mathbf{1}e_1^\top$. For a sufficiently small non-permutation orthogonal rotation Q fixing $\mathbf{1}$, $A' = AQ$ remains positive and row-stochastic, while $B' = Q^\top B = B$. The products and router Gram matrices agree, and $C_i B = 0$ makes every local softmax response term zero.
- Joint router training is material in general. If $\eta_Z = 0$ and $K \geq 3$, choose any positive full-rank router A close enough to uniform, a full-row-rank B , and a small non-permutation orthogonal Q fixing $\mathbf{1}$. Then $(AQ, Q^\top B)$ is valid and product equivalent, while the remaining response $\eta_B A A^\top G$ is unchanged.
- Equality on one task gradient is not operator equality. At a stationary point the observed gradient is zero for every parameterization, regardless of whether their response operators differ. There is also a nonzero blind direction away from stationarity. For a non-permutation exact full-rank pair with common row space S , suppose $D > K$ and $L > 2K$. Choose nonzero $v \in S^\perp$. Since $\text{rank}(AA^\top - A'(A')^\top) \leq 2K < L$, choose nonzero w in its kernel and set the rows of G to $g_j = w_j v$. Both local block differences kill v , while the shared difference kills w , so $(\mathcal{R}_{A,B} - \mathcal{R}_{A',B'})(G) = 0$ even though the two response operators differ by Theorem 1. A passive trajectory probes only its realized directions.
- Strict positivity is automatic for finite softmax logits. The exact algebra above in fact only needs each router row to lie in the closed simplex: positive semidefiniteness and $C_i \mathbf{1} = 0$ make H_i singular. Thus the formula-defined response theorem extends to boundary rows when all stated factor ranks remain full. A uniform interior margin is material only for a quantitative inverse bound.

These are sufficient conditions, not a claim of a jointly minimal assumption set. In particular, the proof uses $\eta_B > 0$ to recover the router Gram matrix, but we have not established that $\eta_B > 0$ is necessary once the complete collection of local softmax terms is available.

Partial audit of the $\eta_B = 0$ boundary. Among the partial subcases closed in the released assumption ledger, we state two simple ones here without promoting them to separate contributions. Write an equivalent chart as $A' = AM$, $B' = M^{-1}B$, with $M\mathbf{1} = \mathbf{1}$. For a router row written as a column p , put $q = M^\top p$. Equality of the remaining local blocks reduces, after cancelling the full-row-rank basis, to

$$C(q)^2 = M^\top C(p)^2 M. \quad (103)$$

If $M \geq 0$, define $K = \text{Diag}(p)M \text{Diag}(q)^{-1}$. Then $\mathbf{1}^\top K = \mathbf{1}^\top$, and equation 103 is equivalent to

$$(K - p\mathbf{1}^\top)^\top (K - p\mathbf{1}^\top) = (I - q\mathbf{1}^\top)^\top (I - q\mathbf{1}^\top).$$

Thus every two columns of K are at distance $\sqrt{2}$. They also lie in the probability simplex, whose diameter $\sqrt{2}$ is attained only by distinct vertices. Hence $K = P$; using $Kq = p$ in its definition gives $M = P$. One positive router row therefore rules out every non-permutation nonnegative gauge.

For $K = 2$, no sign restriction is needed. Up to a candidate column swap, write $M = \begin{bmatrix} x & 1-x \\ y & 1-y \end{bmatrix}$ with $d = x - y > 0$. A row $(p, 1-p)$ maps to first coordinate $q = y + dp$, and equation 103 becomes $q(1-q) = dp(1-p)$. Full router rank supplies two distinct $p_1 < p_2$. Subtracting the two equations gives $q_1 + q_2 = p_1 + p_2 =: S$; adding them and using $q_2 - q_1 = d(p_2 - p_1) = d\Delta$ gives

$$0 = \frac{1-d}{2} \{S(2-S) + d\Delta^2\}.$$

The bracket is positive, so $d = 1$, $y = 0$, and $M = I$, before undoing the possible column swap. For signed gauges with $K \geq 3$, even when both A and AM are positive and full rank, compatibility of the rowwise regular-simplex congruences remains unresolved. Numerical searches that collapse rank, approach a simplex face, or satisfy only one row are not treated as a proof or counterexample. We therefore retain $\eta_B > 0$ in the main theorem while making no necessity claim.

The theorem is specific to the named Euclidean metric and known block rates. Natural gradient, Adam/AdamW or momentum state, finite optimizer steps, factor-specific penalties, and router auxiliary losses define different response objects. Most importantly, $(\Theta, \mathcal{R})$ identifies a present optimizer chart, not the factors or graph that historically generated the data.

D.5 POSITIVE RECOVERY RESULTS

Lemma 1 (Unique barycentric coordinates). *If $b_1, \dots, b_K$ are affinely independent, then $aB = a'B$ for $a, a' \in \Delta_K$ implies $a = a'$.*

Proof. Set $q = a - a'$. The simplex constraint gives $q\mathbf{1} = 0$, and equality of the effective parameters gives $qB = 0$. A nonzero q satisfying both equations would be a nontrivial affine dependence among the rows of B . $\square$

Proof of Proposition 7. Every effective parameter lies in the convex hull of the bases: $\{\theta_i\} \subseteq \text{conv}\{b_j\}$. Separability supplies the reverse containment at the level of convex hulls, because each b_j occurs as an anchor row of Θ . Therefore

$$\text{conv}\{\theta_1, \dots, \theta_L\} = \text{conv}\{b_1, \dots, b_K\}. \quad (104)$$

Affine independence makes the right side a $(K-1)$ -simplex whose vertices are exactly the unordered set $\{b_1, \dots, b_K\}$. Applying the same argument to (A', B') shows that its bases are the same vertex set. Thus $B' = P^\top B$ for a permutation matrix P . Relative to these identified vertices, Lemma 1 gives unique coordinates for every row, hence $A' = AP$.

For the one-hot consequence, $\theta_i = b_{c_i}$. Affine independence implies that bases are pairwise distinct, so $\theta_i = \theta_{i'}$ if and only if $c_i = c_{i'}$. Full basis usage provides all anchors and makes the basis set observable. Without full usage, an unused basis is unconstrained; with duplicate bases, their assignment labels are observationally interchangeable. $\square$

Separability must restrict the candidate model class, not merely the planted truth. If alternative bases may lie outside the observed hull and need not appear as anchors, a larger simplex can contain the same θ_i and assign them different soft barycentric coordinates.

Proof of Theorem 4. If $c_i = c_{i'}$, the triangle inequality gives

$$\|\hat{\theta}_i - \hat{\theta}_{i'}\|_2 = \|e_i - e_{i'}\|_2 \leq 2\varepsilon. \quad (105)$$

If $c_i \neq c_{i'}$, the reverse triangle inequality gives

$$\|\hat{\theta}_i - \hat{\theta}_{i'}\|_2 \geq \|b_{c_i} - b_{c_{i'}}\|_2 - \|e_i\|_2 - \|e_{i'}\|_2 \geq \gamma - 2\varepsilon > 2\varepsilon. \quad (106)$$

The threshold graph therefore contains all and only within-class edges, so its connected components recover the partition. A cluster mean equals its basis plus an average of error vectors. By convexity of the norm, that average has norm at most ε . $\square$

The 4ε condition is a clean sufficient margin, not a necessary one. Stable recovery without some separation condition is impossible: two bases closer than the observation uncertainty can exchange assignments without changing the observed data appreciably.

D.6 FINITE-SAMPLE RECOVERY BY SHARED-INPUT PROBES

For a pair (i, j) write

$$X_s^{ij} = \|Y_i(H_s) - Y_j(H_s)\|_2^2, \quad d_{ij}^2 = \mathbb{E}X_s^{ij}, \quad \hat{d}_{ij}^2 = n^{-1} \sum_{s=1}^n X_s^{ij}. \quad (107)$$

The observation Y_i may include seeded measurement noise. The population distances and their gap must be computed under that same observation model. Noise that differs systematically between layers can close or reverse the gap for the clean functional partition, in which case the theorem does not apply; it does not silently redefine the recovery target.

Proof of Theorem 5. Let $N_p = \binom{L}{2}$. Substitute $u = \delta/4$ into Equation 29. The stated sample-size condition implies for each pair

$$\Pr\left(|\hat{d}_{ij}^2 - d_{ij}^2| > \delta/4\right) \leq \frac{\eta}{N_p}. \quad (108)$$

A union bound gives, with probability at least $1 - \eta$,

$$\max_{i < j} |\hat{d}_{ij}^2 - d_{ij}^2| \leq \delta/4. \quad (109)$$

On this event, every within-class pair obeys

$$\hat{d}_{ij}^2 \leq w + \delta/4 < (w + b)/2, \quad (110)$$

whereas every between-class pair obeys

$$\hat{d}_{ij}^2 \geq b - \delta/4 > (w + b)/2. \quad (111)$$

Thresholding therefore recovers all pairwise equivalence decisions and hence the partition.

For agglomerative clustering, Equation 109 implies that every distance between subsets of one true class is smaller than every distance between subsets of different true classes, for single, complete, or average linkage. While more than K_f clusters remain, at least one true class is split, and the next merge is necessarily within that class. Induction shows that cutting the dendrogram at K_f clusters returns the true partition. This class count is a property of the functional equivalence relation and need not equal the number of router bases. $\square$

A bounded special case. If the observations are independent across probe draws s with the same pairwise expectations, and $0 \leq X_s^{ij} \leq R^2$ almost surely for every pair, Hoeffding's inequality gives

$$\Pr(|\hat{d}_{ij}^2 - d_{ij}^2| > u) \leq 2 \exp(-2nu^2/R^4). \quad (112)$$

Thus it is sufficient that

$$n \geq \frac{8R^4}{\delta^2} \log \frac{2\binom{L}{2}}{\eta}. \quad (113)$$

For unbounded probes, Equation 29 must be justified by the probe distribution, module class, and noise model rather than assumed silently. Clipping or normalizing probe responses supplies a directly checkable bounded version at the cost of changing the target metric.

The released paired sweep uses iid Gaussian inputs and $Y_{is} = r_i(H_s) + \sigma \epsilon_{is}$ with iid Gaussian ϵ_{is} and a fixed absolute σ . For its finite MLP and Transformer blocks, LayerNorm with a positive numerical stabilizer bounds the normalized inputs, and the remaining fixed affine, GELU, softmax-attention, and residual-update maps give a bounded clean update. A bounded vector plus Gaussian noise has a sub-exponential squared norm, so Bernstein constants exist for each frozen configuration. The sweep does not estimate uniform numerical values of (ν, c) or the population gap δ ; consequently its reported empirical gap is descriptive and is not presented as an application of the theorem’s population certificate.

What the certificate means. The theorem targets equality under the chosen P_H and observation model. It cannot certify global functional identity outside the support of P_H . Nor does it identify a historical generative graph when two distinct teacher modules happen to agree on that support.

D.7 OPTIMIZATION AND INITIALIZATION EFFECTS

Proof of Proposition 8. By the chain rule for $\Theta = AB$,

$$\nabla_A \mathcal{L} = GB^\top, \quad \nabla_B \mathcal{L} = A^\top G. \quad (114)$$

Euclidean gradient flow therefore satisfies $\dot{A} = -GB^\top$ and $\dot{B} = -A^\top G$. Differentiating the product gives

$$\dot{\Theta} = \dot{A}B + A\dot{B} \quad (115)$$

$$= -G(B^\top B) - (AA^\top)G. \quad (116)$$

The right-hand side is not determined by AB alone. If two factorizations induce different linear operators on G , choosing G outside the null space of their difference produces different initial effective velocities. Non-orthogonal gauge transformations generically alter both Gram matrices. $\square$

The previous calculation optimizes A directly and is useful because its preconditioning is explicit. The constrained softmax model has the same coordinate dependence. In the next derivation vectors are columns, so $\theta_i = B^\top a_i$ and $g_i = \nabla_{\theta_i} \mathcal{L}$.

Proposition 13 (Exact softmax effective dynamics). *Let*

$$a_i = \text{softmax}(z_i/\tau), \quad J_i = \frac{1}{\tau} (\text{Diag}(a_i) - a_i a_i^\top). \quad (117)$$

Without factor-specific regularizers, Euclidean gradient flow in Z, B obeys

$$\dot{z}_i = -\eta_Z J_i B g_i, \quad (118)$$

$$\dot{B} = -\eta_B \sum_{r=1}^L a_r g_r^\top, \quad (119)$$

$$\dot{\theta}_i = -\eta_Z B^\top J_i^2 B g_i - \eta_B \sum_{r=1}^L \langle a_i, a_r \rangle g_r. \quad (120)$$

Proof. The Jacobian of a_i with respect to z_i is the symmetric matrix J_i . Since $\nabla_{a_i} \mathcal{L} = B g_i$, the chain rule gives $\nabla_{z_i} \mathcal{L} = J_i B g_i$. Similarly, $\nabla_B \mathcal{L} = \sum_r a_r g_r^\top$. Including the block learning rates in the flow therefore gives

$$\dot{a}_i = J_i \dot{z}_i = -\eta_Z J_i^2 B g_i. \quad (121)$$

Substituting $\dot{a}_i$ and $\dot{B}$ into $\dot{\theta}_i = \dot{B}^\top a_i + B^\top \dot{a}_i$ yields Equation 120. $\square$

Proof of Proposition 9. Let $C_i = \text{Diag}(a_i) - a_i a_i^\top$, so $J_i = C_i/\tau$. The matrix C_i is positive semidefinite, hence

$$\|C_i\|_2 \leq \text{tr}(C_i) = 1 - \|a_i\|_2^2 \quad (122)$$

$$\leq 1 - (1 - \varepsilon)^2 \leq 2\varepsilon. \quad (123)$$

Using $\nabla_{z_i} \mathcal{L} = J_i B g_i$ gives

$$\|\nabla_{z_i} \mathcal{L}\|_2 \leq \|J_i\|_2 \|B g_i\|_2 \leq \frac{2\varepsilon}{\tau} \|B g_i\|_2. \quad (124)$$

The exact router velocity additionally obeys

$$\|\dot{a}_i\|_2 \leq \eta_Z \left(\frac{2\varepsilon}{\tau} \right)^2 \|B g_i\|_2. \quad (125)$$

The bound is instantaneous: it accounts for both temperature and the current probability $1 - \varepsilon$, while varying τ at fixed logits changes ε as well. It does not by itself bound cumulative router motion or guarantee preservation of an argmax over a training interval. $\square$

Proposition 14 (Symmetric initialization trap). *If every router row is uniform and every basis is initialized identically, then deterministic gradient flow keeps the router uniform and the bases identical for all time for which the solution exists.*

Proof. If all basis rows equal b , then $B g_i = (b^\top g_i) \mathbf{1}$. Since $J_i \mathbf{1} = 0$, every logit gradient is zero. Under a uniform router, every basis receives the same gradient $K^{-1} \sum_i g_i$. Equality of the bases is therefore invariant, which in turn keeps every router gradient zero. $\square$

The proposition is not a claim that practical floating-point training always remains symmetric: independent random initialization, parameter-specific stochastic perturbations, and implementation nondeterminism can break the equality. Shared minibatch sampling alone preserves this symmetry when every sampled loss depends on the factors only through AB : the same gradient equalities hold separately for each minibatch. It identifies the source of specialization as an explicit or stochastic symmetry breaker rather than information contained in the initial effective model.

D.8 DETAILED CLAIM BOUNDARY

The formal results support the following distinctions.

- Router entropy, smoothness, bandedness, cosine similarity, and argmax graphs are gauge-dependent coordinates. They require gauge-counterfactual or initialization-intervention evidence before receiving a structural interpretation.
- The effective-weight equality partition is gauge invariant. It is still only a parameter-space partition: neural permutation and scaling symmetries can make unequal parameters functionally equivalent.
- Shared-input probes target functional equivalence under a named probe distribution. They do not establish equality outside its support and need not recover the historical graph that generated a teacher.
- A generative graph becomes recoverable under explicit conditions such as one-hot routing, full basis usage, affine independence or separation, and an observation regime strong enough to distinguish the bases. Low task loss alone supplies none of these conditions.
- Affine independence is insufficient for an unknown fully soft factorization. It ensures unique barycentric coordinates only after vertices are known; anchors or another valid simplex-identification condition are still required.
- Factor-specific normalization or regularization may choose a useful gauge. It does not retroactively make that coordinate the data-generating graph. Exact equality constraints can reduce the gauge group but require a fresh identifiability analysis.
- The dynamics results demonstrate coordinate-dependent preconditioning, saturation, and a symmetry trap. They do not prove a universal optimization preference for smooth, monotone, periodic, sparse, or low-entropy routers.
- No strict function-class hierarchy, generalization improvement, or causal performance benefit follows from the algebraic factorization alone. Such claims require separate approximation, statistical, and compute-matched arguments.

E EXTENSIONS TO IMPERFECT PRODUCTS AND NATURAL PROBES

E.1 JOINT PRODUCT AND RESPONSE PERTURBATIONS

The same-product restriction in Theorem 6 can be relaxed for small product perturbations by constructing an intermediate factorization with exactly the reference product. This is an inverse-property result for valid factors; it does not certify the unconstrained numerical recovery algorithm.

Corollary 6 (Joint product–response stability). *Let two factor pairs satisfy all hypotheses of Theorem 6 except equal products, with the same constants $\alpha, s_A, s_B, L_A, L_B$ and known positive rates and temperature. Write*

$$\epsilon_\Theta = \|AB - A'B'\|_F, \quad \epsilon_R = \|\mathcal{R}_{A,B} - \mathcal{R}_{A',B'}\|_{F \rightarrow F}.$$

Assume $\epsilon_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$. Let C_ be the explicit constant of that theorem evaluated at $(\alpha/2, s_A/2, s_B/2, 2L_A, 2L_B)$, and put*

$$\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}, \quad M = 2L_B, \quad L_R = [(2\eta_B\sqrt{L} + 6\lambda_Z M^2)^2 + (2\lambda_Z M)^2]^{1/2}.$$

Then the joint Frobenius distance modulo a common permutation satisfies

$$d_{\text{perm}}((A, B), (A', B')) \leq C_* \epsilon_R + \kappa(1 + C_* L_R) \epsilon_\Theta.$$

Proof. Let $P_A = AA^\dagger$, $\tilde{A} = P_A A'$, and $\tilde{B} = \tilde{A}^\dagger AB$. Since $A\mathbf{1} = \mathbf{1}$, $P_A \mathbf{1} = \mathbf{1}$ and $\tilde{A}\mathbf{1} = \mathbf{1}$. Moreover,

$$(I - P_A)A' = (I - P_A)(A'B' - AB)B'^\dagger, \quad \|\tilde{A} - A'\|_F \leq \epsilon_\Theta / s_B.$$

Thus $\min \tilde{A} \geq \alpha/2$ and $\sigma_{\min}(\tilde{A}) \geq s_A/2$. Its column space is that of A , so $\tilde{A}\tilde{B} = AB$. Also

$$\tilde{B} - B' = \tilde{A}^\dagger P_A (AB - A'B'), \quad \|\tilde{B} - B'\|_F \leq 2\epsilon_\Theta / s_A.$$

Consequently $\sigma_{\min}(\tilde{B}) \geq s_B/2$, and both pairs $(A, B), (\tilde{A}, \tilde{B})$ obey the enlarged domain bounds defining C_* . The joint displacement from (A', B') to the intermediate pair is at most $\kappa\epsilon_\Theta$.

For simplex rows a, a' , let $C(a) = \text{diag}(a) - aa^\top$. The elementary bounds $\|a\|_2 \leq 1$, $\|C(a)\|_2 \leq 1$, and $\|C(a) - C(a')\|_2 \leq 3\|a - a'\|_2$ imply $\|C(a)^2 - C(a')^2\|_2 \leq 6\|a - a'\|_2$. In the response formula, the Gram contribution changes by at most $2\eta_B\sqrt{L}\|\Delta A\|_F$. Each local diagonal block changes by at most $\lambda_Z(6M^2\|\Delta a_i\|_2 + 2M\|\Delta B\|_F)$. The norm of the direct sum is the maximum block norm; therefore the full response is L_R -Lipschitz in the joint factor norm on this domain. The response discrepancy between (A, B) and the intermediate pair is bounded by $\epsilon_R + L_R\kappa\epsilon_\Theta$. Apply the same-product theorem and then the triangle inequality; common permutations preserve the joint norm. $\square$

If two valid feasible pairs each match the same noisy observations within e_Θ, e_R , the corollary applies with $2e_\Theta, 2e_R$, provided the small-product-error condition holds. It does not establish existence of a feasible pair, an efficient constrained solver, or stability near rank loss. The constants are conservative and not numerical accuracy predictions.

E.2 A SUFFICIENT CONDITION FOR TOMOGRAPHY WITH NATURAL GRADIENTS

Let $U \in \mathbb{R}^{K \times D}$ have orthonormal rows spanning the exact product's row space. Write the structured response as $R^{(p)} = QG^{(p)} + \mathcal{T}(G^{(p)}U^\top)U$, where $Q = \eta_B AA^\top$ and $\mathcal{T}$ applies a $K \times K$ matrix T_i to each row coordinate, using column-vector convention. Define $G_\perp^{(p)} = G^{(p)} - (G^{(p)}U^\top)U$ and similarly $R_\perp^{(p)}$.

Proposition 15 (Rank-checked natural-probe reconstruction). *If $H = \sum_p G_\perp^{(p)} G_\perp^{(p)\top}$ is invertible and, for every layer i , the vectors $(G^{(p)}U^\top)_{i,:}$ span $\mathbb{R}^K$, the observations uniquely determine Q and all T_i . They therefore determine the response on all gradients within the assumed response family.*

Proof. Projection eliminates the local terms, giving $R_{\perp}^{(p)} = QG_{\perp}^{(p)}$. Hence

$$Q = \left(\sum_p R_{\perp}^{(p)} G_{\perp}^{(p)\top} \right) H^{-1}.$$

Subtract $QG^{(p)}$ and project onto U ; each layer's remaining linear system has a unique T_i by its spanning assumption. $\square$

This is a sufficient algebraic condition, not a guarantee about task gradients or an optimal-query result. The implementation checks numerical ranks and conditioning, and uses fixed pseudoinverse tolerances. Small singular values can amplify measurement noise. The designed probes avoid relying on empirical task-gradient richness; natural probes trade that design for a condition that must be checked at the checkpoint. Measurements here still use controlled Euclidean responses at a fixed checkpoint, not an observed AdamW trajectory.

F GLOBAL STABILITY FROM FINITE NOISY OBSERVATIONS

This section closes the observation-level gap between an exact-product finite-probe reconstruction and a full-operator inverse bound. It is an explicit measurement-level consequence of Theorem 6 and Corollary 6, not a new general inverse theorem. The conclusion is global over all feasible factors in a specified nondegenerate domain. No numerical recovery algorithm enters its proof.

Domain, observations, and quantifiers. Fix $L \geq K \geq 2$, $D - K \geq L$, and known $\eta_B, \eta_Z, \tau > 0$, with $\lambda = \eta_Z/\tau^2$. For positive α, s_A, s_B, M , let

$$\mathcal{D} = \left\{ (A, B) : \begin{array}{l} A \in \mathbb{R}^{L \times K}, A\mathbf{1}_K = \mathbf{1}_L, A_{ij} \geq \alpha, \sigma_{\min}(A) \geq s_A, \\ B \in \mathbb{R}^{K \times D}, \sigma_{\min}(B) \geq s_B, \|B\|_F \leq M \end{array} \right\}.$$

The margins describe an assumed parameter domain; they are not inferred from one candidate's conditioning. Given any observed product $\hat{\Theta} \in \mathbb{R}^{L \times D}$, choose an orthonormal-row matrix $U \in \mathbb{R}^{K \times D}$ from its top K right singular vectors, and set $\hat{P} = U^\top U$. Choose $W \in \mathbb{R}^{L \times D}$ with $WW^\top = I_L$ and $WU^\top = 0$; this is possible since $D - K \geq L$. All choices depend only on $\hat{\Theta}$ and dimensions, with a fixed tie-breaking rule if needed. Writing $u_p^\top$ for row p of U , define

$$G_p = \frac{W + \mathbf{1}_L u_p^\top}{\sqrt{2L}}, \quad p = 1, \dots, K. \quad (126)$$

Orthogonality gives $\|G_p\|_F = 1$. The complement code W is deliberately repeated in every query. Probes are selected before seeing their responses.

For any observed response matrices $\hat{R}_p$, define

$$\mathcal{F} = \left\{ (A, B) \in \mathcal{D} : \begin{array}{l} \|AB - \hat{\Theta}\|_F \leq \varepsilon_\Theta, \\ (\sum_{p=1}^K \|\mathcal{R}_{A,B}(G_p) - \hat{R}_p\|_F^2)^{1/2} \leq \varepsilon_R \end{array} \right\}. \quad (127)$$

This is the entire feasible set in the original D -dimensional basis space, not a fit restricted to $\hat{P}$. Noise may be deterministic, adversarial, and dependent on the product observation or probes, provided these norm bounds hold.

Explicit constants. Put

$$h = \sqrt{2L}(K^{-1/2} + 1 + \sqrt{L}), \quad \beta = \frac{8\lambda M}{s_A}(h\sqrt{K} + 1), \quad (128)$$

$$\kappa = (s_B^{-2} + 4s_A^{-2})^{1/2}, \quad L_R = \left[(2\eta_B\sqrt{L} + 24\lambda M^2)^2 + (4\lambda M)^2 \right]^{1/2}. \quad (129)$$

To make the inherited inverse constant fully explicit, define

$$L_M = \max\{1, 4\sqrt{L}/s_A\}, \quad c_G = \frac{4L}{\eta_B s_A^2}, \quad c_H = \frac{8}{\lambda s_B^2}, \quad (130)$$

$$\mu_* = \frac{\alpha}{2} \left(1 + \frac{1}{4L_M^2}\right), \quad c_J = \frac{L_M^2(c_G + c_H)}{\mu_*}, \quad c_D = \frac{2\sqrt{LK}}{s_A} c_J, \quad (131)$$

$$c_{\text{row}} = [Kc_D^2 + (c_G + Kc_D^2)^2]^{1/2}, \quad \delta_0 = \min\left\{1, \frac{1}{2c_G}, \frac{1}{2\sqrt{K}c_{\text{row}}}\right\}. \quad (132)$$

Finally set

$$C_* = \max\left\{2\sqrt{L + L_M^2 M^2} \sqrt{K} c_{\text{row}}, \frac{4\sqrt{L + M^2}}{\delta_0}\right\}, \quad (133)$$

$$C_R = 2hC_*, \quad (134)$$

$$C_\Theta = \beta C_* + 2\kappa(1 + C_* L_R), \quad (135)$$

$$\varepsilon_0 = \min\{\alpha s_B/4, s_A s_B/8\}. \quad (136)$$

These constants can be very large. No sharpness or useful numerical accuracy at a particular finite noise level is asserted.

Theorem 7 (Whole-feasible-set diameter from K noisy probes). *Under the preceding definitions, if $0 \leq \varepsilon_\Theta \leq \varepsilon_0$ and $\varepsilon_R \geq 0$, then*

$$\text{diam}_{d_{\text{perm}}}(\mathcal{F}) \leq C_\Theta \varepsilon_\Theta + C_R \varepsilon_R, \quad d_{\text{perm}}(x, x') = \min_{\Pi \in \mathfrak{S}_K} (\|A' - A\Pi\|_F^2 + \|B' - \Pi^\top B\|_F^2)^{1/2}. \quad (137)$$

No smallness restriction on ε_R is needed. Empty feasible sets have diameter zero by convention; recovery interpretations require a nonempty feasible set.

Proof. We give the observation-to-operator bridge first, then apply the explicit global factor inverse to the original, unprojected factors.

Step 1: uniform control outside the observed subspace. For every $(A, B) \in \mathcal{F}$, AB is a rank- K competitor in the best rank- K approximation of $\hat{\Theta}$. Hence

$$\|\hat{\Theta}(I - \hat{P})\|_F \leq \varepsilon_\Theta, \quad \|AB(I - \hat{P})\|_F \leq 2\varepsilon_\Theta.$$

Since A has a left inverse with $\|A^\dagger\|_2 \leq s_A^{-1}$,

$$\|B - B\hat{P}\|_F \leq 2\varepsilon_\Theta/s_A. \quad (138)$$

Set $\bar{B} = B\hat{P}$. This is an analytical device, not an assumption that the true basis lies in the estimated subspace. For $S_i = \text{diag}(a_i) - a_i a_i^\top$, $\|S_i\|_2 \leq 1$ and $\|\bar{B}\|_2, \|B\|_2 \leq M$, so

$$\|B^\top S_i^2 B - \bar{B}^\top S_i^2 \bar{B}\|_2 \leq 2M\|B - \bar{B}\|_F.$$

The shared Gram term is unchanged; the remaining operator difference is block diagonal. Thus, uniformly over $\mathcal{F}$,

$$\|\mathcal{R}_{A,B} - \mathcal{R}_{A,\bar{B}}\|_{F \rightarrow F} \leq t, \quad t = \frac{4\lambda M}{s_A} \varepsilon_\Theta. \quad (139)$$

No rank property of $\bar{B}$ is required.

Step 2: stable decoding on a common support. Consider the larger linear class $\mathcal{S}(G) = QG + \mathcal{T}(GU^\top)U$, where $Q \in \mathbb{R}^{L \times L}$ and $\mathcal{T}$ right-multiplies row i by an arbitrary $K \times K$ matrix T_i . Symmetry is unnecessary. For the difference of two such operators, let $E_p = \Delta\mathcal{S}(G_p)$ and $E = (\sum_p \|E_p\|_F^2)^{1/2}$. Set $c = (2L)^{-1/2}$. Complement projection gives $E_p W^\top = c\Delta Q$, whence

$$\|\Delta Q\|_F \leq \frac{E}{c\sqrt{K}}. \quad (140)$$

Projection onto U gives, row by row,

$$E_p U^\top = c(\Delta Q \mathbf{1}_L) e_p^\top + c[e_p^\top \Delta T_i]_{i=1}^L.$$

Stacking the K equations enumerates every row of every ΔT_i . The triangle inequality and $\|\Delta Q \mathbf{1}_L\|_2 \leq \sqrt{L} \|\Delta Q\|_F$ give

$$\left(\sum_i \|\Delta T_i\|_F^2 \right)^{1/2} \leq E/c + \sqrt{K} \|\Delta Q \mathbf{1}_L\|_2 \leq (1 + \sqrt{L})E/c. \quad (141)$$

The shared term has induced norm $\|\Delta Q\|_2$ and the local direct sum has norm at most $\max_i \|\Delta T_i\|_2$. Equations 140–141 therefore imply

$$\|\Delta S\|_{F \rightarrow F} \leq hE. \quad (142)$$

For a projected factor response, take $Q = \eta_B A A^\top$ and $T_i = \lambda U \bar{B}^\top S_i^2 \bar{B} U^\top$. The lemma applies even if those projected factors fail the domain’s rank margins.

Step 3: return to the original full operators. Choose any $x = (A, B), x' = (A', B') \in \mathcal{F}$. Their observed response stacks differ by at most $2\varepsilon_R$. Each unit-norm probe incurs at most t error under Equation 139; the stack incurs at most $\sqrt{K}t$ for each pair. Applying Equation 142, then adding the two full-operator compression errors, yields

$$\|\mathcal{R}_x - \mathcal{R}_{x'}\|_{F \rightarrow F} \leq h(2\varepsilon_R + 2\sqrt{K}t) + 2t \quad (143)$$

$$= 2h\varepsilon_R + \beta\varepsilon_\Theta. \quad (144)$$

Every pair uses the same observed U, W and probes.

Step 4: the global factor inverse. The two original factors obey the domain bounds with $L_A = \sqrt{L}$ and $L_B = M$, and $\|AB - A'B'\|_F \leq 2\varepsilon_\Theta$. Our threshold ensures $2\varepsilon_\Theta \leq \min\{\alpha s_B/2, s_A s_B/4\}$, exactly the product-error condition of Corollary 6. That corollary’s C_* is the constant of Theorem 6 evaluated at $(\alpha/2, s_A/2, s_B/2, 2\sqrt{L}, 2M)$, which expands to Equations 132–136. It includes the bounded-domain argument for large response differences, not just a local inverse estimate. Consequently,

$$d_{\text{perm}}(x, x') \leq C_* \|\mathcal{R}_x - \mathcal{R}_{x'}\|_{F \rightarrow F} + \kappa(1 + C_* L_R) \|AB - A'B'\|_F.$$

Substituting Equation 144 proves the claimed bound for an arbitrary pair in $\mathcal{F}$, hence for its entire diameter. No connected-component or candidate-neighborhood premise was used. $\square$

Recovery interpretation and boundaries. If the true factors belong to $\mathcal{D}$ and satisfy the declared noise bounds, they lie in $\mathcal{F}$, so every feasible factor estimate obeys the displayed error bound. This does not prove that a particular solver finds a feasible point. It does not verify domain membership from noisy data, establish optimality of K queries, or cover passive task-gradient queries or unknown AdamW state. Each query returns a full $L \times D$ response, not one scalar.

No smooth choice of singular-vector basis is needed: the proof holds for every chosen best rank- K right subspace and orthonormal bases. It does not assume a singular-value gap in the observation. For a nonempty feasible set under the stated threshold, separation at the K th cut also follows from the rank margins and perturbation inequalities, but that fact is not needed in the proof.

For $K = 1$, row stochasticity fixes $A = \mathbf{1}_L$ and the response’s softmax term vanishes. Directly, $d_{\text{perm}}(x, x') = \|B - B'\|_F \leq 2\varepsilon_\Theta/\sqrt{L}$. Thus the boundary case needs no response information or small-noise threshold. The nontrivial theorem concerns $K \geq 2$.

G OBSERVABLE LOCAL SEPARATION FOR NOISY RECOVERY

This section specializes standard Lipschitz-Jacobian Taylor reasoning (Behling et al., 2018, Section 2, equations (5)–(6)) to the response measurement map. It is not a new generic inverse theorem. The distinction between exact diagonalization and sensitivity of approximate diagonalization is already central to Afsari (2008). Throughout the numerical study, $\eta_Z = \eta_B = \tau = 1$.

Exact objective compression in an estimated subspace. Let $U \in \mathbb{R}^{K \times D}$ have orthonormal rows spanning the leading right-singular subspace of the observed product $\hat{\Theta}$. Represent candidates by $B = CU$. For measured pairs $(G_p, \hat{R}_p)$, write $X_p = G_p U^\top$, $G_p^\perp = G_p - X_p U$, $Y_p = \hat{R}_p U^\top$, and $R_p^\perp = \hat{R}_p - Y_p U$. With $Q = AA^\top$ and $S_i = \text{diag}(a_i) - a_i a_i^\top$, the predicted parallel response is $QX_p + [C^\top S_i^2 C X_{p,i}]_i$, and the orthogonal response is $QG_p^\perp$. Set

$$H = \sum_p G_p^\perp (G_p^\perp)^\top, \quad T = \sum_p R_p^\perp (G_p^\perp)^\top.$$

When $H \succ 0$, least-squares orthogonality gives

$$\sum_p \|QG_p^\perp - R_p^\perp\|_F^2 = \|QH^{1/2} - TH^{-1/2}\|_F^2 + \sum_p \|TH^{-1}G_p^\perp - R_p^\perp\|_F^2, \quad (145)$$

$$\|ACU - \hat{\Theta}\|_F^2 = \|AC - \hat{\Theta}U^\top\|_F^2 + \|\hat{\Theta}(I - U^\top U)\|_F^2. \quad (146)$$

The second terms are candidate independent, but are retained in every reported residual. Dividing product and response blocks by their respective observed Frobenius norms gives an exact small least-squares objective. Candidate-dependent differences are preserved isometrically, including their Jacobian singular values. Numerically singular H is rejected. Crucially, a subspace estimated from noisy data need not contain the true basis. This restriction is not justified by the local result below.

An explicit curvature bound. At a strictly interior candidate (A_0, C_0) , use $A = A_0 + s_A ZV^\top$, $C = C_0 + s_C W$, where $s_A = \|A_0\|_F$, $s_C = \|C_0\|_F$, and V is an orthonormal basis for $1^\perp$. Let $x = \text{vec}(Z, W)$ and $r \leq r_0 = \min_{ij} (A_0)_{ij} / (2s_A)$. All routers in this ball are strictly feasible. Set $b = s_C(1+r)$ and $c = \min\{1, \max_i \|S_i(A_0)\|_2 + 3s_A r\}$. The inequalities $\|DS_i[h]\|_2 \leq 3\|h\|_2$ and $\|D^2 S_i[h, k]\|_2 \leq 2\|h\|_2\|k\|_2$ follow by differentiating $\text{diag}(a) - aa^\top$. Differentiating $(S_i C)^\top S_i C$ twice then bounds its bilinear Hessian by the largest eigenvalue $h(r)$ of

$$\begin{pmatrix} (18 + 4c)b^2 s_A^2 & 12cbs_A s_C \\ 12cbs_A s_C & 2c^2 s_C^2 \end{pmatrix}.$$

The matrix is symmetric with nonnegative entries, so its largest eigenvalue equals its spectral norm. For the normalized measurement map F , a conservative Hessian bound is

$$M(r) = \left[\left(\frac{2s_A s_C}{\|\hat{\Theta}\|_F} \right)^2 + \left(\frac{(2s_A^2 + h(r))\sqrt{\sum_p \|G_p\|_F^2}}{\|(\hat{R}_p)_p\|_F} \right)^2 \right]^{1/2}.$$

The $2s_A^2$ term accounts for the Hessian of $AA^\top G_p$. The same derivation applies in full coordinates, with C replaced by B .

Proposition 16 (Conditional local feasible-component separation). *Let $\mu = \sigma_{\min}(DF(0)) > 0$, $\rho = \|F(0) - y\|$, and suppose $\|D^2 F\| \leq M$ on the closed radius- r ball, with $\mu - Mr/2 > 0$. Every x in this ball satisfies*

$$\|F(x) - F(0)\| \geq (\mu - Mr/2)\|x\|.$$

If $\rho \leq \nu$ and $\rho + \nu < \mu r - Mr^2/2$, the connected component of $\{x : \|F(x) - y\| \leq \nu\}$ containing zero lies inside the ball, and each of its points obeys

$$\|x\| \leq \frac{\rho + \nu}{\mu - Mr/2}.$$

Proof. The integral Taylor remainder has norm at most $M\|x\|^2/2$. Combining this with $\|DF(0)x\| \geq \mu\|x\|$ proves the first inequality. No feasible point can lie on the sphere of radius r , because its measurement difference from $F(0)$ would exceed $\rho + \nu$. The norm image of a connected set containing zero cannot reach beyond r without meeting r . The triangle inequality then gives the distance bound. $\square$

The implementation takes $r = \min\{r_0, \mu/M(r_0)\}$ and recomputes $M(r)$. For known relative observation errors bounded by $\sigma < 1$ in both blocks, $\nu = \sqrt{2}\sigma/(1-\sigma)$ is an observable normalized noise upper bound. The separately recorded 10^{-10} numerical floor affects empirical scores only. At zero declared noise, rounding can prevent a numerical local pass.

Scope of the conclusion. This proposition neither excludes distant components nor establishes that the true factors belong to the candidate’s component. It is not a convergence guarantee for the optimizer. Its metric uses candidate norms, whereas benchmark errors use true-factor norms and a common permutation. In a noisy estimated subspace, the true factors can be outside the model class altogether. Floating-point singular values and curvature evaluations are not outward-rounded proofs; we call their successful evaluations *numerical local passes*, not certified global recovery. Sixteen separate small full-coordinate checks place a known feasible point inside the calculated ball and satisfy the distance inequality. These are theorem sanity checks, not sixteen successful noisy-recovery trials.

H DETAILED PRIOR-ART AUDIT

This appendix records the distinctions and exact variable maps behind the condensed audit in Section 2. It separates uniqueness of a factorization, invariance of a computation or decision under refactorization, and recovery of a historical sharing graph. The literature establishes strong results for the first two and contains a few coordinate-to-action systems, but our audit does not support a broad claim that cross-layer compression papers purport to recover a data-generating sharing graph.

Simplex factorization: the static ambiguity is prior art. For $K \geq 2$, after transposition, our parameter-space model is exactly the unconstrained simplex-structured matrix factorization (SSMF)

$$X = WH, \quad X = \Theta^\top, \quad W = B^\top, \quad H = A^\top.$$

Section IV-A of Vu Thanh et al. (2023) already proves that unconstrained SSMF is non-identifiable. In their notation,

$$W(\delta) = W \left((1 + \delta)I - \frac{\delta}{K} \mathbf{1}\mathbf{1}^\top \right), \\ H(\delta) = \left(\frac{1}{1 + \delta} I + \frac{\delta}{(1 + \delta)K} \mathbf{1}\mathbf{1}^\top \right) H.$$

Setting $s = \delta/(1 + \delta)$ makes their coefficient transform $M_s = (1 - s)I + s\mathbf{1}\mathbf{1}^\top/K$, so that $H(\delta) = (AM_s)^\top$ and $W(\delta) = (M_s^{-1}B)^\top$. This is exactly our uniformizing gauge after transpose and substitution, not a distinct neural-network theorem. The entropy consequence uses this uniformizing path, which preserves coordinate ordering and argmax assignments. Our local gauge description and argmax example use the broader family of feasible changes of basis; the argmax example is nonuniform. These specialize known factor ambiguity to router diagnostics, not a new foundational non-identifiability result. Conversely, classical NMF and SSMF work gives recovery under additional support, sufficiently-scattered, facet-coverage, or volume assumptions (Donoho & Stodden, 2003; Fu et al., 2018; Abdolali & Gillis, 2021; Gillis & Rajkó, 2023). These are condition-dependent positive boundaries, not consequences of the simplex constraint alone.

LoRA gauges, invariant processing, and merge consequences. The general low-rank gauge is also established prior art. For full-rank LoRA factors, Putterman et al. (2024) show that effective products separate $GL(r)$ orbits and develop invariant/equivariant processing of the factors. LoRA Done RITE constructs factor-optimization updates invariant to the same change of basis (Yen et al., 2025). LoRA-S equips the full-rank factor quotient with an invariant metric and uses a Sylvester equation to lift preconditioned product-space directions; its invariance comparison uses the same product-space direction in both factor representations (Zheng & Wu, 2026). Thus ordinary factor-gradient chart dependence and quotient-invariant optimizer design are both established precedents. Gauge-aware federated LoRA explicitly shows why raw-factor aggregation is representation dependent and gives an invariant server representation (Chen et al., 2026); W2T obtains gauge-invariant adapter features through product-space QR–SVD processing (Han et al., 2026). Thus neither the existence of the $GL(r)$ gauge, the use of effective products as invariants, nor the general warning against raw-factor aggregation is new here.

TwistedMerge supplies a particularly important trained-model precedent (Gong & Xu, 2026). On trained rank-four residual adapters, its orthogonal, diagonal, and dense gauge scrambles change naive factor-averaging outcomes, whereas whitened synchronization and deterministic full-update SVD

remain stable. It therefore already covers the broad empirical consequence “equivalent trained factors can make naive merging behave differently.” Its action is averaging independently trained adapters in a single residual LoRA-form feature layer, not selecting a directed within-model layer tie from a shared factor; it provides neither an ASLoRA nearest-pair analysis nor the shared-softmax response stabilizer below.

An ICML 2026 workshop poster explicitly studies the $GL(r)$ gauge and LoRA merging (Karuturi et al., 2026). We verified its official title, authors, and poster record, but could not obtain its primary full text during this audit. We therefore leave its coverage of maximal invariants, nearest-pair selection, Loewner-order tests, merge regret, ASLoRA, and optimizer-response stabilizers adjudicated; no priority claim in this paper is inferred from its title.

Structural actions derived from learned coordinates. The closest cases differ in what they claim. Savarese & Maire (2019) represent each convolutional layer as a linear combination of global templates, use coefficient cosine geometry as a sharing measure, infer an approximately recurrent pattern, and then evaluate the folded network. This is architectural-discovery language followed by an empirical intervention, not an assertion that the coefficient plot alone authenticates a historical graph. Neural Parameter Allocation Search learns factor coordinates or embeddings and clusters them after early training to make hard parameter groups (Plummer et al., 2022); its claim is automated allocation under a parameter budget, not recovery of a planted sharing history.

ASLoRA is the most direct target for our decision audit. Its arXiv-v2 method shares one input-side LoRA factor A , ranks layer-specific historical averages of B_i by raw Euclidean distance, and makes the lower layer use the selected upper layer’s factor (Hu et al., 2024). The paper motivates sharing A as removing its and its initialization’s influence and making B -similarity more reasonable and consistent. This is a real coordinate-to-nearest-action rule, but it is not a claim that the selected pair recovers a historical generator. We cite the arXiv version for these equations and algorithm details because the accessible formal journal record supports the broad global- A /selective- B description but did not expose the full text (Hu et al., 2026).

Sharing methods with different structural targets. Three recent systems illustrate the limits of applying our analysis to cross-layer sharing in general. E2LoRA computes proxy entropy and relative mutual information from task-subset gradients of the pretrained model *before* adapter fine-tuning, then fixes contiguous sharing intervals and interval ranks (Li et al., 2026). Although it calls the measured patterns intrinsic and reports stable structural decisions across subset seeds, it neither reads a jointly learned factor coordinate nor claims recovery of a historical graph; its seed/noise checks and downstream evaluation instead make it a positive proxy-to-allocation control. RaSA fixes a global cross-layer rank pool by construction and evaluates reconstruction and task quality (He et al., 2025), so there is no learned partition to interpret. LiSA measures input-dependent attention-score similarity and sensitivity, then trains alignment and difference-compensation modules and directly evaluates the compressed model (Mu et al., 2026); this is nonlinear attention-output reuse, not the linear factor chart studied here. These papers evaluate their structural interventions directly. They do not support a claim that historical sharing graphs are commonly inferred from learned coordinates.

Other cross-layer sharing methods also pursue different objectives. Dynamic Layer Tying searches over explicit copying actions (David-Hay & Wolf, 2024); Basis Sharing and FiPS target compression, reconstruction, and downstream quality (Wang et al., 2025; Üyük et al., 2026); Geo-Sharing selects a low-curvature compression assignment (Zhang et al., 2025). None of these claims a learned soft factor chart recovers a historical truth.

MASA is an especially close mixed boundary case (Zhussip et al., 2026, AAAI pp. 29262–29263; arXiv v2, supplementary p. 14). In its from-scratch path, each attention projection has $\widehat{W} = DC$: shared atoms D and per-layer coefficients C are jointly learned, each coefficient is stated to lie in $\mathbb{R}$, and the paper declines the atom soft constraints it has just described. A block embedding plus three-layer MLP produces C during training before the MLP is discarded. Thus the retained inference factor obeys $(D, C) \mapsto (DR, R^{-1}C)$. Under $\Theta = \widehat{W}^\top$, this maps exactly to $A = C^\top$ and $B = D^\top$, but A is unconstrained rather than row-simplex and the indirect MLP training law is not our Euclidean Z/B response law.

The paper calls coefficients atom contributions and shared atoms carriers of cross-layer statistical regularities. More strongly, its arXiv v2 supplement uses atom cosine (Equation 18) to describe distinct versus redundant atoms and says raw- C heatmaps reveal specialization, redundancy, and layer-wise adaptivity. These are genuine, but weak, chart-level structural interpretations: the paper defines no historical or generative target, reports no structure-recovery metric, and makes no separate hard action from C . For pretrained models it instead groups consecutive blocks using KL divergence between vocabulary-space distributions of block outputs, then applies orthonormal matrix PCA, residual refinement, and downstream validation. That action is function-derived rather than coefficient-derived; the PCA constraint removes general scale/shear freedom but still permits an orthogonal change of basis within the retained subspace. MASA therefore shows that descriptive coordinate reading occurs, while remaining negative evidence for the stronger historical-recovery and coordinate-to-action narrative. Likewise, Yeh et al. (2022) is a positive recovery result for an explicitly defined planted equivariance/partition target under stated assumptions, not an example of the error studied here. The defensible empirical claim is therefore narrow: some systems use learned coordinates to make structural actions, and those actions should be audited for representation dependence and downstream consequence.

A separate 2026 preprint supplies a particularly relevant metric-to-decision warning without involving a learned factor chart. In Pythia experiments, Khilar (2026) report that cross-layer subspace coupling improves two-sided Frobenius reconstruction over a per-layer SVD baseline by (37.2%) at 1.4B scale, yet produces WikiText-2 perplexity (159.14) versus (50.17) for that baseline. Their measured per-layer activation error increases by a matching factor of about three. This is direct prior evidence that a weight-space proxy can rank structural compression choices incorrectly and that deployed consequences or activation-matched controls are needed. It is not a result about gauge-equivalent coordinates, a learned router, or the same-function-to-different-action chain isolated in our ASLoRA audit.

Relation to existing identification results. Jacobian–Gram response operators are standard tangent-kernel objects (Jacot et al., 2018). More generally, Marcotte et al. (2026) write the lifted Euclidean flow as $-D\phi(\theta)D\phi(\theta)^\top \nabla f$; for $\phi(U, V) = UV^\top$, their matrix-factor response acts as $G \mapsto UU^\top G + GV V^\top$. Under $U \leftrightarrow A$, $V \leftrightarrow B^\top$, this is the direct non-softmax factor-response precedent, but it contains neither the rowwise $J(a_i)^2$ blocks nor a common-fiber stabilizer classification. For a single value–softmax output, the derivation in Varre et al. (2026, Lemma D.2) already contains $\dot{\beta} = (\|s\|^2 I + VC(s)^2 V^\top)r$, with $C(s) = \text{Diag}(s) - ss^\top$ and $r = -\nabla \ell$. Thus the local squared-covariance response is also prior art. The reduction of a linear-factor GL gauge to an orthogonal Euclidean isometry and optimizer-compatible softmax permutation/shift symmetries are further precedents (Singh, 2026; Lau & Su, 2026). Gauge-correction work further derives quotient-volume biases and balanced representatives for stochastic dynamics with matrix-factor and attention symmetries (Aladrah et al., 2026); its target is a stationary implicit-bias law, not equality of the finite-width instantaneous response at two fixed charts.

Two earlier identifiability results concern different softmax observations. Tran et al. (2025, Thm. 4.1) prove that, for their ReLU expert class, equality of a dense softmax-MoE function over its input domain, together with pairwise distinct expert functions and pairwise distinct gate-vector differences in each model, identifies its expert functions and gates up to a common gate translation and a permutation. Nguyen et al. (2023, Prop. 1 and Thm. 1) prove translation/label identifiability and a global linear lower bound from conditional-density Hellinger distance to a Voronoi parameter metric for an exact-fitted Gaussian softmax MoE on a compact parameter set. Their observables are respectively an entire MoE function and a conditional density; neither compares two AB charts on a common exact product fiber through their Euclidean instantaneous effective-gradient response. We therefore claim neither the first softmax permutation/translation identifiability result nor the first global inverse bound for a softmax-gated model. The Gram-to-Procrustes step also has a standard quantitative antecedent (Tu et al., 2016, Lem. 5.4); our quantitative addition is only the complete model-specific chain on the displayed uniform nondegenerate set.

A still closer intermediate step is already known for nonnegative representations. Dorrell et al. (2026, Theorem 2 and Appendix D.2) consider a representation matrix Z , its sample Gram $Z^\top Z$, and orthogonal changes $Z' = OZ$. Under two tight-scattering/convex-hull conditions, nonnegativity of Z' forces O to be a permutation. With $Z = A^\top$, $Z' = (A')^\top$, and $O = M^\top$, the Gram step of our proof is exactly in that form. Their scattering hypotheses are additional geometry, however, not

consequences of a strictly positive full-rank router. In fact, their first condition requires the convex hull of the representation columns to contain an ellipsoid that touches every coordinate hyperplane, whereas the convex hull of finitely many strictly positive router rows stays a positive distance from every such hyperplane. Our local $B^\top C_i^2 B$ response blocks, not nonnegativity or the Gram alone, exclude the residual orthogonal ambiguity. The cited theorem supplies no perturbation inverse, and its authors explicitly state that their sufficient scattering conditions are not proved necessary. We therefore credit the orthogonal-to-permutation mechanism while retaining only the response-specific completion and stability chain as candidate additions.

The subsequent step uses established results on diagonal algebras. Afsari (2008, Theorem 2.3) proves essential uniqueness of a nonorthogonal exact joint diagonalizer precisely when no two columns of the matrix of diagonal entries are collinear. Once our proof has obtained $M^\top \text{Diag}(a_i)M = \text{Diag}(a'_i)$ for every layer i , their diagonal-entry matrix is A ; full column rank is stronger than their pairwise noncollinearity condition, so their result yields a monomial M before the row-sum constraint $M\mathbf{1} = \mathbf{1}$ reduces it to a permutation. Their Section 4 and Theorems 4.1–4.2 further give first-order, cost-function-specific perturbation formulas for stationary joint diagonalizers, with sensitivity controlled by diagonalizer conditioning and a modulus of uniqueness. This directly precedes approximate diagonalization in our Steps 3–4. It does not, however, recover the router Gram and covariance families from a response operator, nor give our global compact-set bound, conditioned on a common exact effective product, from response-operator distance to the simultaneous (A, B) permutation orbit. Our quantitative contribution is the complete reduction for this model, building on joint-diagonalization uniqueness and sensitivity.

The same final step also has a global robust tensor precedent. Stack the diagonal family into $\mathcal{T} = \sum_{k=1}^K e_k \otimes e_k \otimes A'_{:,k}$. The approximate congruences in our proof compare its decomposition $[I, I, A']$ with $[M^\top, M^\top, A]$. Under our singular-value and norm bounds, the three reference factors have robust Kruskal ranks K, K, K ; for $K \geq 2$, their sum satisfies $3K \geq 2K + 2$. Consequently, Bhaskara et al. (2014, Theorem 5) already supplies global stability of these bounded tensor factors up to one permutation and diagonal rescalings. With its conditioning parameters fixed, Theorem 5 is linear in the discrepancy: its required tensor accuracy is the requested factor accuracy divided by a fixed polynomial in those parameters. The linear exponent is therefore not an addition of this paper. This is stronger than viewing Afsari’s result as the only robust precedent for Steps 3–4. It still begins from two close tensor decompositions: it does not derive that tensor comparison from the complete response operator, enforce the common exact effective product, or give the tailored explicit linear end-to-end constant for the simultaneous (A, B) orbit. We therefore do not claim the diagonal-tensor robust permutation step, its global character, or its linear error dependence as new.

The optimizer metric is essential. van Oostrum et al. (2023, Thms. 1–2) show that the pushforward of a natural parameter gradient is a tangent-space projection and is invariant under diffeomorphic reparameterization. Along a feasible full-rank factor gauge $A' = AM$, $B' = M^{-1}B$, the two charts span the same product tangent space because

$$\{XB + AY : X\mathbf{1} = 0\} = \{X'B' + A'Y' : X'\mathbf{1} = 0\}, \quad X' = XM, \quad Y' = M^{-1}Y.$$

Thus this response does not reduce $GL(K)$ to permutations. Riemannian quotient optimization explicitly treats (G, H) and $(GM^{-1}, HM^\top)$ as one point under invariant metrics (Mishra et al., 2014). These are exact counter-boundaries to any metric-independent reading of our theorem, which concerns only the stated Euclidean Z/B flow with common known positive rates and temperature.

Two recent softmax-attention analyses provide additional, architecture-level boundaries. Wang & Wang (2026a) characterize the generic function-preserving group of standard multi-head attention as independent query–key and value–output GL actions, together with head permutations; Wang & Wang (2026b) study quotient/Fisher geometry and the horizontal natural gradient, assuming no Fisher degeneracy beyond gauge directions. Their factors are Q/K and V/output projection matrices, not a row-simplex shared router and basis bank, and their observable is the Transformer function or quotient geometry rather than equality of two Euclidean effective-gradient response operators. Conversely, Goel et al. (2026) show that a Gaussian softmax self-attention population loss reduces to a weighted matrix-factorization loss and analyze a preconditioned, regularized optimizer. In that model the Gaussian expectation cancels the softmax normalization before the factor loss is formed; the paper does not classify gauges from complete Jacobian–Gram response. These sources therefore strengthen the prior-art boundary without implying Theorem 1.

Finite operator observations have also been studied. In Novak et al. (2022, Sec. 3.3, Eqs. (7)–(9)), applying NTK–vector products to the O identity columns reconstructs an $O \times O$ finite NTK. The variable map here is $O \mapsto LD$, their NTK to the learning-rate-weighted response Jacobian Gram, and their identity column to $\text{vec}(G)$. More general probing writes an operator in a known linear family and recovers its coefficients from matrix–vector products (Chiu & Demanet, 2012; Halikias & Townsend, 2024; Otto, 2023); the formal COLT result of Amsel et al. (2026) gives broader approximate query bounds, including linear matrix families, with access to both operator directions. Our K -probe construction is only a response-specific use of the known support and shared-block form that compresses the generic LD identity probes under $D - K \geq L$. It is not claimed as a new matrix-probing principle. We also checked the primary full text of Stupariu & Manolache (2026): its Adam/Muon experiments analyze performance, loss geometry, and ranks, but contain no response-stabilizer, inverse, or tomography theorem; workshop metadata alone is not used to infer priority.

Building on these results, we establish the following combination in Theorem 1 and the explicit nondegenerate quantitative inverse in Corollary 1: for a full-rank softmax router and full-row-rank shared bases evaluated under the specified Euclidean Z/B response law, two charts on one common exact product fiber have equal complete instantaneous response for every effective gradient only when they differ by a basis permutation, while response closeness controls distance to that orbit only on the stated uniformly interior, well-conditioned, norm-bounded set. We did not find this combined stabilizer or its model-specific inverse in the audited full texts; generic inverse-function and perturbation tools are not claimed as new. Corollary 3 further combines the response inverse with a uniform bound outside a noisy product’s estimated row space, giving a whole-feasible-set diameter bound from K designed queries under the stated domain, dimension, and product-noise conditions. These results identify the current chart relative to a named optimizer metric; they do not identify an AdamW trajectory, a functional-equivalence partition, or a historical graph.

Our ASLoRA experiment examines the first nearest-pair merge decision. A common $GL(r)$ change of basis preserves every adapted layer while it can change the raw running- B ranking. We distinguish that historical selection proxy from the current directed weight perturbation, layer-specific activation-weighted disturbance, and full-validation consequence of the candidate tie. This experiment connects the established gauge dependence of factors to ASLoRA’s specific selection rule and the consequences of its chosen action.

I PRETRAINED LANGUAGE MODELS: PROTOCOL AND COMPLETE RESULTS

I.1 WHY THE ADDITIVE MODULE IS COVERED BY THE OBSERVATION MODEL

Proposition 17 (Fixed isometric embedding). *Let E be a known linear isometry from the product space $\mathbb{R}^{L \times D}$ to a space of layer-weight perturbations, with $E^*E = I$, and let $W = W_0 + E(AB)$ with fixed known W_0 . For Euclidean updates of Z, B with the rates and temperature in Section 4.1, the effective weight response to a full-space gradient H is*

$$-\dot{W} = E \mathcal{R}_{A,B}(E^*H).$$

Equality of complete full-space response operators is equivalent to equality of the factor response operators. Product and response distances for embedded observations are unchanged. Consequently the identification and stability results apply to this module under their original rank, interior, rate, dimension and noise assumptions.

Proof. The derivative of the affine embedding is E , so the chain rule pulls H back to $G = E^*H$. The specified factor updates therefore produce $-\dot{\Theta} = \mathcal{R}_{A,B}(G)$, and differentiation of the embedding gives the displayed equation. Conversely, any G is realized by $H = EG$, because $E^*EG = G$. Applying E^* to the embedded response recovers $\mathcal{R}_{A,B}(G)$, proving both directions of the operator equivalence. The product coordinates are $E^*(W - W_0) = AB$. Finally, $\|EX\|_F = \|X\|_F$ preserves all stated product/query/response norms. The existing identification and stability results therefore apply with their original assumptions. Efficient solution of the inverse problem remains a separate question. $\square$

Our E inserts 64 evenly spaced output rows into each attention output projection, leaving the other rows and the pretrained backbone fixed. All input channels are retained. Thus $D = 64d_{\text{in}}$, which need not equal 64 times the model’s hidden width when attention head dimensions differ. The table reports actual projection dimensions. Four FP32 bases and FP32 softmax logits are shared over all L matching layers. The fixed-row insertion is an isometry in exact arithmetic. Mixed-precision backbone execution supplies measured directions; FP64 factor-map auditing isolates the mathematical response from rounding in the backbone. The proposition does not say that natural task gradients realize every possible query, nor that nonlinear network-output responses are isometries.

I.2 FROZEN DESIGN, DATA AND ADAPTATION

Model choice and access. The nine exact Hugging Face model IDs and revision commits are stored in `research/LLM.ACCESS_INVENTORY.json`; downloaded snapshot file hashes are separately recorded in `LLM.MODEL_BYTES.json`. Pythia and SmolLM2 use the named base models. Qwen3 uses the named dense non-Base releases, so architecture, pretraining and post-training are confounded across families. Initial unauthenticated requests for Llama3.2 1B and 3B returned HTTP 401; both are recorded as unavailable, with the public SmolLM2 fallback chosen before pilot outcomes. There are no Llama results and no coverage of MoE, multimodal models, or models above 8B.

Development and freeze. Three representative corrected pilots use Pythia160M, Pythia1B and Qwen3-0.6B, seed90, 20 updates. Earlier development records retain a failed keyword call and an incorrect WikiText article separator; these are excluded from the confirmatory evidence. The separator was corrected to top-level article headings before the final pilots and all main runs. The final protocol, 17 source/config/data files and raw corpus hashes were frozen before calibration and evaluation. The four calibration units are Pythia160M and Pythia1B, each seeds101/102. Evaluation uses seeds0/1/2 for every model; no checkpoint or seed is selected by validation performance. Policy thresholds are fixed before evaluation recovery scoring. Evaluation gradient collection can run concurrently with calibration because it does not inspect recovery quality. The full chronology is retained in `research/LLM.DEVELOPMENT_LOG.md`.

Data and disjoint roles. WikiText-2 train supplies adaptation; validation supplies general-text calibration, and test supplies general-text evaluation. GSM8K’s public training JSONL and MBPP’s public JSONL supply held-out math and code text. Dataset repository commits and raw SHA-256 hashes are in `LLM.CORPUS_LOCK.json`. This use of public text is not an official benchmark-accuracy evaluation. We format math as question plus reference answer, and code as comment prompt plus reference program, scoring the entire sequence. We do not execute programs or score generated answers. Public pretraining overlap is unknown.

Within each held-out corpus, alternating documents are assigned to probe and loss roles before tokenization. Each role is EOS-concatenated and divided into nonoverlapping 129-token blocks (128 prediction targets), retaining at most 256 blocks; source document IDs and token hashes are stored. Training draws from up to 4,096 train blocks. Twelve distinct probe blocks are sampled per checkpoint/domain: the first eight are fitting candidates and the last four are reserved for response prediction. Loss evaluation uses the first 32 loss-role blocks, in 16 batches of two. The same documents can contribute multiple probe blocks, so query cases are not independent statistical replicates. Corpus partition names for code/math describe study roles, not official benchmark test splits.

Adaptation budget. Each run trains only the $4D + 4L$ adapter parameters for 256 updates, batch size 4 and sequence length 128, or 131,072 prediction tokens. AdamW uses basis learning rate 10^{-3} and router rate 10^{-2} , zero weight decay, gradient norm clipping at 1, 16 linear warmup steps, then cosine decay to 0.1 of the initial rate. Initial logits are standard normal, and initial bases are nonzero Gaussian matrices scaled by $0.01/\sqrt{d_{\text{in}}}$. The last checkpoint is used. This fixed budget is not a convergence certificate or an optimization comparison. Mean router ℓ_1 movement is 0.250–0.850 across checkpoints; minimum router entries are 6.86×10^{-4} –0.0114, minimum router singular values 0.415–2.39, and minimum basis singular values 0.624–6.52. These are measured properties, not acceptance-policy inputs derived from hidden truth. Table 6 includes adverse domain shifts. The noise grid is not restricted to the global theorem’s domain-dependent product-noise threshold, and

Table 6: Adapter-induced change in held-out token NLL (adapted minus frozen backbone), averaged over three adapter seeds. Negative is lower. Domains use 32 fixed blocks per checkpoint. Tokenizers differ, so changes are within-model comparisons; these are not answer-accuracy or code-execution scores.

Model	General	Code	Math
Pythia 160M	-0.0481	+0.0303	+0.1146
Pythia 1B	-0.4439	+0.0485	+0.0812
Pythia 2.8B	-0.4220	+0.0301	+0.0256
Qwen3 0.6B	-0.7086	-0.0392	+0.0036
Qwen3 1.7B	-0.5666	-0.1779	-0.2103
Qwen3 4B	-0.6006	-0.1514	-0.2646
Qwen3 8B	-0.4193	-0.1729	-0.2005
SmolLM2 360M	-0.3016	+0.0013	+0.0216
SmolLM2 1.7B	-0.3080	+0.0071	+0.0461

the numerical solver does not enforce its entire prescribed domain. Small observed errors therefore do not measure the theorem’s universal feasible-set diameter or certify its constants.

I.3 OBSERVATIONS, ESTIMATORS AND TRANSFER DIAGNOSTICS

Observation grid and arithmetic. Each evaluation checkpoint supplies five families, three query budgets and four noise levels: 60 cases, or 1,620 total. Calibration uses only designed, Gaussian and general-text queries, giving 36 cases per checkpoint and 144 total. Designed queries are the repeated-complement unit construction; $q = 8$ repeats its four-query cycle. Gaussian and natural queries are individually normalized. Designed-query holdouts are independent Gaussian directions; the other families use their last four measured directions. Product and each response receive fixed-relative-Frobenius-norm Gaussian perturbations with common random directions across noise levels. The product row space is recomputed from its noisy observation.

The observation code measures each response by reverse-mode factor differentiation and a forward-mode JVP, rather than by the analytic response formula. Audit temperature and both Euclidean block rates are set to one, independently of the AdamW adaptation learning rates. All formula checks have relative error at most 1.62×10^{-16} . A common factor permutation preserves the response; a fixed non-permutation transform preserves the product to rounding while changing the response by 0.103–0.481 on tested directions. This is a finite witness, not a test of every alternative. Same-forward-graph chain checks pass, but separately repeated BF16 backbone backward passes differ by up to 0.0126 relatively; measured directions are treated as fixed inputs afterward. At finite step 10^{-5} , FP64 response errors are 1.01×10^{-9} – 6.88×10^{-9} , versus the much larger FP32/BF16 errors reported in the main text. These casts are not an experiment on mixed-precision optimizer steps with FP32 master weights.

Recovery and scoring. The three estimators are spectral recovery, simplex-projected spectral recovery with a basis refit, and constrained joint least squares with three starts and at most 80 function evaluations per start. All operate on compressed observations. Compression retains the product component outside the fitted row space and the orthogonal response residual, so an estimated-subspace fit cannot silently discard unexplained energy. Ground truth enters only afterward in the permutation-aligned error

$$e = \min_P \left(\frac{\|\hat{A} - AP\|_F^2}{\|A\|_F^2} + \frac{\|\hat{B} - P^T B\|_F^2}{\|B\|_F^2} \right)^{1/2}.$$

Basis error includes the true basis component outside the estimated subspace. No compression failures occur in the main grid. All 540 $q = 1$ cases fail the sufficient per-layer query rank and skip joint fitting; spectral outputs remain recorded. All 1,080 eligible joint fits return. Raw/projected spectral methods have 580/531 errors above 0.10 over all 1,620 cases; these counts include their insufficient-query outputs. The joint-fit median is conditional on the 1,080 returned estimates.

Table 7: Unchanged calibration policies transferred to 1,620 held-out cases per estimator. Coverage includes all 540 insufficient-rank cases at $q = 1$; joint fitting is not run there. Bad means permutation-aligned factor error > 0.10 . An empty accepted set has undefined bad fraction. Raw spectral estimates fail the strict simplex gate in calibration, yielding an all-reject policy.

Estimator	Score	Accepted	Coverage	Bad	Bad / accepted
Spectral	Residual	0	0.0%	0	—
Spectral	Condition	0	0.0%	0	—
Spectral	Combined	0	0.0%	0	—
Projected	Residual	1037	64.0%	20	1.93%
Projected	Condition	960	59.3%	48	5.00%
Projected	Combined	1056	65.2%	43	4.07%
Joint	Residual	868	53.6%	0	0.00%
Joint	Condition	1080	66.7%	0	0.00%
Joint	Combined	1080	66.7%	0	0.00%

Acceptance and uncertainty. The common gate requires positive simplex entries, row-sum error below 10^{-8} and sufficient query rank. Scores are residual-only, inverse candidate tangent singular value, and a combined residual, conditioning, subspace and competitive-start-disagreement score with a 10^{-10} numerical floor. On calibration, each method/score threshold maximizes accepted count subject to at most 5% bad outputs and at least 20 accepted cases, keeping ties together; no feasible threshold means all reject. Good error is at most 0.05, bad error exceeds 0.10, and intermediate cases stay in the denominators. These are empirical policies. Spectral estimates often have row-sum errors of order 10^{-8} – 10^{-7} despite small factor errors, so their strict gate explains the all-reject policy; simplex projection is an explicitly reported separate estimator.

The released summary also includes descriptive intervals from 2,000 bootstrap resamples of the three adapter checkpoints within each fixed model, preserving all query/noise/domain variants within a checkpoint. They are not population intervals over arbitrary LLMs. In particular, resampling zero observed bad joint fits still gives zero; this supplies no upper confidence bound on rare or out-of-distribution failures. Full model/domain breakdowns and all 4,860 method outcomes are retained in `results/llm/recovery-cases.csv`.

I.4 SECONDARY CONDITIONING STRESS

A separate prospectively recorded secondary sweep transforms the learned Pythia1B, Qwen3-4B and SmoLM2-1.7B factors at seed0. For $t \in \{1, 0.1, 0.01, 0.001\}$ it either contracts routers toward the uniform row (router-rank stress), interpolates toward balanced one-hot rows (interior stress), or scales the smallest basis singular value by t (basis-rank stress). Four designed queries, noise0 or 10^{-4} , and unchanged estimators/policies give 72 cases. These are artificial factor modifications, not additionally trained models or real task gradients at the modified checkpoint.

I.5 MATCHED STRUCTURAL DECISIONS AND REPEAT CONTROLS

The predeclared decision units are Pythia1B, Qwen3-4B and SmoLM2-1.7B, each at seeds0/1/2. Every action uses four nonempty equal-sized groups. The native and recovered router actions maximize balanced assignment scores; the effective-parameter control uses balanced k -means with deterministic farthest-point initialization and 50 iterations; the random control uses a fixed seeded balanced partition. Recovered actions use general-text $q = 4$ joint fits at noise 10^{-4} and 10^{-2} . Every hard group receives the mean of the same canonical FP32 effective tensors, followed by 64 updates of only the four bases with fresh AdamW states, rate 10^{-4} , zero decay, the same batch indices and clipping. All 45 actions complete. Rejected recovery would retain the native action; all 18 recovered decision inputs are accepted by the combined policy.

All recovered and effective-clustering partitions have zero co-membership difference from the native partition. General-text random-minus-native post-recovery NLL is 0.0383–0.1098 across the nine checkpoints; held-out code and math differences have mixed signs. This establishes no advantage over the effective-parameter control. Equivalent partitions on Pythia exhibit small numerical NLL differences under the original BF16 execution. A labeled post hoc diagnostic repeats the *identical*

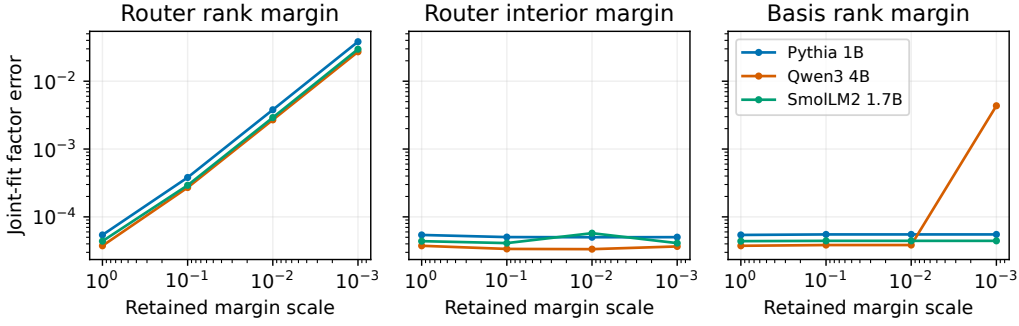


Figure 4: Secondary stress at observation noise 10^{-4} . Shrinking router rank amplifies error from roughly $4\text{--}5 \times 10^{-5}$ to $0.027\text{--}0.038$. The other tested paths do not show comparable monotone growth; the absence of failure on a selected path does not remove the theorem’s nondegeneracy assumptions.

Table 8: Post-recovery NLL minus native-router partition, averaged over three seeds. All actions use four balanced groups, a common effective-tensor refit and 64 matched updates. Both recovered and effective-clustering partitions equal the native partition at all nine checkpoints. Small Pythia discrepancies under equivalent partitions are numerical execution variation, not structural improvements. Per-seed values and immediate losses are retained in the released CSV.

Model	Action	General	Code	Math
Pythia 1B	Effective clustering	-0.0004	-0.0005	-0.0001
Pythia 1B	Random balanced	+0.0629	-0.0196	-0.0291
Pythia 1B	Recovered, 10^{-4}	-0.0004	-0.0005	-0.0002
Pythia 1B	Recovered, 10^{-2}	+0.0001	-0.0007	-0.0003
Qwen3 4B	Effective clustering	+0.0000	+0.0000	+0.0000
Qwen3 4B	Random balanced	+0.0801	+0.0096	+0.0010
Qwen3 4B	Recovered, 10^{-4}	+0.0000	+0.0000	+0.0000
Qwen3 4B	Recovered, 10^{-2}	+0.0000	+0.0000	+0.0000
SmoLLM2 1.7B	Effective clustering	+0.0000	+0.0000	+0.0000
SmoLLM2 1.7B	Random balanced	+0.1050	-0.0041	-0.0238
SmoLLM2 1.7B	Recovered, 10^{-4}	+0.0000	+0.0000	+0.0000
SmoLLM2 1.7B	Recovered, 10^{-2}	+0.0000	+0.0000	+0.0000

native action three times per Pythia1B seed with identical tensors, labels, batches and optimizer settings. The largest within-seed/domain NLL range is 0.00152, confirming execution variation on the scale of these differences. We do not interpret them as structural improvements. There is no measured compression speedup: the pretrained backbone remains stored and executed in full.

I.6 COMPUTE, REPLAY AND ARTIFACT SCOPE

The experiments used eight RTX 5090 GPUs with about 32 GiB each. Model training ran in a dedicated environment with Transformers 4.57.6 and PyTorch 2.7.0a0 (NVIDIA build); the previous project environment was retained. The largest allocated GPU footprint was 19.08 GiB. Per-checkpoint timing includes explicit synchronization for training; total training-stage timing additionally includes loading, evaluation and natural-gradient collection. Recovery is performed on CPU, with up to 12 jobs in parallel. The complete environment, source freeze, corpus lock, checkpoint hashes, observations and per-case solver records are retained.

An isolated PyTorch 2.7.1 CPU environment replays 81 selected cases: every evaluation checkpoint, $q = 4$, designed/noiseless and general-text natural queries at noise 10^{-4} and 10^{-2} . Across 243 estimator outputs, the largest permutation-aligned estimate difference is 1.39×10^{-10} , and all acceptance decisions agree. Three additional full-dimensional natural-query checks compare explicit analytic residuals against the compressed objective for deliberately wrong candidates; absolute discrepancies

Table 9: Per-checkpoint costs: median over three seeds; maximum allocated GPU memory across training and observation. Training excludes loading, evaluation and gradient collection; observation covers the entire noise/query sweep; recovery is single-thread CPU time elapsed for all three estimators. Timings are local measurements under concurrent jobs.

Model	Adapter (M)	Train (s)	Observe (s)	Recover (s)	Peak (GiB)
Pythia 160M	0.197	3.2	14.9	10.4	0.75
Pythia 1B	0.524	5.8	35.6	9.7	2.65
Pythia 2.8B	0.655	11.8	45.3	34.4	6.68
Qwen3 0.6B	0.524	10.4	36.3	25.3	2.82
Qwen3 1.7B	0.524	12.4	36.1	26.7	5.26
Qwen3 4B	1.049	20.0	72.4	43.3	10.95
Qwen3 8B	1.049	34.3	72.4	43.3	19.08
SmolLM2 360M	0.246	9.9	18.9	36.0	1.50
SmolLM2 1.7B	0.524	9.4	35.8	19.6	4.49

are below 3.8×10^{-15} . This is independent arithmetic replay, not independent pretrained-model training or external peer review. The source package contains the manuscript; the research artifact separately contains observations, checkpoints, protocols and commands. Upstream model weights and raw gradient caches are not bundled with the manuscript. Scripts regenerate them from pinned inputs.

J EARLIER SYNTHETIC AND BYTE-LANGUAGE-MODEL ILLUSTRATIONS

These experiments preceded the pretrained-model study in Section 6. They check the response formula and finite reconstruction, and measure the effects of precision and conditioning. The global claims rely on the theory. They use the exact-product design of Corollary 2 or rank-checked natural/Gaussian probes, *not* the repeated-complement design of Corollary 3. Thus they do not empirically evaluate its whole-feasible-set diameter bound. All numerical tables come from validated result records; protocols, negative results, source hashes, and replay limitations are retained in Appendices K–S.

Independent full-dimensional response measurements. Three soft-router byte-LM checkpoints have $L = 12$, $K = 4$, and $D = 787,712$. We measure responses by reverse-mode differentiation of a linear effective loss followed by a forward-mode JVP of the packed factor-to-product map. The observation code does not call Equation 7. For each native, permutation, and fixed non-permutation chart, four designed probes reconstruct the response and two unseen Gaussian probes test its predictions. Table 10 shows numerical agreement with the formula, successful held-out prediction, and separation of the fixed alternative. The experiment checks this fixed alternative; separation of every feasible alternative follows from the theorem under its assumptions.

Table 10: Full-dimensional autodiff response audit. Errors are relative Frobenius norms. Formula and held-out columns take the maximum over native, permutation, and non-permutation charts; each chart uses four designed and two unseen probes.

Seed	Formula error	Held-out error	Permutation gap	Non-perm. gap
0	3.52e-16	4.68e-14	9.73e-17	0.9734
1	3.37e-16	5.07e-14	1.05e-16	0.9731
2	3.71e-16	4.12e-14	1.11e-16	0.9735

Natural directions and finite precision. Actual byte-LM cross-entropy gradients with respect to independent effective layer weights supply 16 directions per checkpoint. Controlled autodiff responses are measured at a fixed checkpoint; the first $q \in \{1, 4, 8, 12\}$ directions fit the structural estimator and the last four test prediction. With the exact product row space, four directions meet the sufficient reconstruction ranks at all six checkpoints in Table 11. Factor errors use permutation alignment only for evaluation. One direction has insufficient local rank and gives errors 0.258–1.34 at the three longer-trained checkpoints.

Table 11: Four natural task-gradient probes with known response structure. Original and new checkpoints use the separately registered 1,500- and 6,000-step configurations. Recovery error is the combined relative factor error after evaluation-only permutation alignment. These are FP64 fixed-checkpoint measurements, not observed training trajectories.

Configuration	Seed	Max. held-out response error	Factor error
Original	0	9.57×10^{-14}	3.02×10^{-8}
Original	1	8.22×10^{-14}	5.37×10^{-9}
Original	2	8.63×10^{-14}	1.36×10^{-8}
New	10	9.39×10^{-14}	2.31×10^{-8}
New	11	6.45×10^{-14}	1.35×10^{-8}
New	12	9.95×10^{-14}	1.48×10^{-8}

Finite Euclidean steps reveal an arithmetic limit. At step size 10^{-5} , the longer-trained checkpoints have FP64 response errors 3.3×10^{-8} – 5.2×10^{-8} , but FP32 subtraction errors 0.0208–0.0260 and BF16 errors about one. These tests cast factors and arithmetic to the stated precision; they do not test mixed-precision training with FP32 master weights. The step-size sweep is in Appendix R.

Noisy recovery and diagnostic failures. A separate frozen development/calibration/evaluation protocol compares spectral recovery, simplex projection with a basis refit, and three-start constrained joint least squares. Both the product and measured responses receive relative noise up to 10^{-2} ; the estimated product subspace is recomputed. On 1,920 held-out synthetic variants across 40 seeds, joint fitting reduces the median factor error from 3.34×10^{-3} for projected spectral recovery to 1.01×10^{-4} , and errors above 0.10 from 566 to 227. All 48 variants at four fresh byte-LM checkpoints have joint-fit errors below 0.0107.

The largest joint-fit error is 10.54, and a combined diagnostic falsely accepts an error of 0.82. Its accepted-bad fraction is $92/1782 = 5.16\%$, with seed-cluster bootstrap interval $[4.50\%, 5.79\%]$; residual-only rejection yields $68/1757 = 3.87\%$. The combined score therefore shows no advantage here, and its point estimate exceeds the 5% calibration target. The candidate-local checks also do not exclude distant solutions or establish truth membership in the fitted subspace. Appendix S retains all methods, tail errors, failed solver statuses, and independent CPU replay differences. None of these diagnostics is the global certificate above.

Auxiliary structural decisions. Identifying coordinates under an optimizer does not make actions derived from them product invariant. In a documented reimplement of ASLoRA’s first RoBERTa/MRPC merge rule, a fixed grid changes the primary action in 11/30 seed-gauge instances. Only the selected layer indices are carried back to a common canonical model; validation-loss differences have both signs. A separate controlled byte-LM partition audit finds changes at all three longer-trained checkpoints, but matched recovery training greatly reduces the performance gaps. These are bounded decision witnesses, not applications of the softmax theorem to LoRA, universal harm claims, or full-training regret evaluations. Detailed results, failed signed-local searches, and the earlier initialization-dominated experiment remain in Appendices L and R.

K DETAILED AUDIT PROTOCOLS AND HISTORICAL SETTINGS

This appendix preserves the protocols for the original response and decision audits. The current numerical overview is in Section 6; the later controlled partition experiments and noisy-recovery studies have separate protocols in Appendices R and S. They use different training configurations and must not be pooled as one replication rate.

K.1 FROZEN-CHECKPOINT RESPONSE AUDIT

We first measure the observation used by the identification theorem: the instantaneous Euclidean response of the *current* softmax-router chart. At each of three independently trained soft-router byte-LM checkpoints, we verify the theorem’s assumptions and exact gauge construction. On a predeclared 4×32 extraction we check Equation 7 against autograd and finite differences; on the

full $12 \times 787,712$ factor model we evaluate eight seeded Gaussian directions plus a permutation control. Separately, the $K = 4$ interventions of Corollary 2 reconstruct the structured response. Their full-dimensional outputs are evaluated analytically, with the extraction providing the implementation check. They are not four full-model JVP calls. The Gaussian test is fixed-alternative; tomography assumes exact common product, rank/dimension conditions, and this response family. Neither is an AdamW-trajectory or historical-graph claim.

An additive full-dimensional audit replaces the analytic observations with reverse-mode factor gradients followed by forward-mode JVPs of the packed effective-parameter map. It retains the same three checkpoints and fixed gauge, uses two unseen response directions per chart to test reconstruction, and records a five-level fixed-norm observation-noise sensitivity grid. Its prospective local protocol, source bindings, and numerical gates are released separately from the original audit; no retraining is involved.

K.2 ASLoRA FIRST-MERGE DECISION AUDIT

This experiment uses a documented reimplement of the accessible ASLoRA-v2 first-merge rule (Hu et al., 2024). We train rank-eight, $\alpha = 16$ adapters on RoBERTa-base/MRPC to the first scheduled merge opportunity, optimizer step 560. Query and value select separate pairs; a joint pair is sensitivity analysis. Because the source leaves adjacent-versus-all-pairs candidates ambiguous, we report both. A frozen bank contains identity, orthogonal, diagonal, and dense gauges. Each float64 gauge selects only a layer-index action by running- B distance; we then apply the same temporary directed tie in the unchanged canonical model and evaluate the full validation split.

The selection proxy and the executed action are different estimands. Writing $c = \alpha/r$, the common change of rank coordinates $B_{i,Q} = B_i Q$, $A_Q = Q^{-1} A$ leaves $c B_{i,Q} A_Q = c B_i A$ fixed, but scores a candidate $p = (\ell \leftarrow u)$ by the order-equivalent squared score

$$s_p(Q) = \|(\bar{B}_\ell - \bar{B}_u)Q\|_F^2. \quad (147)$$

Executing that index action instead replaces the *current* B_ℓ^t by B_u^t ; its immediate weight perturbation is $c(B_u^t - B_\ell^t)A$, and its lower-layer local RMS disturbance is $\{\mathbb{E}\|c(B_u^t - B_\ell^t)Ax_\ell\|_2^2\}^{1/2}$. Without assumptions linking historical averages, current factors, and the lower-layer input moment, the running score is not guaranteed to estimate action cost.

Three comparisons separate coordinate dependence from decision quality: current effective-update Frobenius distance is gauge invariant and uses only the prevalidation snapshot; current local-activation RMS is gauge invariant but post-hoc and label-free; validation loss is a labeled oracle never used for selection. We exhaust query-only, value-only, joint-same-pair, and selected composite actions, not their full Cartesian product. Product invariance, restoration, and canonical-action checks gate the run; passing live-float32 re-execution remains a numerical diagnostic, not the causal estimand.

K.3 HISTORICAL BYTE-LM SAME-BUDGET PARTITION-FOLD SEARCH

The original three-checkpoint experiment asked whether a controlled downstream consequence found at one checkpoint replicated. At each of the same three byte-LM checkpoints, a predeclared finite router-only search seeks a positive-stochastic gauge that changes the argmax partition at the same four-group budget. Both partitions are refit from the same effective tensors and compared on paired held-out batches; literal factor hardening is only a confounded diagnostic. Fixed-budget failures remain negative results, not proofs of non-existence, so one successful checkpoint establishes possibility rather than prevalence. All three routers meet the protocol’s initialization-dominated flag. Signed-local and same-sorted-group-size searches were unexecuted post-run amendments to this original protocol. They were subsequently executed at three fresh, more extensively trained checkpoints under the separate protocol in Appendix R.

L DETAILED NUMERICAL RESULTS AND AUXILIARY DECISIONS

The audit and extension tables are generated from validated summary JSON artifacts, including negative seeds and failed searches; exact paths and hashes are recorded in Appendix M.

L.1 A NON-PERMUTATION CHART IS VISIBLE IN OPTIMIZER RESPONSE

Table 12: Frozen-checkpoint response audit. The effective-product column checks the fixed non-permutation gauge; $\Delta\mathcal{R}$ compares Euclidean instantaneous response operators on fixed Gaussian effective-gradient probes. The permutation column is the label-swap control.

Seed	Effective product rel. error	$\min A$	$\sigma_{\min}(A)$	$\sigma_{\min}(B)$	Perm. max rel. $\Delta\mathcal{R}$	Nonperm. median rel. $\Delta\mathcal{R}$	Detected probes
0	1.58×10^{-16}	2.88×10^{-4}	1.7291	27.445	8.82×10^{-17}	0.5045	8/8
1	1.70×10^{-16}	2.72×10^{-4}	1.7291	27.339	8.82×10^{-17}	0.5051	8/8
2	1.63×10^{-16}	2.61×10^{-4}	1.7290	27.371	8.82×10^{-17}	0.5058	8/8

Table 12 checks the response formula and separates a fixed alternative at frozen checkpoints under Theorem 1’s assumptions. All stated rank conditions hold. The non-permutation reparameterization preserves the effective model to at most 1.70×10^{-16} relative error. On the declared 4×32 extraction, the analytic response agrees with autograd within numerical precision, and the worst finite-difference error across native and transformed charts is 9.57×10^{-13} . The maximum permutation-control discrepancy is 8.82×10^{-17} after rounding, whereas all 24/24 checkpoint–probe evaluations detect a change (eight per checkpoint, with overlapping seed windows and 10 unique Gaussian draws; per-seed medians $[0.5045, 0.5058]$). The separate structured audit reduces the all-gradient quantifier to 4 designed analytic response evaluations, which reconstruct all response components with worst relative error 6.85×10^{-11} ; the permutation control stays below 9.44×10^{-17} ; the minimum across seeds of each seed’s maximum fixed non-permutation difference over the four designed probes is 0.9731 after rounding. This does not say that every individual probe reaches that value. These are analytic evaluations of Equation 7, validated on the extraction, not four full-model JVP calls. The eight Gaussian probes remain a fixed-alternative test; the designed certificate assumes exact common product and the stated response family. Neither audit describes AdamW or a historical graph.

Independent full-dimensional response measurements. A subsequent audit obtains the responses by differentiating the complete packed map $(Z, B) \mapsto \text{softmax}(Z/\tau)B$ at all three released checkpoints. Reverse-mode differentiation of a linear probe loss gives the factor gradients; a forward-mode JVP then measures the induced effective-parameter response. The observation code does not call the analytic response formula. Each native, permutation, and fixed non-permutation chart uses four designed probes and two unseen Gaussian probes. Table 10 reports formula agreement and prediction of the unseen responses from the reconstructed operator. The maximum relative formula discrepancy is 3.71×10^{-16} , and the maximum held-out response error is 5.08×10^{-14} . A five-level observation-noise grid from zero to 10^{-4} relative Frobenius norm satisfies the deterministic component error bounds, with recorded floating-point slack. These measurements concern the full shared-parameter map, not an end-to-end language-model loss or an AdamW step; operator reconstruction does not itself implement noisy factor recovery. The additive records preserve the original analytic audit.

Exploratory reconstruction of the factors. The construction in Appendix D.4 recovers factors from the effective product and measured components, without supplying the original factors to the reconstruction. We test it on the native charts of the same three checkpoints, using fresh autodiff observations and one fixed Gaussian weight vector for joint diagonalization. Table 13 reports the errors after a common permutation chosen only for evaluation. The noiseless combined relative errors are below 3.7×10^{-9} . All five noise levels complete without a rank or eigengap failure, but the unprojected noisy estimates need not have exact unit row sums. These are three checkpoint diagnostics with exact effective products, not independent training replications, adversarial-noise tests, or a general stability result for the PSD-clipped recovery algorithm.

Table 13: Exploratory factor recovery from full-dimensional autodiff observations. Orbit error combines relative router and basis errors after one common permutation. Noisy rows use relative observation noise 10^{-4} and no simplex projection.

Seed	Noiseless orbit error	Noisy orbit error	Noisy row-sum error
0	3.66e-09	1.81e-05	6.06e-05
1	3.23e-09	1.61e-05	6.68e-05
2	3.28e-09	1.59e-05	5.99e-05

Table 15: Prospective equal-size folding audit. Signed BPB differences are gauge-selected minus native-selected folds after matched recovery updates. Each row is one independent training seed. Positive-stochastic search succeeds in 3/3 seeds; signed-local search fails in 3/3, each with 5,000 router-only candidates.

Seed	Router movement	Δ_0	Δ_{100}	Δ_{500}
10	0.593	-0.00112	+0.00224	+0.00009
11	0.720	+0.87112	+0.03574	+0.01823
12	0.729	+0.14873	+0.00580	+0.00343

L.2 GAUGE-EQUIVALENT ASLoRA STATES CHANGE THE FIRST MERGE DECISION

Table 14: ASLoRA first-merge decision audit over 10 gauges per seed. Changes are relative to the identity-gauge raw-running- B action. The loss interval is the canonical validation difference (selected action minus identity action), not a gauged float32 re-execution; accuracy and F1 intervals are in Appendix Table 18. The chain column requires a changed action with a nonzero validation consequence in every seed.

Candidates	Scope	By seed	Changed	Seeds	Δ loss	Chain
Adjacent	per-proj.	(2, 6, 3)	11/30	3/3	$[-0.0531, +0.0089]$	yes
Adjacent	joint pair	(0, 5, 0)	5/30	1/3	$[+0.0000, +0.0077]$	no
All pairs	per-proj.	(4, 7, 3)	14/30	3/3	$[-0.0531, +0.0027]$	yes
All pairs	joint pair	(0, 0, 0)	0/30	0/3	$[+0.0000, +0.0000]$	no

The primary per-projection action changes under both interpretations of the source’s candidate set. Among 30 predeclared gauge instances, the raw-running- B action changes in 11 instances for adjacent candidates and 14 instances for all pairs. Every seed has at least one changed action with a nonzero canonical validation consequence under both candidate readings. The signs in Table 14 go in both directions: the experiment demonstrates representation-dependent decisions, not that every gauge-induced change is harmful or that the identity action is optimal. The joint-same-pair sensitivity does not replicate (one seed for adjacent candidates and none for all pairs), and Appendix Table 19 shows that neither gauge-invariant rule uniformly improves on raw running B across all seeds under both candidate readings. The causal contrast uses a float64-equivalent snapshot only to select layer indices, then executes the same temporary tie in the unchanged canonical model; factorized-float32 re-execution is a passing secondary diagnostic. The experiment links the change in coordinates to action selection and validation outcomes for this first merge, task, and fixed gauge grid.

L.3 PROSPECTIVE CONTROLS: LEARNED ROUTERS AND RECOVERY TRAINING

The historical example in Appendix R.1 motivated a separate extension whose protocol was recorded before execution. We train three fresh width-128, twelve-layer, four-basis byte LMs for a fixed 6,000 steps with random router initialization and a separate router learning rate. Router L1 movement is 0.593–0.729; all three models pass the unigram-improvement and router-movement gates. Both gauge families receive 5,000 router-only trials, now requiring the same sorted group-size multiset as the native partition. Positive-stochastic search finds a first valid candidate in every seed; signed-local search fails in all three. Failures are retained and never replaced by larger searches.

Every fold uses the same native effective tensors. Hard routes remain fixed while all remaining parameters receive identical minibatches, optimizer initialization, update counts, and learning rates. Table 15 shows substantial immediate differences in two seeds, but much smaller differences after 500 recovery updates. Conditional paired 95% bootstrap intervals at that stage are $[-0.00094, 0.00112]$, $[0.01654, 0.01989]$, and $[0.00244, 0.00437]$ BPB, respectively; they describe the fixed evaluation blocks, not uncertainty over training populations. The same-size effective-parameter local-search baseline selects the native partition in two seeds and performs slightly worse after recovery in the third. The decision remains sensitive to coordinates after controlling for the earlier initialization and group-size differences. The results do not establish persistent large losses or a consistently

Table 16: Frozen held-out noisy recovery. Each synthetic instance contributes 48 correlated variants; each language checkpoint contributes 12. Acceptance uses the calibrated empirical score. Bad means factor error > 0.10 ; good means ≤ 0.05 . Local passes concern only a numerical restricted-subspace component.

Domain	Method	Median error	Bad/total	Accept	Bad/accept	Local
Synthetic	Spectral	0.0041	564/1920	196	0/196	0
Synthetic	Projected	0.0033	566/1920	1396	56/1396	0
Synthetic	Joint fit	0.0001	227/1920	1782	92/1782	140
Byte-LM	Spectral	0.00017	1/48	0	—	0
Byte-LM	Projected	0.0002	1/48	48	1/48	0
Byte-LM	Joint fit	3.3e-05	0/48	48	0/48	2

better invariant rule. This is one architecture and corpus, with a fixed training budget rather than a convergence certificate.

L.4 NATURAL TASK GRADIENTS AND FINITE ARITHMETIC

We differentiate actual byte-LM cross-entropy with respect to independent layer-effective weights, holding other parameters fixed. A separate forward and chain-rule check agrees with the original model. At each of the three original and three new checkpoints, 16 fixed training minibatches supply natural directions. Controlled reverse/forward autodiff measures the response; the first $q \in \{1, 4, 8, 12\}$ directions fit the estimator, and the last four test prediction. The estimator uses the exact product row space and known response structure, not the true factors.

At $q = 4$, all six checkpoints meet the rank conditions of Proposition 15. The largest held-out response error is below 10^{-13} and the factor error below 3.1×10^{-8} . This is stronger than separation of a single alternative chart, but remains conditional on the known Euclidean response family and precise measurements. A single direction has insufficient local rank; at the new checkpoints its factor errors range from 0.258 to 1.34. Generic linear prediction from the observed gradient span also differs from the structural estimator. These directions are measurements at fixed checkpoints, not observations of an uncontrolled training trajectory.

Explicit finite Euclidean steps expose a numerical boundary. At step size 10^{-5} , the new checkpoints' FP64 response errors are 3.3×10^{-8} – 5.2×10^{-8} , whereas FP32 subtraction errors are 0.0208–0.0260 and BF16 errors are about one. Here factors and arithmetic are cast to the stated precision; this is not a test of BF16 training with FP32 master weights. Appendix Figure 9 reports the step-size tradeoff and the separate joint-noise conditioning diagnostic.

L.5 NOISY RECOVERY AND THE LIMITS OF OBSERVABLE REJECTION

We fix a prospective development/calibration/evaluation split before measuring the final outcomes. Forty held-out synthetic seeds each contribute 48 correlated noise/probe/regime variants; four freshly trained byte-LM checkpoints each contribute 12. Both product and autodiff response observations receive relative noise from zero to 10^{-2} , and the rank- K product row subspace is re-estimated from the noisy product. Three fixed methods use the same observations: spectral recovery, simplex projection with a basis refit, and three-start constrained joint least squares. The estimator never receives the true factors. Appendix S gives splits, budgets, calibration, and all diagnostic comparators.

Joint fitting reduces the synthetic median factor error from 3.34×10^{-3} for projected spectral recovery to 1.01×10^{-4} , and the number of errors above 0.10 from 566 to 227 out of 1920. At the four byte-LM checkpoints, all 48 joint-fit errors are below 0.0107, including the 1% observation-noise cases. These are controlled Euclidean measurements at checkpoints, not noisy AdamW trajectory observations.

We also test whether a diagnostic based on observations can identify unreliable estimates. With its threshold fixed on separate calibration data, the combined residual/conditioning/subspace/multi-start score accepts 1782/1920 synthetic outputs, including 92 bad outputs. Its 5.16% accepted-bad fraction has a seed-cluster bootstrap interval of [4.50%, 5.79%] and has a point estimate above the 5% calibration target; the interval includes that target. Residual-only rejection accepts 1757/1920

with $68/1757 = 3.87\%$ bad. The combined score therefore shows no advantage over residual-only rejection. The full evaluation retains a joint-fit error of 10.54, and a falsely accepted error of 0.82. These failures prevent a general reliability claim.

The explicit local separation bound in Proposition 16 addresses a different question. Only 140/1920 synthetic and 2/48 language outputs pass its numerical local-component conditions. These passes neither establish truth membership in the restricted subspace nor rule out distant solutions.

M EXPERIMENTAL DETAILS

M.1 ORIGINAL AUDIT INVENTORY

Table 17 records the original audit units. The three response checkpoints and three byte-LM decision checkpoints are paired by training seed. The ASLoRA runs are a separate RoBERTa-base/MRPC experiment.

All numerical entries in the three original audit tables are generated directly from `results/response_summary.json`, `results/aslora_summary.json`, and `results/router_decisions_summary.json` by `experiments/generate_paper_audit_tables.py`. Each generated file records the SHA256 of its source summary, retaining negative seeds and failed searches in the reported aggregates; the same script emits the appendix metric and search-detail tables. The additive tomography numbers are generated separately from `results/response_tomography_summary.json` by `experiments/generate_response_tomography_latex.py`; this leaves the source-attested older response table unchanged. The subsequent full-dimensional autodiff and factor-recovery tables are generated by their separate summarizers from `results/autograd_tomography/` and `results/factor_recovery/`. Both validate generation-source and checkpoint hashes. They reuse the same three checkpoints and therefore do not increase the number of independent training seeds.

Table 17: Original primary audit inventory. Each audit has three seeds; counts include failed searches and controls. Additive audits reuse the response checkpoints, as detailed below.

Audit	Per-seed protocol	Required checks
Optimizer response	8 Gaussian and 4 designed probes; fixed non-permutation and permutation control	Rank, formula, exact-gauge product, reconstruction
ASLoRA first merge	10 gauges; adjacent/all pairs and per-projection/joint actions	Coverage, canonical action, restoration
Byte-LM partition fold	5,000 trials; four-group, same-effective re-fits	Fixed search and paired evaluation

The response audit uses a fixed gauge strength of 0.35, unit Euclidean learning rates for both parameter blocks, and probe seeds determined before evaluation. Each result records its checkpoint and source hashes. The best finite-difference scale on a predeclared four-row, 32-coordinate checkpoint extraction is selected only to validate the closed-form response; the eight fixed Gaussian directions on the full $12 \times 787,712$ packed model provide the separate non-permutation detection test. The additive four-probe audit constructs the Corollary 2 design from Θ and reconstructs the response components. Its full-dimensional responses are evaluations of the already checked analytic formula, not four additional full-model autograd/JVP calls. Its version-2 artifacts separately record the reference rank/dimension gate, the floating-point product residual, and exact common-product validity from the explicit invertible-gauge construction.

The subsequent autodiff audit instead differentiates the packed factor map. For each checkpoint it measures four designed and two unseen responses in each of three charts: native, permutation, and a fixed non-permutation gauge. Its five-level observation-noise diagnostic uses the native chart. The factor-recovery pilot makes four fresh native-chart measurements, reuses one fixed diagonalization weight seed across checkpoints, and reports all five noise levels without selecting a favorable weight vector. True factors enter the observation oracle and post-reconstruction error calculation, but not the

reconstruction routine. Neither experiment differentiates the language-model task loss or simulates an AdamW trajectory.

The ASLoRA protocol follows the paper’s defined configuration: 30 epochs, effective batch size 16, rank eight, $\alpha = 16$, $T_s = 320$, and merge interval 240. Appendix Table 5 of the source additionally records an update ratio $\lambda = 0.5$, but the accessible method text and Algorithm 1 define no operation involving that quantity. We record λ as `update_ratio_record_only`; the reimplementation assigns it no operation because the source defines none. With the one-indexed rule $t > T_s$ and $(t - T_s) \bmod 240 = 0$, the first audit point is $t = 560$. Running B is observed after every optimizer update. The three artifacts contain 204, 208, and 200 canonical action evaluations, respectively: all 66 query-only, 66 value-only, and 66 joint-same-pair actions plus every additional composite selected by a declared baseline or gauge. There are no missing or undeclared candidates. Every temporary tie is restored, the trainable digest and parameter identities agree before and after the audit, and no merge is committed.

The amendment chronology in `research/ASLORA_EMPIRICAL_PROTOCOL.md` records the superseded nondeterministic 12/30 and 16/30 batch, its source hashes, the outcome-blind deterministic protocol lock, and the canonical replacement by the 11/30 and 14/30 results reported here.

The ASLoRA artifacts contain two numerical paths. The primary path transforms the frozen running factors in float64, chooses a layer-index action, and evaluates that action in the canonical model. All canonical gates pass. The diagnostic path re-executes the live factorized float32 model and also passes its tolerance in all three deterministic runs. No main-paper metric is taken from that diagnostic path. Both candidate interpretations are retained because Algorithm 1 and the prose in the source differ on whether all or only adjacent layer pairs are sorted.

Table 18: Canonical validation accuracy and F1 differences for the four ASLoRA candidate/scope cells in Table 14. Intervals range over all 30 selected gauge actions in each cell, including unchanged selections, and are selected-action minus identity-action differences.

Candidates	Scope	Δ accuracy	Δ F1
Adjacent	per-projection	$[-0.0025, +0.0098]$	$[-0.0022, +0.0057]$
Adjacent	joint pair	$[-0.0049, +0.0000]$	$[-0.0051, +0.0000]$
All pairs	per-projection	$[-0.0025, +0.0098]$	$[-0.0022, +0.0057]$
All pairs	joint pair	$[+0.0000, +0.0000]$	$[+0.0000, +0.0000]$

Table 19: Declared alternatives for the primary per-projection ASLoRA decision. Effective-update and local-activation rules are gauge invariant. Their cells give the selected action and canonical validation-loss difference from the identity raw- B action (negative is lower; exact zero is a tie). Across the six rows, effective has 2 negative/2 positive/2 zero differences and local activation has 2 negative/1 positive/3 zero differences; neither rule uniformly improves across all seeds under both candidate readings. Validation minima are separate query-only and value-only enumerations; they are not a full Cartesian per-projection oracle.

Seed	Candidates	Raw running B	Current effective	Local activation	Single-target minima
0	Adjacent	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 6 \leftarrow 7$
		$V : 0 \leftarrow 1$	$V : 10 \leftarrow 11$ $\Delta L = -0.0336$	$V : 0 \leftarrow 1$ $\Delta L = +0.0000$	$V : 10 \leftarrow 11$
0	All pairs	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 6 \leftarrow 9$
		$V : 0 \leftarrow 1$	$V : 1 \leftarrow 11$ $\Delta L = -0.0075$	$V : 0 \leftarrow 11$ $\Delta L = -0.0008$	$V : 10 \leftarrow 11$
1	Adjacent	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 4 \leftarrow 5$
		$V : 2 \leftarrow 3$	$V : 10 \leftarrow 11$ $\Delta L = +0.0018$	$V : 0 \leftarrow 1$ $\Delta L = +0.0001$	$V : 3 \leftarrow 4$
1	All pairs	$Q : 0 \leftarrow 2$	$Q : 0 \leftarrow 3$	$Q : 0 \leftarrow 3$	$Q : 4 \leftarrow 11$
		$V : 0 \leftarrow 2$	$V : 10 \leftarrow 11$ $\Delta L = +0.0028$	$V : 0 \leftarrow 11$ $\Delta L = -0.0005$	$V : 3 \leftarrow 5$
2	Adjacent	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 8 \leftarrow 9$
		$V : 0 \leftarrow 1$	$V : 0 \leftarrow 1$ $\Delta L = +0.0000$	$V : 0 \leftarrow 1$ $\Delta L = +0.0000$	$V : 8 \leftarrow 9$
2	All pairs	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 0 \leftarrow 1$	$Q : 8 \leftarrow 11$
		$V : 0 \leftarrow 1$	$V : 0 \leftarrow 1$ $\Delta L = +0.0000$	$V : 0 \leftarrow 1$ $\Delta L = +0.0000$	$V : 8 \leftarrow 11$

The byte-LM search makes 5,000 predeclared positive-stochastic trials per checkpoint and preserves a four-group action budget. Failed searches retain their attempted, invertible, condition-passing, simplex-passing, and group-passing counts. The successful contrast refits both partitions from common effective tensors. Literal hardening is excluded from the primary comparison because it changes basis coordinates and the partition jointly. All three routers satisfy the initialization-dominated flag. Only seed 0 reaches decision evaluation: its k-means and Ward effective-tensor baselines both have BPB 2.1753145 and centroid SSE 5.46896×10^{-5} . Seeds 1 and 2 store no folds or baseline evaluation because their searches exit first. Signed-local and same-sorted-group-size searches were not executed in that historical audit; they were post-run, non-primary protocol amendments. In particular, the observed group-size change $3/3/3/3 \rightarrow 5/3/3/1$ remains a balance confound. The reported interval resamples the 256 paired evaluation-batch differences 10,000 times. It describes evaluation variability conditional on this checkpoint and selected contrast; it does not cover training-seed or gauge-search uncertainty.

The protocol defines the initialization-dominated flag as mean row-wise router L_1 movement from initialization strictly below 0.01. The three observed movements round to 0.000313–0.000401, so every checkpoint meets that declared threshold; this flag describes training movement, not recovery of a historical graph.

Table 20: Byte-LM checkpoint and finite-search diagnostics omitted from the compact main table. The funnel is attempted/invertible/condition-pass/simplex-pass/group-pass/partition-change.

Seed	Soft BPB	Mean row router L_1 move	Search funnel A/I/C/S/G/P
0	2.186	3.35×10^{-4}	5000/5000/4586/4586/3762/1
1	2.188	3.13×10^{-4}	5000/5000/4601/4601/3790/0
2	2.168	4.01×10^{-4}	5000/5000/4581/4581/3769/0

Table 21: Geometry and gauge-invariant comparator details for the sole completed byte-LM witness. The k-means and Ward folds coincide here; the unigram baseline is included only as a scale reference.

Seed	$\kappa(Q)$	Group sizes ref. $\rightarrow$ cand.	Disagreement	k-means/Ward BPB	Centroid SSE	Unigram BPB
0	7.93	3/3/3/3 $\rightarrow$ 5/3/3/1	0.121	2.175/2.175	5.47×10^{-5}	4.624

N SECONDARY PLANTED-DATA AND PROBE SANITY CHECKS

The original TYINGBENCH and SHAREPROBE experiments below check the implementation and illustrate the theory in planted settings. They are secondary to the main experiments and do not estimate the prevalence of router misinterpretation in deployed systems.

N.1 TYINGBENCH PROTOCOLS

Exact factorization. We generate K orthogonal basis vectors with controlled condition number, choose one of five depth graphs (cycle, contiguous, palindrome, balanced random, or imbalanced), and expose the resulting effective layer vectors. We jointly fit soft bases and router coefficients from neutral, random, correct-pattern, and deliberately wrong-pattern initializations. Hard straight-through, vertex-regularized, k -means, and oracle estimators provide comparisons. This setting removes sampling error: failure is factorization or optimization ambiguity, not insufficient task data.

Functional distillation. A planted one-hot teacher and a jointly trained soft student share the same architecture and basis budget. We use an eight-layer residual MLP and an eight-layer decoder-only Transformer. The Transformer mixes parameters before applying each block, evaluates one block per layer, and uses an explicit upper-triangular causal mask. Students observe teacher outputs but not the teacher graph. Test MSE or KL measures function fit; ARI measures recovery of the planted graph. These labels are legitimate for the planted teacher, not for the natural-language experiment.

Language-model stability arm. We train a 12-layer, four-basis causal Transformer on WikiText-2 bytes. The task has no privileged sharing graph, so this arm has no recovery label. BPB, router movement, and cross-run graph agreement diagnose path dependence only. It is separate from the original same-budget partition-fold audit.

Table 22: Secondary sanity-sweep sizes. Every listed seed is crossed with the conditions defined by the corresponding executable sweep.

Family	Depth	Bases	Width	Seeds	Budget / grid
Factorization	12	4	128	10	2,000 steps; 5 graphs
Residual MLP distillation	8	4	64	5	1,500 steps; 4 graphs
Causal Transformer distillation	8	4	64	5	1,000 steps; 4 graphs
Functional probes	12	4	64	10	$n \in \{1, 2, 4, 8, 16, 32, 64\}$; 6 noise levels
WikiText-2 stability	12	4	256	3	1,500 steps; context 128

All models use parameter-space mixing: each effective tensor is formed once from the same layer router row, and each block is evaluated once. This avoids the compute mismatch of an output-space mixture that evaluates all K basis blocks. Teacher graphs and datasets are deterministic functions of recorded seeds. The causal Transformer test changes future tokens while asserting that all earlier logits are unchanged.

N.2 SHAREPROBE AS A CONDITIONAL DIAGNOSTIC

Native activations can confound a layer transition with the distribution created by preceding layers. SHAREPROBE draws $H_1, \dots, H_n \sim P_H$ once and evaluates every residual transition r_ℓ on the same inputs. In the revised paired protocol, observations satisfy $Y_{\ell s} = r_\ell(H_s) + \sigma \epsilon_{\ell s}$ with iid standard-Gaussian noise at a fixed absolute scale σ . Its signature and empirical squared distance are

$$\phi_\ell = n^{-1/2}[Y_{\ell 1}; \dots; Y_{\ell n}], \quad \hat{d}_{\ell m}^2 = \|\phi_\ell - \phi_m\|_2^2. \quad (148)$$

Counterfactual and native estimators use the same initial probes, noise array, and optional random projection. The native trajectory advances only with clean residual updates. Thus each method has the stated marginal observation model and their difference uses common random numbers; unlike the superseded implementation, noise is not rescaled by a realized batch standard deviation. For example, layers using $f(h) = h$ and $g(h) = -h$ in the true pattern (f, g, f, g) yield the wrong native-output groups $\{1, 2\} \mid \{3, 4\}$ on inputs $(1, -1, -1, 1)$, whereas the common input $h = 1$ recovers $\{1, 3\} \mid \{2, 4\}$. This is a counterexample to an unqualified native-trajectory interpretation, not evidence that native signatures are generally inferior.

Table 31 shows no consistent empirical advantage for SHAREPROBE. Shared-input versus native mean ARI is 0.6764 versus 0.6751 for the MLP and 0.5006 versus 0.5009 for the Transformer. The paired win/tie/loss counts are 65/1570/45 and 61/1558/61, respectively, and the seed-averaged difference changes sign (six of ten MLP seeds but only three of ten Transformer seeds favor shared inputs). A simple baseline clusters the concatenated visible effective weights and biases of every mixed linear module; it obtains ARI 1.000 in every planted cell. That baseline requires parameter access and parameter distance is *not* functional equivalence in a neural network with parameter symmetries. It nevertheless shows that the probe is unnecessary in this benchmark when effective block tensors are visible.

We therefore retain SHAREPROBE only as a conditional functional diagnostic. Its graph is invariant to factor coordinates and targets equivalence on P_H , but the probe distribution must cover the claimed domain, K must be supplied to the clusterer, and the observed planted-partition gap becomes a certificate only under the population-separation and uniform-tail assumptions of Appendix D.6. The experiment does not estimate those population quantities.

O SECONDARY SANITY RESULTS

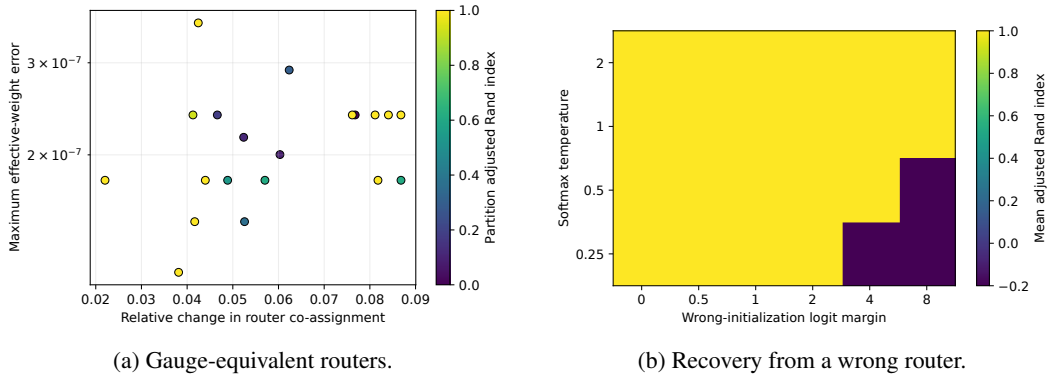


Figure 5: Synthetic theory checks. Effective weights remain equal at numerical precision while router summaries can change; saturation can suppress recovery from a deliberately wrong initialization.

The exact-factorization and distillation sweeps show that, for the tested planted teachers and budgets, similar reconstruction or task loss need not select the same router graph. They are possibility and path-dependence checks, not estimates of real-world frequency. Discrete clustering succeeds when its one-hot model matches the teacher, which is a positive control rather than a general recovery guarantee.

Table 23: Gauge counterfactuals. Values are mean $\pm$ sample standard deviation across trials.

Metric	Value	n
Effective max error	$2.11 \times 10^{-7} \pm 5.41 \times 10^{-8}$	20
Argmax-partition ARI	0.686 ± 0.380	20
Co-assignment relative change	0.059 ± 0.019	20
Post-step effective relative difference	$0.002 \pm 9.90 \times 10^{-4}$	20

Table 24: Softmax-saturation sweep. Values are aggregated across seeds for each temperature and wrong-initialization margin.

Temperature	Margin	Initial grad.	Final MSE	ARI	n
0.250	0	$0.008 \pm 4.18 \times 10^{-5}$	$8.07 \times 10^{-8} \pm 2.96 \times 10^{-10}$	1.000 ± 0	5
0.250	0.500	$0.002 \pm 1.04 \times 10^{-10}$	$7.88 \times 10^{-8} \pm 2.08 \times 10^{-13}$	1.000 ± 0	5
0.250	1.000	$4.62 \times 10^{-5} \pm 1.63 \times 10^{-12}$	$6.00 \times 10^{-8} \pm 1.86 \times 10^{-13}$	1.000 ± 0	5
0.250	2.000	$1.55 \times 10^{-8} \pm 7.94 \times 10^{-16}$	$1.94 \times 10^{-8} \pm 9.27 \times 10^{-14}$	1.000 ± 0	5
0.250	4.000	$9.14 \times 10^{-16} \pm 1.16 \times 10^{-22}$	$0.021 \pm 2.28 \times 10^{-9}$	$-0.222 \pm 3.10 \times 10^{-17}$	5
0.250	8.000	0 ± 0	$0.021 \pm 2.28 \times 10^{-9}$	$-0.222 \pm 3.10 \times 10^{-17}$	5
0.500	0	$0.004 \pm 8.64 \times 10^{-6}$	$2.52 \times 10^{-7} \pm 6.31 \times 10^{-10}$	1.000 ± 0	5
0.500	0.500	$0.004 \pm 2.65 \times 10^{-10}$	$2.68 \times 10^{-7} \pm 4.13 \times 10^{-13}$	1.000 ± 0	5
0.500	1.000	$0.001 \pm 5.21 \times 10^{-11}$	$2.76 \times 10^{-7} \pm 2.96 \times 10^{-13}$	1.000 ± 0	5
0.500	2.000	$2.31 \times 10^{-5} \pm 8.13 \times 10^{-13}$	$2.49 \times 10^{-7} \pm 3.02 \times 10^{-13}$	1.000 ± 0	5
0.500	4.000	$7.74 \times 10^{-9} \pm 3.97 \times 10^{-16}$	$1.44 \times 10^{-7} \pm 9.62 \times 10^{-14}$	1.000 ± 0	5
0.500	8.000	$4.57 \times 10^{-16} \pm 5.80 \times 10^{-23}$	$0.021 \pm 2.28 \times 10^{-9}$	$-0.222 \pm 3.10 \times 10^{-17}$	5
1.000	0	$0.002 \pm 1.73 \times 10^{-6}$	$7.25 \times 10^{-7} \pm 7.43 \times 10^{-10}$	1.000 ± 0	5
1.000	0.500	$0.002 \pm 2.85 \times 10^{-10}$	$7.17 \times 10^{-7} \pm 4.67 \times 10^{-13}$	1.000 ± 0	5
1.000	1.000	$0.002 \pm 1.33 \times 10^{-10}$	$7.58 \times 10^{-7} \pm 5.92 \times 10^{-13}$	1.000 ± 0	5
1.000	2.000	$5.41 \times 10^{-4} \pm 2.60 \times 10^{-11}$	$7.66 \times 10^{-7} \pm 3.52 \times 10^{-13}$	1.000 ± 0	5
1.000	4.000	$1.16 \times 10^{-5} \pm 4.07 \times 10^{-13}$	$6.68 \times 10^{-7} \pm 6.33 \times 10^{-13}$	1.000 ± 0	5
1.000	8.000	$3.87 \times 10^{-9} \pm 1.99 \times 10^{-16}$	$4.91 \times 10^{-7} \pm 1.19 \times 10^{-12}$	1.000 ± 0	5
2.000	0	$9.76 \times 10^{-4} \pm 1.81 \times 10^{-7}$	$2.13 \times 10^{-6} \pm 4.74 \times 10^{-10}$	1.000 ± 0	5
2.000	0.500	$9.90 \times 10^{-4} \pm 8.23 \times 10^{-11}$	$2.10 \times 10^{-6} \pm 1.15 \times 10^{-12}$	1.000 ± 0	5
2.000	1.000	$0.001 \pm 1.43 \times 10^{-10}$	$2.09 \times 10^{-6} \pm 1.19 \times 10^{-12}$	1.000 ± 0	5
2.000	2.000	$9.09 \times 10^{-4} \pm 6.64 \times 10^{-11}$	$2.19 \times 10^{-6} \pm 2.68 \times 10^{-12}$	1.000 ± 0	5
2.000	4.000	$2.71 \times 10^{-4} \pm 1.30 \times 10^{-11}$	$2.17 \times 10^{-6} \pm 3.48 \times 10^{-12}$	1.000 ± 0	5
2.000	8.000	$5.78 \times 10^{-6} \pm 2.03 \times 10^{-13}$	$1.82 \times 10^{-6} \pm 9.90 \times 10^{-13}$	1.000 ± 0	5

Table 25: Exact-factorization results for cycle teachers. Values are mean $\pm$ sample standard deviation across seeds.

Method	Init relation	Raw init	Rel. error	ARI	Assignment acc.	n
k-means	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
oracle	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
soft	matched	cycle	$0.003 \pm 8.36 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
soft	mismatched	contiguous	$0.004 \pm 7.92 \times 10^{-4}$	0.636 ± 0.158	0.725 ± 0.142	10
soft	neutral	neutral	$0.004 \pm 9.08 \times 10^{-4}$	0.894 ± 0.171	0.925 ± 0.121	10
soft	random	random	0.005 ± 0.002	0.894 ± 0.171	0.925 ± 0.121	10
straight-through	matched	cycle	0.004 ± 0.001	1.000 ± 0	1.000 ± 0	10
straight-through	mismatched	contiguous	0.009 ± 0.025	1.000 ± 0	1.000 ± 0	10
straight-through	neutral	neutral	$8.44 \times 10^{-4} \pm 9.05 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
straight-through	random	random	0.006 ± 0.015	1.000 ± 0	1.000 ± 0	10
vertex-hard	matched	cycle	0.005 ± 0.002	1.000 ± 0	1.000 ± 0	10
vertex-hard	mismatched	contiguous	0.854 ± 0.001	$-0.222 \pm 2.93 \times 10^{-17}$	0.333 ± 0	10
vertex-hard	neutral	neutral	$4.93 \times 10^{-4} \pm 3.89 \times 10^{-5}$	1.000 ± 0	1.000 ± 0	10
vertex-hard	random	random	0.475 ± 0.271	0.523 ± 0.333	0.783 ± 0.172	10

Table 26: Exact-factorization results for contiguous teachers. Values are mean $\pm$ sample standard deviation across seeds.

Method	Init relation	Raw init	Rel. error	ARI	Assignment acc.	n
k-means	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
oracle	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
soft	matched	contiguous	$0.003 \pm 8.04 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
soft	mismatched	cycle	$0.004 \pm 6.87 \times 10^{-4}$	0.636 ± 0.158	0.725 ± 0.142	10
soft	neutral	neutral	0.005 ± 0.001	0.929 ± 0.150	0.950 ± 0.105	10
soft	random	random	0.006 ± 0.003	0.929 ± 0.150	0.950 ± 0.105	10
straight-through	matched	contiguous	0.004 ± 0.001	1.000 ± 0	1.000 ± 0	10
straight-through	mismatched	cycle	0.009 ± 0.025	1.000 ± 0	1.000 ± 0	10
straight-through	neutral	neutral	0.006 ± 0.016	1.000 ± 0	1.000 ± 0	10
straight-through	random	random	0.004 ± 0.005	1.000 ± 0	1.000 ± 0	10
vertex-hard	matched	contiguous	0.005 ± 0.002	1.000 ± 0	1.000 ± 0	10
vertex-hard	mismatched	cycle	0.855 ± 0.002	$-0.222 \pm 2.93 \times 10^{-17}$	0.333 ± 0	10
vertex-hard	neutral	neutral	$5.21 \times 10^{-4} \pm 6.09 \times 10^{-5}$	1.000 ± 0	1.000 ± 0	10
vertex-hard	random	random	0.492 ± 0.220	0.525 ± 0.297	0.792 ± 0.153	10

Table 27: Exact-factorization results for palindrome teachers. Values are mean $\pm$ sample standard deviation across seeds.

Method	Init relation	Raw init	Rel. error	ARI	Assignment acc.	n
k-means	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
oracle	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
soft	neutral	neutral	0.007 ± 0.006	0.940 ± 0.126	0.967 ± 0.070	10
soft	random	random	0.005 ± 0.002	0.910 ± 0.145	0.950 ± 0.081	10
soft	structured-other	contiguous	$0.004 \pm 6.92 \times 10^{-4}$	0.970 ± 0.095	0.983 ± 0.053	10
soft	structured-other	cycle	$0.004 \pm 7.73 \times 10^{-4}$	0.940 ± 0.126	0.967 ± 0.070	10
straight-through	neutral	neutral	$9.97 \times 10^{-4} \pm 5.35 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
straight-through	random	random	0.046 ± 0.128	0.969 ± 0.098	0.975 ± 0.079	10
straight-through	structured-other	contiguous	0.010 ± 0.026	1.000 ± 0	1.000 ± 0	10
straight-through	structured-other	cycle	$0.002 \pm 9.25 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
vertex-hard	neutral	neutral	0.092 ± 0.201	0.936 ± 0.163	0.958 ± 0.106	10
vertex-hard	random	random	0.604 ± 0.136	0.359 ± 0.237	0.708 ± 0.132	10
vertex-hard	structured-other	contiguous	0.907 ± 0.001	-0.243 ± 0	0.333 ± 0	10
vertex-hard	structured-other	cycle	$0.795 \pm 4.82 \times 10^{-4}$	$0.139 \pm 2.93 \times 10^{-17}$	0.500 ± 0	10

Table 28: Exact-factorization results for random balanced teachers. Values are mean $\pm$ sample standard deviation across seeds.

Method	Init relation	Raw init	Rel. error	ARI	Assignment acc.	n
k-means	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
oracle	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
soft	neutral	neutral	0.005 ± 0.002	0.858 ± 0.183	0.900 ± 0.129	10
soft	random	random	0.007 ± 0.003	0.856 ± 0.254	0.900 ± 0.175	10
soft	structured-other	contiguous	$0.004 \pm 9.89 \times 10^{-4}$	0.818 ± 0.311	0.875 ± 0.212	10
soft	structured-other	cycle	$0.003 \pm 6.11 \times 10^{-4}$	0.965 ± 0.112	0.975 ± 0.079	10
straight-through	neutral	neutral	$5.97 \times 10^{-4} \pm 2.50 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
straight-through	random	random	0.105 ± 0.209	0.919 ± 0.173	0.942 ± 0.125	10
straight-through	structured-other	contiguous	0.004 ± 0.007	1.000 ± 0	1.000 ± 0	10
straight-through	structured-other	cycle	0.002 ± 0.001	1.000 ± 0	1.000 ± 0	10
vertex-hard	neutral	neutral	$5.00 \times 10^{-4} \pm 4.46 \times 10^{-5}$	1.000 ± 0	1.000 ± 0	10
vertex-hard	random	random	0.493 ± 0.279	0.497 ± 0.343	0.775 ± 0.162	10
vertex-hard	structured-other	contiguous	0.796 ± 0.061	-0.039 ± 0.172	0.475 ± 0.125	10
vertex-hard	structured-other	cycle	0.785 ± 0.050	-0.019 ± 0.127	0.492 ± 0.100	10

Table 29: Exact-factorization results for imbalanced teachers. Values are mean $\pm$ sample standard deviation across seeds.

Method	Init relation	Raw init	Rel. error	ARI	Assignment acc.	n
k-means	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
oracle	neutral	neutral	0 ± 0	1.000 ± 0	1.000 ± 0	10
soft	neutral	neutral	0.006 ± 0.004	0.997 ± 0.010	0.992 ± 0.026	10
soft	random	random	0.004 ± 0.002	0.960 ± 0.086	0.958 ± 0.044	10
soft	structured-other	contiguous	$0.004 \pm 9.57 \times 10^{-4}$	1.000 ± 0	1.000 ± 0	10
soft	structured-other	cycle	$0.289 \pm 2.68 \times 10^{-5}$	0.413 ± 0.028	0.617 ± 0.043	10
straight-through	neutral	neutral	0.008 ± 0.013	1.000 ± 0	1.000 ± 0	10
straight-through	random	random	0.216 ± 0.150	0.698 ± 0.275	0.800 ± 0.181	10
straight-through	structured-other	contiguous	0.030 ± 0.050	1.000 ± 0	1.000 ± 0	10
straight-through	structured-other	cycle	$0.289 \pm 1.44 \times 10^{-4}$	0.440 ± 0.017	0.658 ± 0.026	10
vertex-hard	neutral	neutral	0.662 ± 0.070	0.090 ± 0.101	0.458 ± 0.044	10
vertex-hard	random	random	0.658 ± 0.064	0.078 ± 0.098	0.500 ± 0.039	10
vertex-hard	structured-other	contiguous	0.722 ± 0.003	$-0.031 \pm 7.31 \times 10^{-18}$	0.500 ± 0	10
vertex-hard	structured-other	cycle	0.696 ± 0.002	0.026 ± 0	0.417 ± 0	10

Table 30: Full functional-distillation results by architecture, teacher pattern, and raw initialization. Values are mean $\pm$ sample standard deviation.

Student / init relation	Raw init	Test loss	ARI	n
MLP: contiguous / matched	pattern_contiguous	$0.006 \pm 8.56 \times 10^{-5}$	1.000 ± 0	5
MLP: contiguous / mismatched	pattern_cycle	$0.006 \pm 7.00 \times 10^{-5}$	-0.167 ± 0	5
MLP: contiguous / neutral	neutral	$0.012 \pm 1.99 \times 10^{-4}$	0.283 ± 0.082	5
MLP: cycle / matched	pattern_cycle	$0.005 \pm 6.21 \times 10^{-5}$	1.000 ± 0	5
MLP: cycle / mismatched	pattern_contiguous	$0.006 \pm 7.65 \times 10^{-5}$	-0.167 ± 0	5
MLP: cycle / neutral	neutral	$0.012 \pm 1.85 \times 10^{-4}$	-0.226 ± 0.056	5
MLP: palindrome / neutral	neutral	$0.011 \pm 2.24 \times 10^{-4}$	-0.077 ± 0.062	5
MLP: palindrome / structured_other	pattern_contiguous	$0.005 \pm 1.27 \times 10^{-4}$	-0.189 ± 0	5
MLP: palindrome / structured_other	pattern_cycle	$0.005 \pm 9.60 \times 10^{-5}$	0.075 ± 0	5
MLP: random_balanced / neutral	neutral	$0.012 \pm 1.91 \times 10^{-4}$	-0.053 ± 0.130	5
MLP: random_balanced / structured_other	pattern_contiguous	$0.006 \pm 9.98 \times 10^{-5}$	-0.108 ± 0.130	5
MLP: random_balanced / structured_other	pattern_cycle	$0.006 \pm 6.30 \times 10^{-5}$	-0.050 ± 0.261	5
Transformer: contiguous / matched	pattern_contiguous	$4.03 \times 10^{-4} \pm 9.10 \times 10^{-5}$	1.000 ± 0	5
Transformer: contiguous / mismatched	pattern_cycle	$4.09 \times 10^{-4} \pm 9.89 \times 10^{-5}$	-0.167 ± 0	5
Transformer: contiguous / neutral	neutral	$5.24 \times 10^{-4} \pm 1.30 \times 10^{-4}$	0.205 ± 0.050	5
Transformer: cycle / matched	pattern_cycle	$3.90 \times 10^{-4} \pm 6.52 \times 10^{-5}$	1.000 ± 0	5
Transformer: cycle / mismatched	pattern_contiguous	$4.04 \times 10^{-4} \pm 7.90 \times 10^{-5}$	-0.167 ± 0	5
Transformer: cycle / neutral	neutral	$4.98 \times 10^{-4} \pm 9.52 \times 10^{-5}$	-0.032 ± 0.122	5
Transformer: palindrome / neutral	neutral	$4.99 \times 10^{-4} \pm 8.18 \times 10^{-5}$	-0.091 ± 0.032	5
Transformer: palindrome / structured_other	pattern_contiguous	$4.20 \times 10^{-4} \pm 6.29 \times 10^{-5}$	-0.189 ± 0	5
Transformer: palindrome / structured_other	pattern_cycle	$3.93 \times 10^{-4} \pm 5.14 \times 10^{-5}$	0.075 ± 0	5
Transformer: random_balanced / neutral	neutral	$4.72 \times 10^{-4} \pm 1.11 \times 10^{-4}$	0.007 ± 0.159	5
Transformer: random_balanced / structured_other	pattern_contiguous	$3.91 \times 10^{-4} \pm 7.13 \times 10^{-5}$	-0.108 ± 0.130	5
Transformer: random_balanced / structured_other	pattern_cycle	$3.76 \times 10^{-4} \pm 7.10 \times 10^{-5}$	-0.050 ± 0.261	5

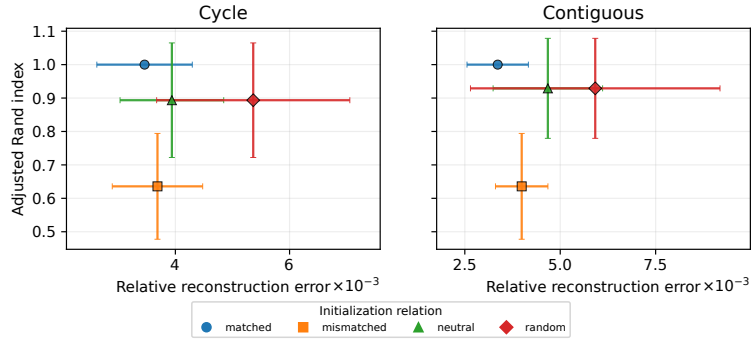


Figure 6: Soft-router exact-factorization runs for cycle and contiguous teachers. Reconstruction error and planted-graph recovery are separate axes.

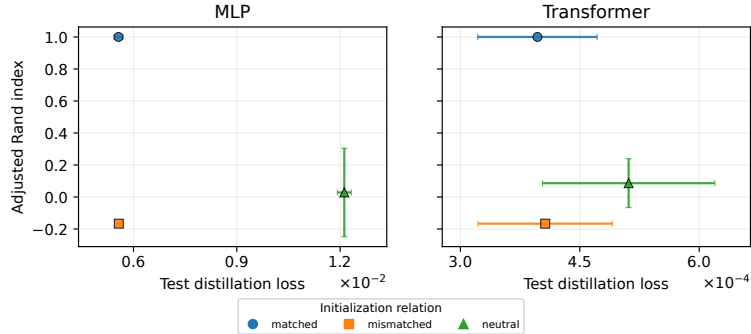


Figure 7: Distillation loss versus planted-graph ARI for the residual MLP and causal Transformer sanity checks.

Table 31: Paired probe recovery aggregated over probe counts, fixed additive-noise levels, graphs, and seeds. Functional modes share inputs, noise, and projection. The visible-effective-parameter baseline uses neither observation noise nor projection and is not a test of functional equivalence. Values are mean $\pm$ sample standard deviation.

Architecture	Signature	ARI	Empirical gap	n
MLP	shared input	0.676 ± 0.432	-27.192 ± 66.232	1680
MLP	native	0.675 ± 0.434	-27.293 ± 66.173	1680
MLP	effective parameter	1.000 ± 0	$0.008 \pm 5.82 \times 10^{-5}$	1680
Transformer	shared input	0.501 ± 0.471	-88.706 ± 190.527	1680
Transformer	native	0.501 ± 0.470	-88.758 ± 190.490	1680
Transformer	effective parameter	1.000 ± 0	$0.009 \pm 2.08 \times 10^{-5}$	1680

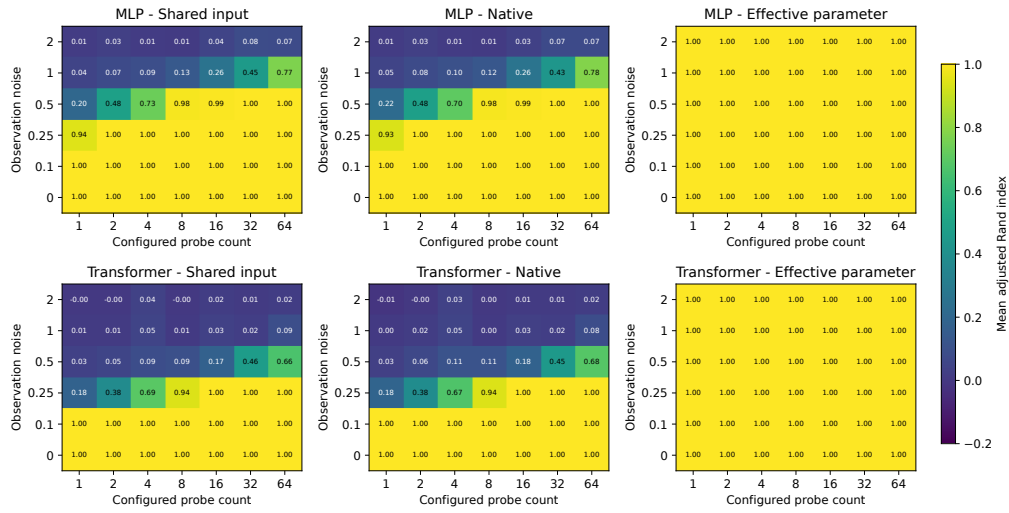


Figure 8: Shared-input, native-signature, and visible-effective-parameter recovery as probe count and fixed additive observation noise vary. The parameter baseline is repeated across the grid only for comparison and is not a functional-equivalence test. This is a planted-partition sanity check, not evidence of superiority on natural systems.

Functional-signature recovery generally improves with probe count and degrades with observation noise. The empirical within/between gap is computed against the planted partition, rather than against the predicted clusters, but remains only a finite-sample diagnostic unless the theorem’s population gap and uniform concentration assumptions are justified for this observation model.

P LANGUAGE-MODEL STABILITY RESULTS

Table 32: Language-model stress test. BPB is estimated from 100 seeded, sampled test batches. Values are mean $\pm$ sample standard deviation across seeds; pairwise ARI descriptively summarizes the three dependent run pairs.

Initialization	Sampled test BPB	L1 movement	Pairwise ARI	n
neutral	2.415 ± 0.009	0.012 ± 0.003	0.099 ± 0.157	3
pattern_contiguous	2.111 ± 0.008	$3.10 \times 10^{-4} \pm 1.35 \times 10^{-5}$	1.000 ± 0	3
pattern_cycle	2.181 ± 0.011	$3.50 \times 10^{-4} \pm 4.60 \times 10^{-5}$	1.000 ± 0	3
random	2.319 ± 0.015	0.065 ± 0.008	-0.045 ± 0.112	3

Distinct structured initializations can retain their partitions while reaching competitive BPB. With no privileged natural-language graph, this is evidence of initialization dependence, not evidence that either retained partition is false or historically correct. The one-of-three partition-fold witness in Table 33 is reported separately and does not convert this stability arm into a prevalence result.

Q REPRODUCIBILITY CHECKLIST

The response summary validates source hashes, checkpoint hashes, rank assumptions, analytic/native formula agreement, the permutation control, and all fixed probe outcomes. The separate tomography summary validates three newly generated raw artifacts, exact-gauge/common-design gates, rank and dimension conditions, component reconstruction, and the permutation and non-permutation controls without changing the older response generator. The ASLoRA summary validates a common protocol hash, generation-time source hashes, complete declared candidate coverage, the canonical equivalence gate, and exact restoration for every seed and gauge. The router-decision summary retains both exhausted searches, their full funnel counts, checkpoint hashes, and the paired successful-seed evaluation; unlike the other three summaries, it does not itself contain a generation-source-hash attestation, so we do not claim equal provenance strength.

The artifact additionally includes deterministic seed handling, raw JSON records, executable summarizers, the script that generates the three original audit LaTeX tables and their appendix details from validated summaries, tests for summary-gate failures, and unit tests for causal masking, permutation-invariant metrics, gauge equivalence, response formulas, structured response reconstruction and its approximate-product rejection boundary, restoration, and probe recovery. Dataset download is separated from training and accesses no private data.

R ADDITIONAL EXPERIMENTS: PROTOCOLS AND NUMERICAL LIMITS

R.1 HISTORICAL BYTE-LM WITNESS

Table 33: Finite same-group-budget partition-fold search at trained, initialization-dominated byte-LM checkpoints. “Not found” means the predeclared finite search exhausted; it is not a non-existence claim. Fold BPB compares partitions refit from the same effective tensors. The sole measurable consequence was not replicated under the same search (one witness in three seeds).

Seed	Search	Fold BPB ref. $\rightarrow$ cand.	Δ BPB [95% CI]	Replicated?
0	found	2.175 $\rightarrow$ 4.815	+2.640 [+2.629, +2.651]	no (1/3)
1	not found	—	—	no (1/3)
2	not found	—	—	no (1/3)

The original fixed-budget search succeeds at one of three checkpoints; all three routers are initialization-dominated. The successful candidate changes group sizes from 3/3/3/3 to 5/3/3/1, so its large BPB difference does not isolate membership from balance. Failed searches store no downstream fold or baseline evaluation. The detailed historical protocol and conditional interval interpretation remain in Appendix M. The equal-size experiments below follow a separately recorded protocol.

Training and matched decision budgets. The extension uses fresh seeds 10–12, width 128, MLP width 512, 12 layers, four bases, eight heads, context 128 and batch size 16. AdamW uses base learning rate 3×10^{-4} , router multiplier ten, 300 warmup steps, cosine decay to one tenth, weight decay 0.1 for weights and zero for routers, and BF16 autocast. There is no routing regularizer or temperature annealing. Only the fixed final step 6,000 is audited; a separate 100-step seed-90 validation-only pilot checked implementation and timing. This configuration differs from the original checkpoints and is not a controlled comparison of training lengths. Router movement and validation curves do not alone prove convergence or task-essential routing.

The positive-stochastic and signed-local families retain the previously specified distributions and condition-number bound 30; the new legality gate requires identical sorted group sizes. Selection

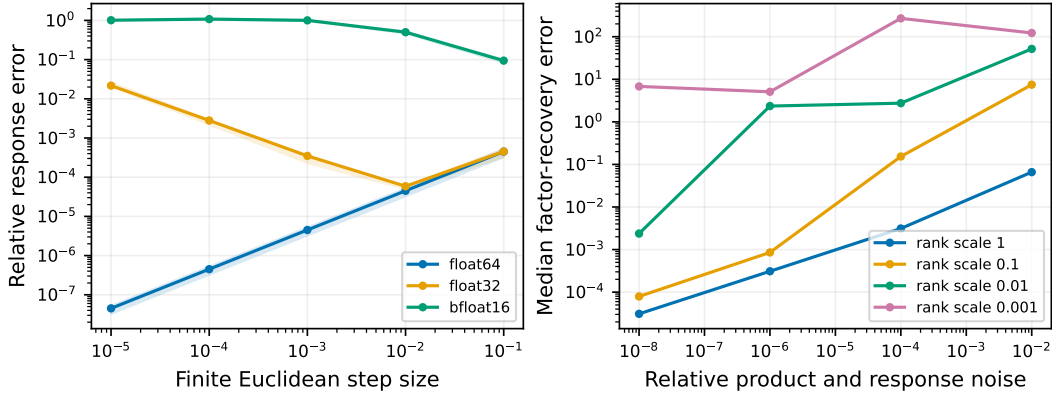


Figure 9: Two numerical limits. Left: finite-step response error at the three new checkpoints; curves show medians and shading the observed seed range. Smaller steps reduce FP64 truncation error but amplify low-precision subtraction error. Right: median factor error over ten synthetic seeds as both product and response observations are perturbed. The rank scale contracts the router toward a rank-one uniform matrix; it is not a noise level. These finite grids are diagnostic rather than worst-case guarantees.

sees router probabilities only. Test evaluation uses 64 disjoint batches of eight 128-token sequences, with identical stored starts for every action. Recovery uses 500 updates at learning rate 10^{-4} , fresh AdamW state, weight decay 0.1 and clipping norm one, evaluated at steps 0,100,500. The hard routing assignments are frozen; inactive stored slots are retained, so no latency or realized storage saving is claimed. The fixed-size effective-parameter comparator uses 16 seeded restarts and at most 50 improving pair swaps per restart to minimize centroid SSE. It is a local search, not a global optimum or a task-loss optimizer. Unconstrained Ward is included descriptively, since its group sizes may differ.

Natural-gradient measurement. Each checkpoint contributes 16 training minibatches of two 32-token sequences. The effective-weight forward matches the native model, and its gradients, pulled back through the factor map, match the native factor gradients. For numerical conditioning, each direction and its measured response receive the same normalization. The complement Gram and local linear systems in Appendix E.2 use a fixed relative pseudoinverse cutoff 10^{-12} . Factor recovery retains the original fixed diagonalization weights. All ranks, spectra, component errors, held-out predictions, unsuccessful low-probe reconstructions and finite-step measurements are stored. The last four directions are held out for every q ; larger fits are nested, so these measurements are not independent training replicates.

Joint noise and degeneracy. A prospective 450-cell FP64 synthetic grid uses ten seeds with $L = 6$, $K = 3$, $D = 16$, interior factors and contractions toward router rank loss or simplex vertices. Relative product and response noise levels are 0, 10^{-8} , 10^{-6} , 10^{-4} , 10^{-2} ; common noise directions couple levels within each cell. The observed product is truncated to rank K , its row space supplies the designed probes, and the true response to those probes is measured before adding response noise. This tests geometry estimation as well as response error. All runs return factors. Near rank loss, however, relative factor errors can exceed 60 even at 10^{-6} noise, and returned factors need not be feasible. Corollary 6 requires valid, uniformly nondegenerate factors and does not certify these unconstrained outputs.

Artifacts and scope. Versioned protocols and source-bound raw records are supplied under `research/` and `results/enhancement/`, including checkpoints, seeds and corpus hashes. Each seed’s paired bootstrap uses 10,000 resamples of its evaluation blocks; intervals are conditional and do not substitute for independent training seeds. The original raw records and negative findings remain available alongside these additional experiments.

S NOISY RECOVERY AND REJECTION UNDER A FIXED PROTOCOL

Prospective splits and observations. The local protocol and source freeze are `research/TRUSTED_RECOVERY_PROTOCOL.md` and `research/TRUSTED_RECOVERY_FREEZE.json`. They are timestamped execution records, not an independent preregistration. Development uses synthetic seeds 0–3 and existing byte-LM seed 10. Calibration uses synthetic seeds 100–119 and newly trained byte-LM seeds 20–21; evaluation uses synthetic seeds 200–239 and new byte-LM seeds 22–25. The six fresh checkpoints use the unchanged width-128, $L = 12$, $K = 4$, 6000-step configuration of the preceding controlled training study. There is no checkpoint selection on recovery outcomes.

Synthetic factors have $L = 6$, $K = 3$, $D = 16$, random-normal bases, and four router regimes: interior softmax, contractions toward the uniform router by factors 0.1 and 0.01, and a contraction toward balanced one-hot rows by 0.01. Each instance supplies 16 normalized Gaussian gradient directions. Language directions are actual effective-parameter cross-entropy gradients from 16 training-split minibatches of size 2×32 , sampled with fixed seed offset 3100000. Four or twelve initial directions fit the estimator; the last four assess held-out response fit. Response observations come from automatic differentiation of the factor map, without calling its closed-form response formula.

Product and each response receive independently drawn Gaussian perturbation directions rescaled to relative Frobenius norms $\sigma \in \{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}\}$. Directions are shared across noise levels within an instance. The supplied bound σ is known; the true perturbation vector and factors are not supplied to recovery. The estimated rank- K product row subspace is used by all methods, and evaluation includes basis error outside it. This observation-noise experiment does not model biased, adversarial, quantized, or unknown optimizer-state errors.

Fixed estimators and observable scores. The projected spectral baseline clips router entries below 10^{-8} , normalizes rows, and refits the basis by least squares. Joint fitting uses simplex logits in orthonormal zero-sum coordinates and normalized basis coordinates, with SciPy trust-region reflective least squares and exact forward-mode Jacobians. Three starts are fixed: projected spectral, a 0.1 logit perturbation, and a random simplex router. Each has a budget of 80 function evaluations; the smallest observed training residual selects the output. There is one developed solver variant, with no tuning on calibration or evaluation factors.

All acceptance rules require a numerically feasible router and finite score. The combined score is the maximum of (i) training plus held-out residual plus twice the normalized noise budget, divided by the simplex-tangent Jacobian’s smallest singular value; (ii) an observed product-subspace perturbation ratio; and (iii) permutation-aligned disagreement among competitive starts for the joint fit. A separately disclosed 10^{-10} numerical floor enters the empirical noise budget. The comparison scores use only residual or inverse Jacobian singular value. Their implementation and thresholds are frozen, and no score uses truth at prediction time.

For each method/score pair, one threshold is chosen on pooled calibration data to maximize accepted count, subject to at least 20 accepted synthetic outputs and at most 5% bad among all accepted outputs. Ties are accepted together. An infeasible threshold would reject all and be reported as a failure. The same threshold transfers to both evaluation domains. Good errors are at most 0.05, bad errors exceed 0.10, and intermediate errors are reported separately. This empirical calibration is not a distribution-free risk guarantee.

Results, including unsuccessful diagnostics. All 1920 synthetic joint fits return; there are 1639 good, 54 intermediate, and 227 bad outputs. Projected spectral recovery has three failed outputs, all retained in the 1920-case denominator. Seventy selected joint-fit runs exhaust their 80-evaluation budget; a returned estimate is not a convergence claim. Competitive-start agreement is also not a uniqueness test, especially if only one start meets the residual cutoff. Joint fitting improves error in 1916 of the 1917 paired successful synthetic comparisons and all 48 language comparisons. This comparison counts correlated variants, not independent replications.

The combined joint-fit diagnostic rejects 59.47% of bad synthetic outputs, with 0.18% false rejection of good outputs and 92.81% coverage. Seed-cluster bootstrap 95% intervals, resampling the 40 independent seeds and retaining all their variants, are respectively [54.79%, 63.82%], [0, 0.49%], and

Table 34: Synthetic held-out diagnostic comparison (percent). Coverage is the accepted fraction; bad rejection is recall on errors > 0.10 ; good rejection is false rejection on errors ≤ 0.05 ; accepted bad is the bad fraction among accepted outputs. Each threshold was frozen using calibration only.

Method	Score	Coverage	Bad reject	Good reject	Accepted bad
Spectral	Combined	10.21	100.00	85.04	0.00
Spectral	Residual	10.21	100.00	85.04	0.00
Spectral	Condition	10.26	100.00	84.96	0.00
Projected	Combined	72.71	90.11	0.00	4.01
Projected	Residual	72.66	92.23	0.00	3.15
Projected	Condition	21.30	98.06	70.48	2.69
Joint fit	Combined	92.81	59.47	0.18	5.16
Joint fit	Residual	91.51	70.04	0.18	3.87
Joint fit	Condition	24.32	91.63	72.73	4.07

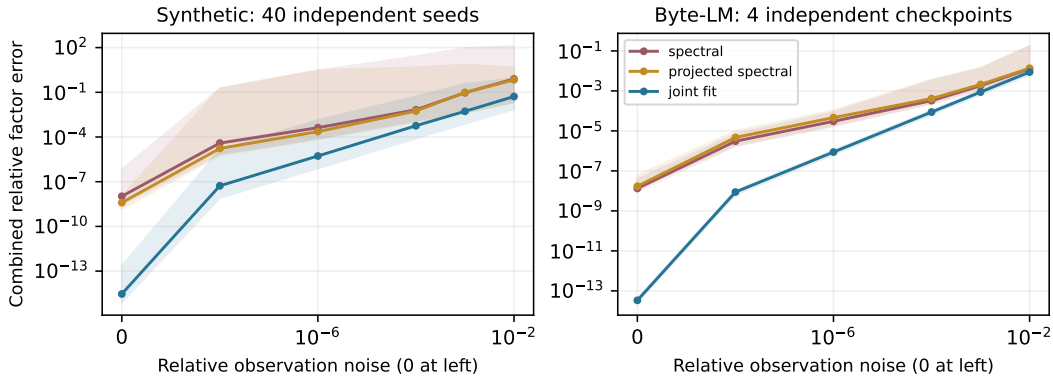


Figure 10: Held-out recovery under simultaneous product and response noise. Lines show medians and shading the 10th–90th percentiles of the fixed grid, not confidence intervals. Synthetic aggregation includes all four regimes and both probe counts; language aggregation includes four checkpoints and both probe counts. Zero noise is placed at the leftmost plotting coordinate. Tail failures and diagnostic errors are reported separately.

[92.08%, 93.54%]. Residual-only rejection instead rejects 70.04% of bad outputs with the same good false-rejection rate, at 91.51% coverage. Condition-only rejection discards 72.73% of good outputs. The raw spectral method’s feasibility gate rejects almost all noisy candidates. Its high rejection recall therefore comes with very low coverage.

Failures concentrate near router rank collapse. At contraction 0.01, joint fitting has 165/480 bad outputs; the combined score accepts 352, including 37 bad outputs. At contraction 0.1, it accepts 473 of 480, including 55 bad outputs. Thus aggregate calibration masks substantial regime-specific error rates. Interior and near-boundary regimes have no errors above 0.10 on this grid. The largest joint-fit error is 10.54; the largest falsely accepted error, 0.820, occurs at seed 210, contraction 0.01, $\sigma = 10^{-3}$, $q = 4$. These cases remain in the released records and were not used to retune the primary thresholds.

All 48 language joint fits are good and accepted. Maximum error at seeds 22, 23, 24, and 25 is respectively 0.00752, 0.01063, 0.01002, and 0.00650. There are only four independent training runs, and no bad joint-fit output in this cohort: bad-output rejection recall is undefined, not demonstrated to be perfect. The projected baseline has one bad language output and accepts it. The condition-only joint-fit rule rejects every language output. Rejecting every output provides no usable recovery estimate.

Numerical local separation and reproducibility. The 140 synthetic local passes occur in the interior (60) and near-boundary (80) regimes; there are none in the two rank-contracted regimes. Only two language outputs pass, both at seed 22. The pass concerns an estimated-subspace feasible component and has no unconditional truth-error interpretation. Raw observations, estimates, starts,

failures, diagnostics, source hashes, calibration thresholds, and per-seed summaries are released under `results/trusted_recovery/`. Full-coordinate theorem checks are distinct from the recovery experiment. A separate CPU environment replays the frozen observations; its results are recorded separately because floating-point optimization need not be bitwise identical. Across all 1968 cases, good/intermediate/bad classifications agree for every method. Combined-score acceptance changes in four spectral and two joint-fit cases; the largest joint-fit error difference is 0.0171. This replay measures agreement between environments directly. The release verification rebuilds the reported tables and checks source and threshold bindings.